MRC National Institute for Medical Research

WHO INFLUENZA CENTRE

LONDON

Report prepared for the WHO annual consultation on the composition of influenza vaccine for the Southern Hemisphere

September 2010

WHO Collaborating Centre for
Reference and Research on Influenza
National Institute for Medical Research
The Ridgeway
Mill Hill
London, NW7 1AA

Dr. John McCauley (Director)
Dr. Rodney Daniels (Deputy Director)
Dr. Yi Pu Lin (Assistant Director)
Dr. Xiang Zheng
Dr. Ting Hou
Ms Victoria Gregory
Ms Lynne Whittaker
Mr Nicholas Cattle
Ms Chandrika Halai
Dr. Karen Cross
Mr Johannes Kloess

Tel: 020 8816 2149; 020 8816 2152
Fax: 020 8906 4477
Email: jmccaul@nimr.mrc.ac.uk; whocc@nimr.mrc.ac.uk

Acknowledgements

We thank all who have contributed information and viruses, and associated data, to the WHO Global Influenza Surveillance Network, which provide the basis for our current understanding of recently circulating influenza viruses, and this summary.

CONTENTS

Southern Hemisphere Vaccine Recommendation Meeting: September 2010.
Summary of NIMR Results

Influenza Activity October 2009 to September 2010.

Influenza activity continued to be dominated by the pandemic A(H1N1) virus over this period, but in recent months there has been a resurgence of H3N2 and influenza B (both lineages) activity. Across Europe and the wider WHO European Region influenza activity has been low following the end of the 2009-10 season.

Figure 1 (page **9**) shows the number of viruses characterised through the sentinel and non-sentinel surveillance systems within the WHO European Region (as reported to WHO EuroFlu). Since October 2009, ~170,000 viruses have been typed/subtyped with 98.88% of the subtyped influenza A viruses being H1N1 pandemic and 0.44% being H3N2. The apparent 0.68% of seasonal H1N1 viruses has not been confirmed and we have not received these samples. Influenza B detections have been low, representing just 1.37% of the total number of viruses typed. The bulk of these virus detections/typings were during the European influenza season and were reported by EU/EAA countries (ECDC data not shown). Latest figures on deaths related to infection by H1N1 pandemic viruses since the beginning of the pandemic are from the end of May 2010 and since week 4/2010 there have been approximately 382 (1910 in total) reported to ECDC and 1231 (4880 in total) reported to EuroFlu.

In previous seasons there has been evidence for West to East transmission of influenza across the WHO European Region and this may have had a role in the relative timings of influenza seasons and related deaths. In **Figure 2** (page **10**) we show individual country data for a Central European country (Germany, which experienced high intensity of infection in the 'seasonal' wave) and countries to the east of Poland in the WHO EURO Region. There is some indication of a delay in peak activity moving West to East, but across the WHO EURO Region the influenza season was effectively over by week 6/2010. Thereafter low influenza activity has been reported in all countries although there has been some sustained level of influenza B activity notably in Latvia and the Russian Federation.

Even within a small geographic region like the UK distinct patterns have been observed; Northern Ireland and, to a lesser extent, Scotland had more intense 'seasonal' peaks, compared to England and Wales, than those reported in the summer waves (**Figure 3**, page **11**). Such outcomes may relate to the different timings of school terms between England/Wales and Northern Ireland/Scotland. Since the end of the 2009/10 influenza season, influenza activity has been low across all of the UK and levels of ILI reporting have been spread equally across all age groups.

The overall effects of the H1 pandemic in Europe (EuroFlu data) has been an early start to the season, bringing peak activity forward by some 10-12 weeks, approximately a 3-fold higher number of influenza detections and suppression of all seasonal influenza types/subtypes, compared to the three previous European (Northern Hemisphere) seasons (**Figure 4**, page **12**). The suppression of type is particularly noticeable for influenza B compared to the 2007/8 and 2008/9 seasons, although there has been sustained low level circulation in some countries following the 'seasonal' wave (**Figure 2**, page **10**). An additional effect has been on the timing of Respiratory Syncytial Virus (RSV) activity compared to the preceding three seasons (**Figure 5**, page **13**). RSV activity preceded influenza activity in the three previous seasons but in 2009/10 RSV activity was delayed although numbers of detections ultimately approached those observed in the three previous seasons and in all four seasons RSV detections effectively ceased by week 20. The reasons for this are not clear.

Antigenic and Genetic Characteristics of pandemic A(H1N1) and seasonal influenza viruses isolated January to September 2010.

Table 1 (page **14**) summarises the clinical specimens and virus isolates received for samples taken between 01 January 2010 and the end of August 2010 in 29 countries of Europe, Africa and Asia. Propagation success rates varied from 31% (H3N2) to 77% (B/Yamagata lineage) and were 50% overall. The majority of isolates or clinical samples were pandemic A(H1N1) viruses, but influenza B/Victoria-lineage and H3N2 viruses also were well represented. The last seasonal H1N1 virus received was collected in October 2009.

Pandemic A(H1N1) Virus Analyses

Table 2 (page **15**) summarises the samples received in which pandemic A(H1N1) viruses were detected. One hundred and eighty samples were propagated for antigenic analysis by HI assay and assessment of drug resistance. (Isolation and propagation of viruses from clinical samples with high Ct values estimated by qRT-PCR was not attempted).

Tables 3, 4 and 5 (pages **16-18**) show the results of HI assays using a panel of anti-sera with turkey red blood cells. The vast majority were closely related antigenically to the prototype and vaccine viruses (A/California/4/2009 and A/California/7/2009). Only a single virus, 1 out of 180, showed an 8-fold drop in HI titre with the ferret antiserum directed against the vaccine virus (A/California/7/2009). Yellow and green highlighting in the Tables indicate viruses subject to HA and NA sequence analyses, those in yellow are shown in the phylogenetic trees presented in **Figures 6** (page **19**) and **9** (page **23**).

Sequence alignments of a subset of the viruses shown in **Figure 6** (HA, page **19**) and **Figure 9** (NA, page **23**) are presented in **Figure 7** (pages **20-21**) and **Figure 10** (pages **24-25**) for the HA and NA protein sequences respectively. Amino acid residues defining sub-clades are shown on the phylogenetic trees.

Several sub-clades can be seen for the HA gene in **Figure 6** (page **19**). Many recent isolates have grouped within two sub-clades. One sub-clade is defined by a substitution at amino acid 125 (with additional substitutions at amino acid residues 94 and 250 being observed in several viruses, represented by A/Christchurch/16/2010). The second sub-clade carries substitutions at residues 128, 199 and 295 of the HA, represented by A/Hong Kong/2213/2010, with additional substitutions at residues 163 and 271 in a sub-set of viruses.

The amino acids associated with the two sub-clades of the HA are mapped onto an H1 HA structure in **Figure 8** (page **22**). Several of the residues are exposed on the surface of the HA trimer and some are present in the globular head. However, viruses from both of the sub-clades show no significant difference in their antigenic properties.

As for the HA gene, many recent isolates group within two sub-clades for the NA gene (**Figure 9**, page **23**) defined by changes at positions 365 and 386 (viruses isolated in Hong Kong) and 15 and 189 (viruses of more diverse origin), with the same grouping as seen for the HA. Any significance of this co-variation between the HA and the NA is not overt.

Figure 25 (page **54**) **and Figure 26** (page **54**) present a summary of the testing and results of drug sensitivity assays of the propagated pandemic (and seasonal) H1N1 viruses. The assays were carried out using MUNANA as substrate in the presence of oseltamivir or zanamivir. **Figure 27** (page **55**) shows the total number of pandemic viruses received at NIMR observed to be resistant to oseltamivir since the start of the pandemic. The last resistant virus was collected in December 2009. **Figure 28** (page **55**) summarises the number of NA genes sequenced and represents those determined for both virus isolates and clinical samples.

The data obtained from the viruses resistant to oseltamivir and zanamivir has also been analysed to examine whether viruses with distinct inhibition profiles could be observed. A total of 4 out of 617 viruses were identified as having elevated IC_{50} values to oseltamivir or zanamivir and amino acid substitutions in the NA gene associated with the altered profile are shown (**Figure 29**, page **56**), and mapped onto the NA (**Figure 30**, page **57**).

Conclusion:
The A(H1N1) pandemic 2009 viruses propagated at NIMR remain antigenically similar to the vaccine virus A/California/7/2009. Fewer low reactors have been detected in 2010 than was observed in 2009. New genetic sub-clades have been detected but they do not appear antigenically distinct from the majority of A(H1N1) pandemic 2009 viruses collected since the start of the pandemic.

Seasonal Influenza A Virus Analyses

H3N2 viruses

Table 6 (page **26**) shows a summary of the number of H3N2 clinical samples or isolates collected between January 2010 and August 2010 received and propagated at NIMR. Viruses were obtained from a small number (9) of geographically diverse centres.

As described previously and published (Lin et al. J. Virol. 84, 6769-6781, 2010), the NA glycoprotein of many H3N2 viruses can contribute to the agglutination of red blood cells of some species. Such activity can result in interpretation of HI assays being unreliable unless virus binding to red cells is known to be exclusively dependent on the HA. Some red blood cells are less susceptible to agglutination by the NA and for some viruses reliable results can be obtained by using guinea pig red cells in the HA and HI assay. However, many viruses in our hands no longer bind to guinea pig red blood cells predominantly through the HA glycoprotein. For many H3N2 viruses agglutination of guinea pig red blood cells was shown to be oseltamivir sensitive, which implies that many H3N2 viruses bind to guinea pig red cells through their NA glycoprotein. Our preferred option is to carry out HI assays using guinea pig red blood cells in the presence of 20nM oseltamivir. Lin et al. (*ibid*) described a polymorphism at residue 151 of the NA with Asp combined with Gly or Asn to be associated with the NA-dependent agglutination of red blood cells. (The location of residue 151 in the NA glycoprotein is shown in **Figure 16**, page **36**).

HI assays have been carried out for viruses showing agglutination of guinea pig red cells in the presence (**Table 7A**, **Table 8** pages **27** & **28**) or absence (**Table 7B**, page **27**) of oseltamivir. **Table 7A** (page **27**) shows that four viruses from Hong Kong isolated in March and April have good reactivity with sera raised against A/Perth/16/2009 and the other related viruses in HI assays carried out **in the presence of oseltamivir**. These results confirm that these viruses remain antigenically similar to A/Perth/16/2009. Two other viruses from Hong Kong and viruses from Stockholm and Ghana are analysed using guinea pig red blood cells (**Table 7B**, page **27**) **in the absence of oseltamivir** and these generally show a good reaction with the A/Perth/16/2009 serum and other sera raised against related viruses but the results shown in **Table 7B** (page **27**) cannot be conclusively interpreted in terms of the characteristics of the HA.

Two of the viruses analysed in **Table 7B** (page **27**) were re-analysed by HI assay carried out **in the presence of oseltamivir** along with three viruses received from other CCs (**Table 8** page **28**). Both test viruses showed reduced reactivity with reference sera in the HI assays. The newly received viruses from the Atlanta and Melbourne CCs showed low activities against one or more of the reference anti-sera used.

Phylogenetic analysis has been used to correlate alterations in the HI assay with genetic changes in the HA and NA gene. Phylogenetic trees of the HA gene (**Figure 11**, page **29**) and the NA gene (**Figure 14**, page **33**) are shown. Alignment of the amino acid sequences of the HA and NA are shown in **Figures 12 and 15** (pages **30-31** & **34-35**), respectively. Most recently collected viruses belonged to the Victoria/208 clade of the HA. Within the Victoria/208 clade three sub-clades can be presented; one is defined by substitutions at amino acid residues T48A and K92R along with N312S; another by S45N; a third sub-clade with recent viruses from Hong Kong, South America, Europe, North America and New Zealand is evident. Viruses in the latter sub-clade carry the changes: D53N, I230V, E280A, with a sub-set having an additional substitution at 94 and some of these also carrying substitution at residue 208.

Of the viruses in the sub-clade with substitutions at 53, 230 and 280, represented by A/Hong Kong/2121/2010, we have carried out HI assays **in the presence of Oseltamivir** for only A/Hong Kong/1737/2010, A/Hong Kong/1838/2010 and A/Stockholm/1/2010, (these carry **only** D53N, I230V and E280A). The two viruses from Hong Kong reacted well with the panel of anti-sera but the Stockholm virus showed a reduced level of reactivity against some members of the panel. The virus received from CDC, A/Alabama/5/2010, belonged to the same sub-clade and also carried the additional changes Y94H, R208K and showed good reactivity with the A/Perth/16/2009 anti-serum but low reactivity with some of the other anti-sera in the panel. The correlation of antigenic properties of viruses in this sub-clade with the HA sequence is complex.

We have no HI results using guinea pig red blood cells **in the presence of oseltamivir** for viruses carrying the S45N mutation. For viruses in the clade defined by T48A, K92R and N312S, only one (A/Rhode Island/01/2010) was assayed **in the presence of oseltamivir.** In the assay presented in **Table 8** (page **28**), A/Rhode Island/01/2010 showed varied reactivity across the panel of sera: it showed good reactivity with the A/Perth/16/2009 anti-serum but lower activity against the A/Victoria/208/2009 and A/Victoria/210/2009 anti-sera used in the assays.

The locations of the amino acid substitutions in the emergent sub-clades of the HA is shown in **Figure 13** (page **32**). Some of the changes are located in the globular head of the HA with residues 92, 94 and 208 being exposed at the surface of the trimer. Residue 230 lies within the receptor binding pocket.

A phylogenetic tree for the NA gene is shown in **Figure 14** (page **33**). Indicated on the tree are those sequences that have residue 151 substitutions (D151G or D151x, with x indicating a mixture). Many of the NAs from recent viruses display polymorphism at residue 151, whether there is minor polymorphism in other viruses is not known. It is striking that the recently collected viruses have an extensive number of changes in the NA: residues 367, 369 and 464 defining a sub-clade with other substitutions also accumulating.

Of 167 viruses tested for resistance to oseltamivir or zanamivir (**Figures 26 and 28**, pages **54** & **55**), no viruses showing resistance have been detected.

Conclusion:
Interpretation of antigenic analyses of the H3N2 viruses as assessed by HI is uncertain. The incorporation of oseltamivir in the assay can focus the assay on the role of the HA in neutralisation but this has proved less valuable in the analysis of recently isolated viruses with a large number of viruses showing little/no NA-independent binding to red blood cells of any species tested.

Several new genetic sub-clades of the Victoria/208 clade have emerged but we have no conclusive evidence to indicate that the observed genetic changes have resulted in a change in antigenicity.

Seasonal Influenza B Virus Analyses

B Victoria lineage

We have received 100 influenza B viruses of the Victoria lineage from 13 countries collected between January 1 2010 and August 31 2010; seventy four viruses were propagated (**Table 9**, page **37**). Many of the viruses showed a significant reduction in titre compared with the anti-serum produced against the **egg-isolated** vaccine virus B/Brisbane/60/2008.

Antigenic analysis by HI assays using turkey red blood cells is shown in **Tables 10 and 11** (pages **38** & **39**). Whilst the majority of viruses showed reduced reactivity with anti-serum raised against the egg grown virus B/Brisbane/60/2007, most showed good reactivity with sera raised against more recently collected influenza B viruses isolated and propagated in tissue culture cells. One virus showed an increased reactivity with anti-sera raised against older viruses and 9 viruses showed lower reactivity against the anti-sera raised against 2009 reference viruses. (Note that the cell propagated reference viruses – B/Paris/1762/2008, B/Madagascar/7002/2009 - each had two amino acid substitutions compared with the vaccine virus B/Brisbane/60/2008).

A phylogenetic tree of the HA gene is presented in **Figure 17** (page **40**) with an amino acid sequence alignment presented for a sub-set of the viruses in **Figure 18** (pages **41-42**). Few amino acid substitutions are evident compared with the current vaccine virus B/Brisbane/60/2008. No clear indication of particular amino acid substitutions being associated with consistent change in antigenicity was obtained as viruses carrying the same substitutions gave different patterns of HI reactivity, some showing low reactivity while others reacted well. The group of viruses from Madagascar showed slight reductions in HI activity with the reference anti-sera, possibly associated with the substitution A202V in the HA. A/England/63/2010 was unusual, and may represent a new clade of virus, falling close to the recently emerged clade represented by B/Bolivia/104/2010.

A phylogenetic tree of the NA gene is presented in **Figure 19** (page **43**) with an amino acid sequence alignment presented for a sub-set of the viruses in **Figure 20** (pages **44-45**). Numerous amino acid substitutions have been noted in the tree. The significance of these changes is unknown.

B Yamagata lineage

A summary of the B/Yamagata-lineage viruses received from January 1 2010 to August 31 2010 is shown in **Table 12** (page **46**). Only 13 viruses were received from a total of 6 countries. Ten were propagated in tissue culture cells and all propagated viruses showed good reactivity with anti-sera raised against the **egg grown** previous vaccine virus B/Florida/4/2006 (now a reference virus).

Antigenic analysis by HI assays using turkey red blood cells is shown in **Table 13** (page **47**). All viruses showed good reactivity to anti-sera raised against B/Florida/4/2006 virus as well as to anti-sera raised against egg-grown (B/England/145/2010, B/Bangladesh/3333/2007) or cell grown (B/Valladolid/18/2008) reference viruses.

Phylogenetic analysis of the HA gene (**Figure 21**, page **48**) showed that all of the recently collected viruses belonged to the B/Bangladesh/3333/2007 clade. An amino acid alignment for a subset of viruses is shown in **Figure 22** (pages **49-50**). Amino acid substitutions were observed at residues 121 and 183; 181; and 202 and 146. In our HI assays the viruses that carried T121A in combination with G183R showed no reduction in their reactivity with anti-sera raised against B/Florida/4/2006 or reference viruses, similarly those that carried N202S either in combination with 146S or with A146, also showed good reactivity with anti-sera raised against B/Florida/4/2006 or reference viruses. Likewise, the substitution T181K had no marked affect on antigenicity.

A phylogenetic tree of the NA gene for the B/Yamagata-linage viruses is presented in **Figure 23** (page **51**) with an amino acid sequence alignment presented for a sub-set of the viruses in **Figure 24** (pages **52-53**). Amino acid substitutions have been noted in the tree but there have been few amino acid substitutions observed within the viruses belonging to the B/Bangladesh/3333/2007 clade.

A total of 100 influenza B viruses have been tested for resistance to the antiviral drugs oseltamivir and zanamivir; none was found to be resistant (**Figure 26**, page **54**).

Conclusion:
Influenza B viruses of the Victoria-lineage continued to predominate over those of the Yamagata-lineage.

In the Victoria-lineage many viruses yielded reduced titres with anti-sera raised against the egg-grown vaccine virus. Anti-sera raised against reference strains isolated and propagated in mammalian cells continued to react well with recently isolated viruses. There have been only a small number of amino acid substitutions in recently isolated viruses in comparison with the current vaccine virus.

There was little evidence of antigenic change compared with reference strains in the small number of viruses of the B/Yamagata-lineage that were analysed.

Antiviral Resistance Screening

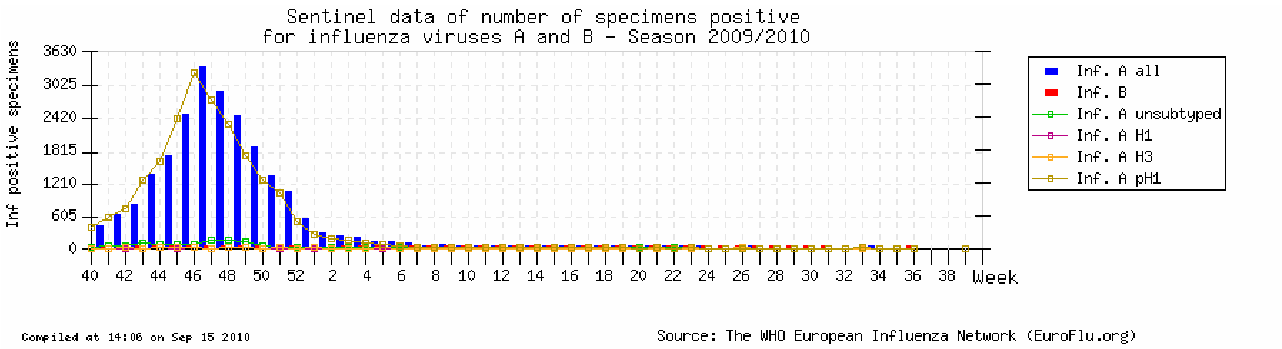
A specimen collection date time-line for viruses subjected to Neuraminidase Inhibitor (NI – both oseltamivir and zanamivir) susceptibility testing since January 2009, using a MUNANA fluorescent assay, is shown in **Figure 25** (page **54**). This covers viruses isolated since April 2009 for H1N1 pandemic, March 2009 for H3N2 and January 2009 for H1N1 seasonal and influenza B, with the last H1N1 seasonal virus being isolated in October 2009. **Figure 26** (page **54**) shows the number of viruses assayed by type/subtype and WHO region. In total 1766 viruses were assayed, 1477 H1N1 pandemic, 167 H3N2, 22 H1N1 seasonal and 100 influenza B. Of these just 7 (0.4%) H1N1 pandemic and 22 (100%) H1N1 seasonal viruses were oseltamivir resistant, all due to H275Y substitution in NA, and all viruses (1766) remained sensitive to zanamivir. Details of the 7 H1N1 pandemic oseltamivir-resistant viruses are shown in **Figure 27** (page **55**), all were related to drug use in either prophylaxis or treatment settings.

Additional to NI susceptibility phenotypic assays over 1000 full length NA gene sequences have been determined, some as part of full genome determinations (**Figure 28**, page **55**). These sequences confirmed NA H275Y substitution in all the H1N1 seasonal viruses analysed and only the 7 H1N1 pandemic viruses shown to be oseltamivir resistant by phenotypic testing. No other resistance markers were observed in NA genes of all influenza types/subtypes and, for H1N1 pandemic viruses, all NA genes encoded I223 indicative of retained susceptibility to oseltamivir and zanamivir (I223R substitution has been associated with reduced susceptibility to both oseltamivir and zanamivir). Among the 1477 H1N1 pandemic viruses assayed phenotypically, just 4 viruses showed decreased susceptibility to oseltamivir and/or zanamivir (**Figure 29**, page **56**). These 4 viruses were derived from countries with little/no NI drug use. A number of the amino acid substitutions occurring in the NAs of these 4 viruses were in the stalk region of NA, but all contained at least one substitution that could result in the reduced susceptibility: K262R and I396V (A/Georgia/4032/2009), Y274H (A/Livadia/1851/2010), S105N and I263V (A/Argos/2231/2010), D199G and R257K (A/Volgda/3/2010) (**Figures 29** & **30**, pages **56** & **57**). Substitutions at positions 105 and 262 have been associated with reduced NI susceptibility in N1 containing viruses of avian origin.

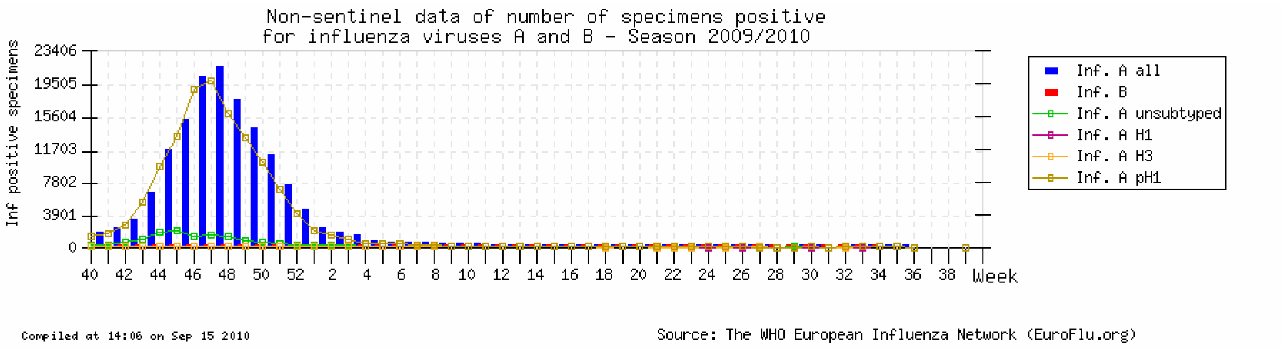
In terms of resistance to the M2-inhibitors (amantadine and rimantadine), full M gene sequencing has shown 100% resistance in H1N1 pandemic and H3N2 viruses due to the S31N substitution in the M2 protein, while H1N1 seasonal viruses remained susceptible.

Figure 1: Current European Situation (EuroFlu.org: 15.09.2010)

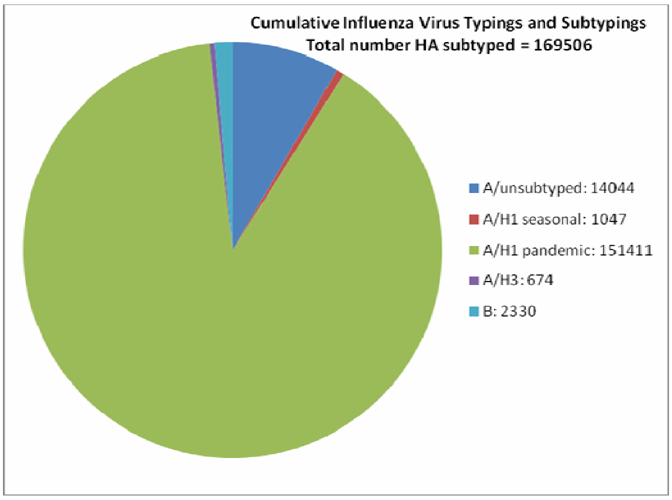
Sentinel Data



Non-sentinel Data



Virus typing/subtyping and H1pdm-associated deaths



Europe: H1pdm-associated deaths since the beginning of the pandemic (confirmed/probable).

	Wk 04/2010		
EuroFlu:	**3649**	**4880**	**(30.05.10)**
WHO/ud86:	**~3605**	**~4879**	**(01.08.10)ud112**
ECDC/WISO:	**1528**	**1910**	**(30.05.10)**

Figure 2: Eastern WHO Euro Country situations – EuroFlu.org (week 35/2010)

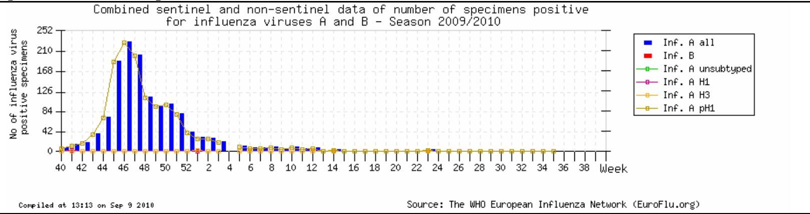
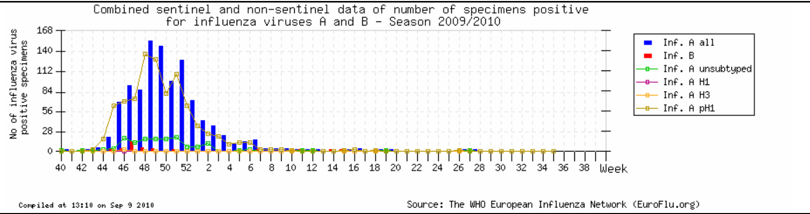
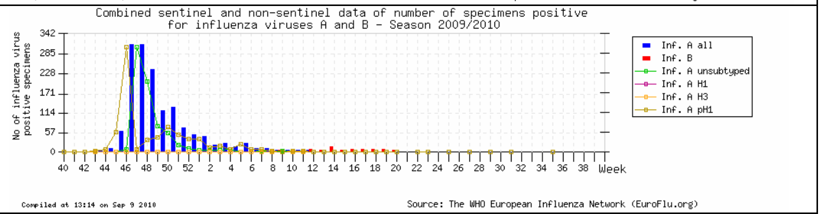
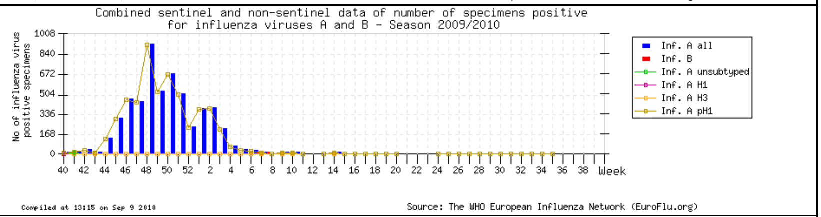
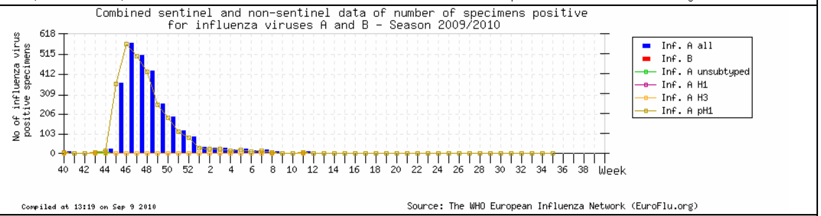
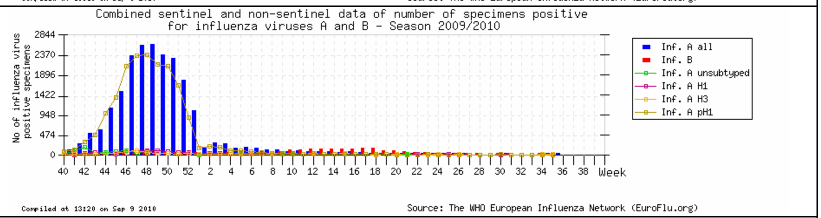
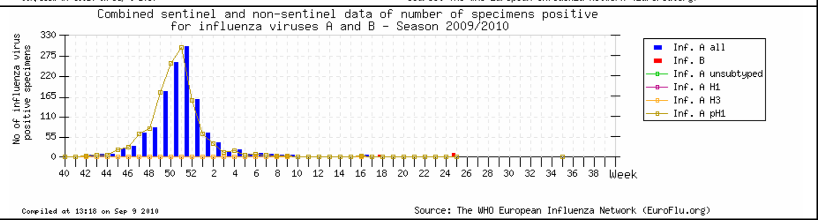
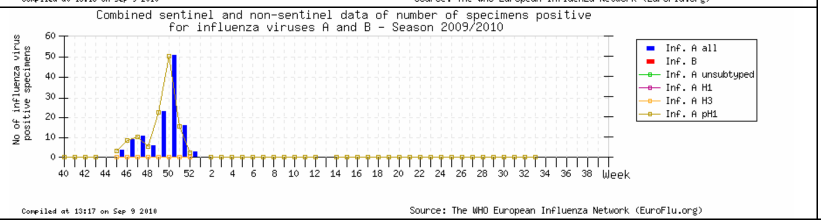
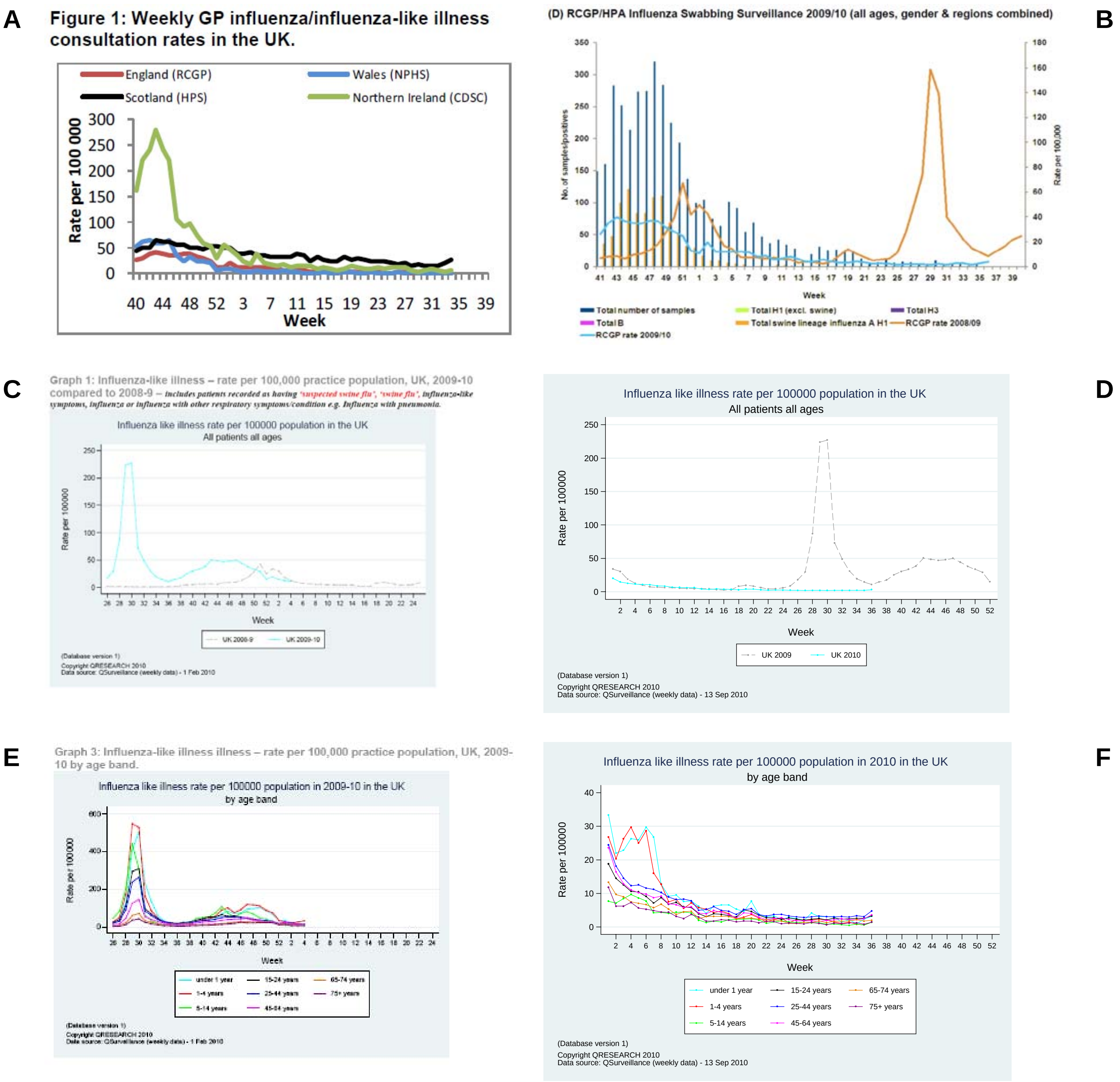
<table>
<tr><th>Country</th><th>Combined Sentinel/Non-sentinel influenza positive specimens</th><td rowspan="9">Despite some delay in peak influenza detections moving West to East across the WHO EURO Region, the Northern Hemisphere season was effectively over in all countries by week 6/2010. Thereafter low influenza activity has been reported.</td></tr>
<tr><td>Germany</td><td>
</td></tr>
<tr><td>Estonia</td><td>
</td></tr>
<tr><td>Latvia</td><td>
</td></tr>
<tr><td>Romania</td><td>
</td></tr>
<tr><td>Moldova</td><td>
</td></tr>
<tr><td>Russian Federation</td><td>
</td></tr>
<tr><td>Georgia</td><td>
</td></tr>
<tr><td>Armenia</td><td>
</td></tr>
</table>

Figure 3: UK situation – Figures from HPA/RCGP to week 36/2010



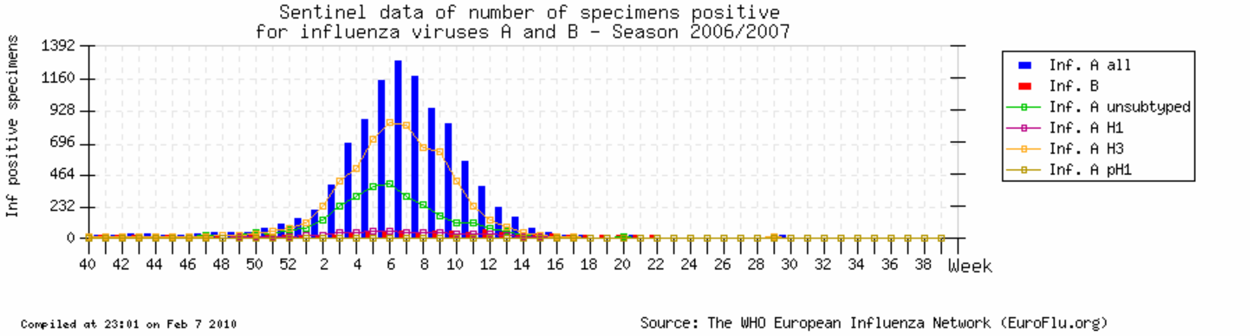
Based on the different national sentinel schemes, Northern Ireland had a more intense influenza season than the other three countries. All have had low levels of ILI since the end (wk6/2010) of the Northern Hemisphere [European] season (A). Within England and Wales, decreasing trend for influenza virus detection has been observed since wk6/2010 and H1 pandemic viruses have been dominant throughout (B).

Based on Q-surveillance that covers 19.4 million people in England, Wales, Scotland and Northern Ireland (2996 Medical practices), for ILI the epidemic threshold was crossed in the Summer 2009 wave of H1pdm activity and was much reduced in the autumn/winter 'seasonal' wave (C). Following the 'seasonal' wave influenza activity has been at /below the 'out of season' baseline level (D).

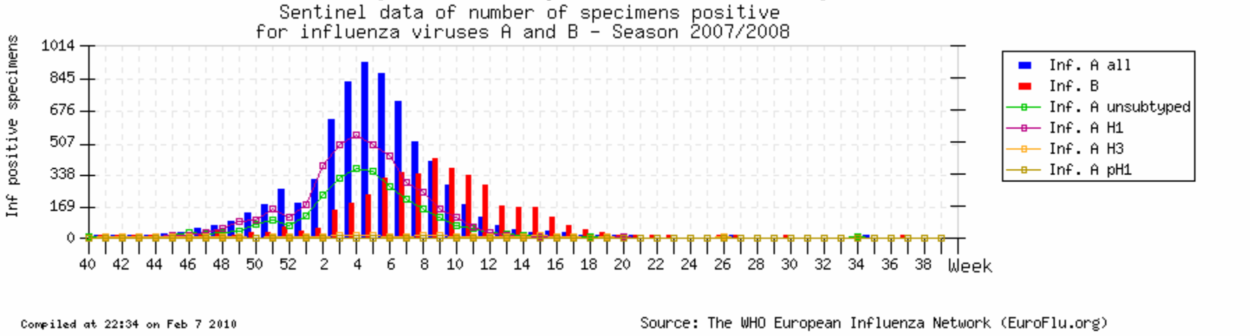
When age-stratified, the Q-surveillance data show that all groups aged <1-44 years presented at epidemic threshold crossing levels in the Summer 2009 wave of H1pdm activity (E). ILI was still pronounced in those aged <4 years at the end of the' seasonal' wave but subsequently has been equivalent across all age groups (F).

Figure 4: Influenza Season 'timing' by year (EuroFlu.org: 15.09.2010) Sentinel Data

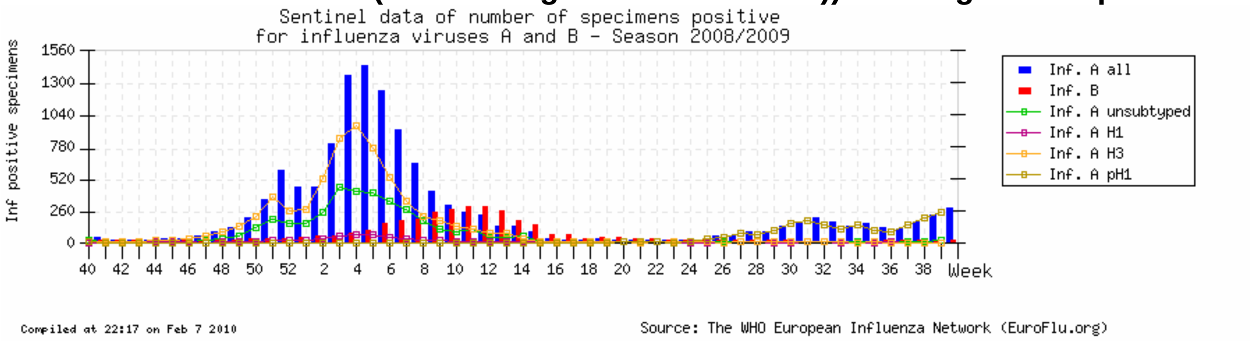
H3N2 Dominant 2006/07



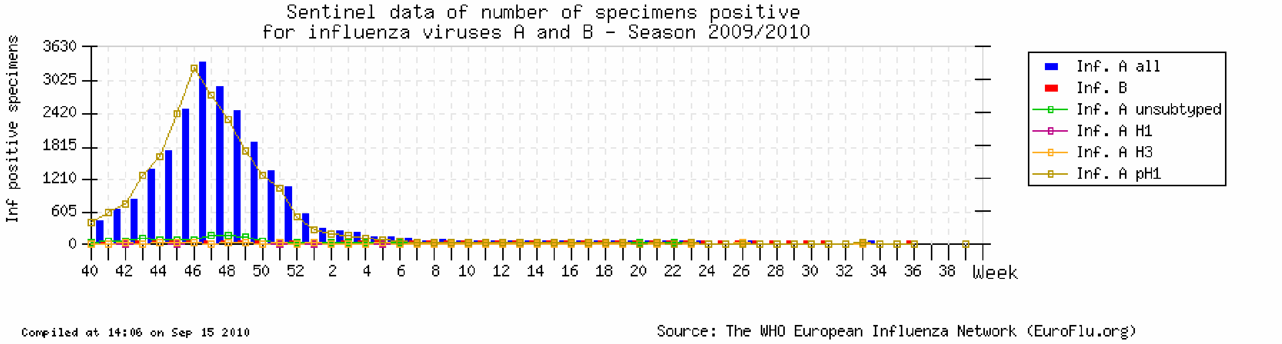
H1N1 Dominant 2007/08 (Flu B rising at end of season)



H3N2 Dominant 2008/09 (Flu B rising at end of season)) + emergence H1pdm



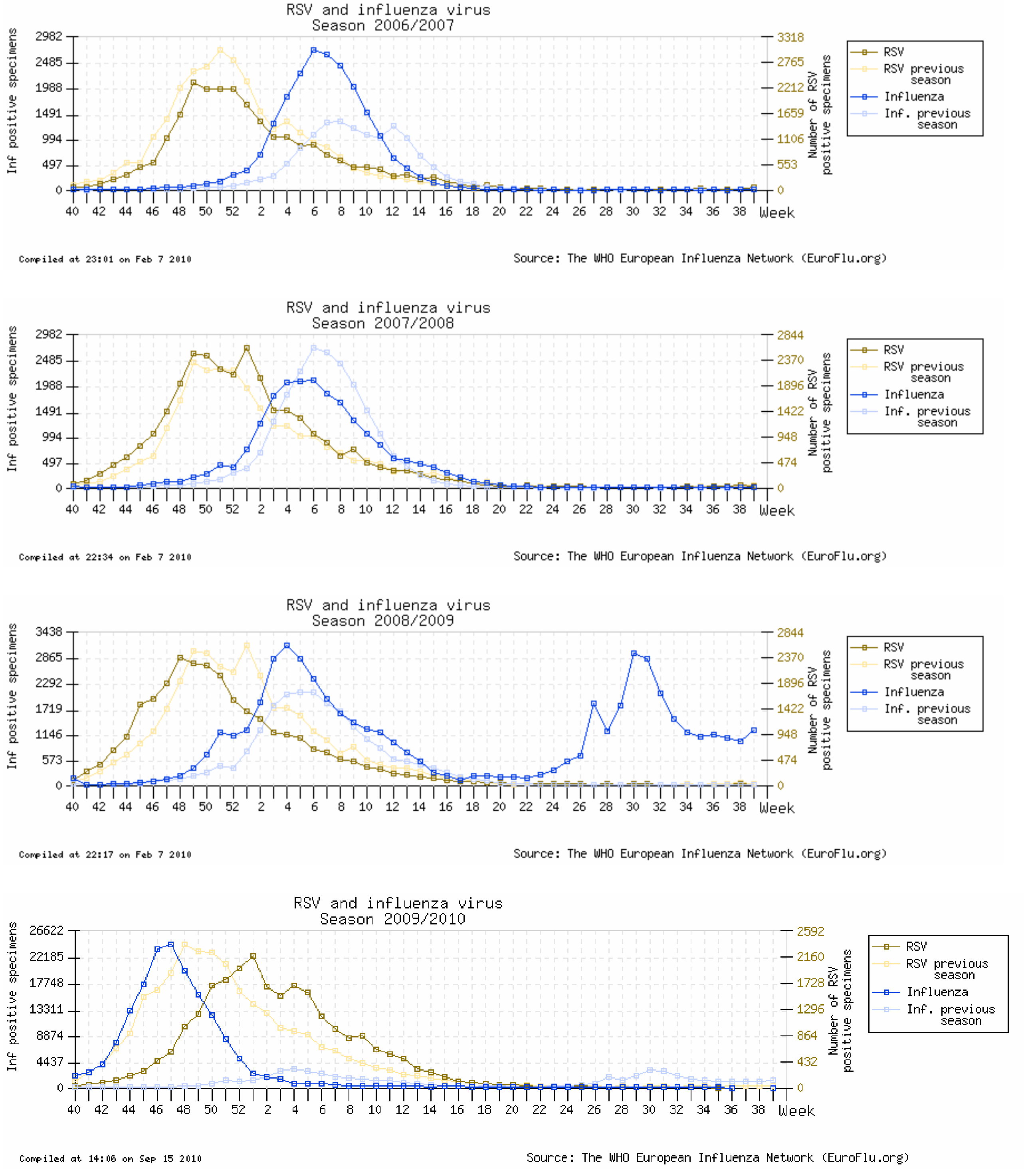
H1pdm Dominant 2009/10



2-3 fold higher number of positive specimens compared to previous three seasons.
'Season' started early, effectively over by wk6/2010, H1pdm suppressed seasonal influenza.

Figure 5: RSV and influenza – activity timings (EuroFlu.org: 15.09.2010)

Combined sentinel and non-sentinel data for numbers of positive specimens.



In the three previous seasons RSV peaks have preceded those for influenza.
'Early' H1pdm in 2009/10 was followed by RSV activity somewhat later than in previous seasons.

Table 1. Summary of clinical samples and isolates received, collected between January and August 2010

MONTH		H1N1pdm		H3N2		B Victoria lineage		B Yamagata lineage	
Continent	Number Countries	Number received	Number propagated	Number received	Number propagated	Number received	Number propagated	Number received	Number propagated
JANUARY									
Africa	**2**	17	7	3	0	5	1		
Asia									
Europe	**19**	201	76	1	1	4	1		
FEBRUARY									
Africa	**4**	20	15	5	1				
Asia									
Europe	**13**	94	35			4	3		
MARCH									
Africa	**2**	10	5	8	3	3	2		
Asia	**1**			3	3	2	1	2	2
Europe	**10**	16	12			29	26	4	2
APRIL									
Africa	**2**	7	4			2	1	1	1
Asia	**1**	4	4	1	1				
Europe	**10**	7	7	5	2	16	9	1	0
MAY									
Africa	**1**			2	0				
Asia									
Europe	**6**	3	3			13	11		
JUNE									
Africa	**3**	2	2	15	0	6	6	2	2
Asia	**1**			1	1				
Europe	**4**	1	1	2	0	5	3		
JULY									
Africa	**2**	4	2	6	2	4	4	1	1
Asia	**1**	4	4	3	3	2	2	2	2
Europe	**2**					2	1		
AUGUST									
Africa	**1**	4	3			3	3		
Asia									
Europe									
Total	**29**	**394**	**180 (46%)***	**55**	**17 (31%)**	**100**	**74 (77%)**	**13**	**10 (77%)**

*** Propagation success rates are shown in parentheses - the overall success rate was 50%**

Table 2. Summary of influenza A (H1N1pdm) samples received, collected between January 2010 and present

MONTH Country	H1N1pdm Number received	 Number propagated	 ≤2 fold*	 4 fold*	 ≥8 fold*
JANUARY					
Belgium	11	10	10		
Cyprus	4	3	3		
Czech Republic	1	1	1		
France	12	12	12		
Georgia	3	2	2		
Greece	2	1	1		
Latvia	6	6	6		
Luxembourg	15	14	10	4	
Macedonia	9	0			
Madagascar	6	3	2	1	
Moldova	98	9	6	3	
Norway	3	3	3		
Portugal	6	6	5	1	
Russia	5	4	2	2	
Senegal	11	4	4		
Spain	1	1	1		
Tajikistan	20	0			
Ukraine	5	4	4		
United Kingdom					
FEBRUARY					
Belgium	7	6	6		
France	2	2	2		
Georgia	1	1		1	
Ghana	1	1	1		
Greece	24	12	12		
Latvia	3	2	2		
Luxembourg	2	2	2		
Madagascar	6	6	6		
Moldova	32	0			
Niger	4	4	4		
Norway	6	5	3	2	
Russia	4	4	3	1	
Senegal	9	4	4		
Tajikistan	12	0			
Ukraine	1	1	1		
MARCH					
Ghana	6	5	4	1	
Madagascar	4	0			
Moldova	2	0			
Norway	2	1	1		
Russia	7	6	1	5	
Slovakia	3	3	3		
United Kingdom	2	2	2		
APRIL					
Ghana	4	4	3		1
Hong Kong	4	4	3	1	
Madagascar	3	0			
Norway	1	1	1		
Slovakia	3	3	3		
Sweden	1	1		1	
United Kingdom	2	2	2		
MAY					
Sweden	1	1	1		
United Kingdom	2	2	2		
JUNE					
Norway	1	1	1		
South Africa	2	2	1	1	
JULY					
Hong Kong	4	4	4		
Madagascar	2	1	1		
South Africa	2	1	1		
AUGUST					
South Africa	4	3	3		
Total	**394**	**180**	**155**	**24**	**1**

*** Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine strain)**

Table 3. Antigenic analyses of pandemic influenza A(H1N1) viruses

			Haemagglutination inhibition titre[1]					
			Post infection ferret sera					
Viruses	Collection date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 C4/31/09	A/Eng 195/09 NIBSC F17/09	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09
REFERENCE VIRUSES								
A/California/4/2009		C1,E3	2560	5120	2560	5120	2560	2560
A/California/7/2009		E8	1280	2560	640	2560	1280	2560
A/England/195/2009		MDCK5	1280	2560	1280	2560	640	1280
A/Auckland/3/2009		Ex+3	2560	2560	1280	5120	1280	2560
A/Bayern/69/2009		MDCK4/SIAT2	160	320	40	160	640	640
A/Lviv/N6/2009		MDCK4	640	1280	160	320	1280	2560
TEST VIRUSES								
A/Cyprus/5800/2010[2]	2010-01-07	SIAT2	1280	2560	1280	2560	1280	1280
A/Cyprus/5867/2010	2010-01-19	SIAT2	1280	2560	1280	2560	1280	1280
A/Cyprus/5870/2010[3]	2010-01-20	SIAT2	1280	2560	1280	2560	1280	1280
A/Ostrava /221/2010	2010-01-25	E1	2560	2560	1280	2560	1280	2560
A/Extremadura/RR6530/2010	2010-01-29	MDCK2	1280	2560	2560	5120	1280	2560
A/Georgia/54/2010	2010-01-05	SIAT2	1280	1280	640	2560	640	1280
A/Georgia/308/2010	2010-01-09	SIAT2	1280	1280	640	2560	640	1280
A/Georgia/670/2010	2010-02-04	SIAT3	320	640	160	320	640	1280
A/Livadia/1851/2010	2010-01-26	SIAT3	1280	2560	1280	2560	1280	1280
A/Samos Island/2161/2010	2010-02-02	SIAT3	1280	2560	1280	5120	1280	2560
A/Argos/2231/2010	2010-02-03	SIAT4	1280	1280	1280	2560	640	1280
A/Pirgos/2274/2010	2010-02-04	SIAT3	1280	1280	1280	2560	640	1280
A/Livadia/2292/2010	2010-02-05	SIAT4	1280	1280	1280	2560	640	1280
A/Livadia/2298/2010	2010-02-06	SIAT4	1280	1280	1280	2560	1280	2560
A/Athens/2316/2010	2010-02-06	SIAT3	1280	2560	1280	2560	1280	2560
A/Argos/2369/2010	2010-02-08	SIAT3	1280	2560	1280	2560	1280	2560
A/Thessaloniki/730/2010	2010-02-12	E3	2560	1280	2560	2560	1280	2560
A/Thessaloniki/790/2010	2010-02-16	E3	2560	2560	1280	5120	2560	2560
A/Thessaloniki/788/2010	2010-02-17	E3	2560	2560	1260	5120	2560	5120
A/Thessaloniki/791/2010	2010-02-17	E3	1280	2560	1280	2560	1280	2560
A/Latvia/1-358/2010	2010-01-06	MDCKx/SIAT1	1280	1280	1280	1280	1280	1280
A/Latvia/1-597/2010	2010-01-13	MDCKx/SIAT1	1280	1280	1280	1280	1280	2560
A/Latvia/1-1321/2010	2010-01-28	MDCKx/SIAT1	1280	2560	1280	2560	1280	2560
A/Latvia/1-1346/2010	2010-01-27	MDCKx/SIAT1	1280	2560	1280	2560	1280	1280
A/Latvia/1-1403/2010	2010-01-28	MDCKx/SIAT1	1280	1280	1280	2560	1280	2560
A/Latvia/2-37/2010	2010-01-29	MDCKx/SIAT1	2560	1280	1280	2560	1280	2560
A/Latvia/2-764/2010	2010-02-15	MDCK3	2560	2560	1280	2560	640	2560
A/Latvia/2-953/2010	2010-02-17	MDCK3	1280	2560	640	1280	640	1280
A/Luxembourg/8/2010	2010-01-04	MDCKx/SIAT1	1280	640	320	1280	640	640
A/Luxembourg/22/2010	2010-01-05	MDCKx/SIAT3	1280	1280	1280	2560	640	1280
A/Luxembourg/24/2010	2010-01-05	MDCKx/SIAT1	2560	2560	1280	2560	1280	2560
A/Luxembourg/42/2010	2010-01-06	MDCKx/SIAT1	640	640	320	640	320	640
A/Luxembourg/48/2010	2010-01-07	MDCKx/SIAT2	640	640	640	1280	320	640
A/Luxembourg/93/2010	2010-01-13	MDCKx/SIAT1	640	1280	640	1280	640	1280
A/Luxembourg/110/2010	2010-01-15	MDCKx/SIAT1	1280	1280	640	1280	640	1280
A/Luxembourg/112/2010	2010-01-15	SIAT2	1280	1280	1280	2560	1280	2560
A/Luxembourg/137/2010	2010-01-19	MDCKx/SIAT1	1280	1280	640	2560	640	1280
A/Luxembourg/162/2010	2010-01-20	MDCKx/SIAT1	1280	1280	1280	2560	640	1280
A/Luxembourg/184/2010	2010-01-25	MDCKx/SIAT1	640	1280	640	1280	640	1280
A/Luxembourg/185/2010	2010-01-25	MDCKx/SIAT1	640	1280	640	2560	640	1280
A/Luxembourg/194/2010	2010-01-26	SIAT4	1280	1280	640	1280	640	1280
A/Luxembourg/200/2010	2010-01-26	SIAT4	640	640	320	1280	640	1280
A/Luxembourg/251/2010	2010-02-02	MDCKx/SIAT1	2560	2560	2560	5120	1280	2560
A/Luxembourg/315/2010	2010-02-22	SIAT3	1280	1280	640	1280	640	1280
A/Belgium/G3/2010	2010-01-04	SIAT3	1280	2560	1280	2560	1280	2560
A/Belgium/G26/2010	2010-01-07	SIAT3	1280	2560	1280	2560	1280	2560
A/Belgium/G132/2010	2010-01-11	SIAT4	2560	2560	2560	2560	1280	2560
A/Belgium/G47/2010	2010-01-14	SIAT3	1280	1280	1280	2560	640	1280
A/Belgium/G51/2010	2010-01-14	SIAT3	1280	1280	1280	2560	640	1280
A/Belgium/G67/2010	2010-01-18	SIAT3	1280	1280	1280	2560	640	1280
A/Belgium/G71/2010	2010-01-18	SIAT2	1280	1280	640	1280	640	1280
A/Belgium/G85/2010	2010-01-20	SIAT2	1280	1280	1280	2560	1280	1280
A/Belgium/G82/2010	2010-01-21	SIAT3	1280	1280	1280	1280	1280	1280
A/Belgium/G110/2010	2010-01-26	SIAT2	1280	1280	1280	2560	640	1280
A/Belgium/G151/2010	2010-02-01	SIAT2	2560	2560	1280	2560	1280	2560
A/Belgium/G161/2010	2010-02-04	SIAT4	1280	1280	1280	2560	1280	1280
A/Belgium/G158/2010	2010-02-05	SIAT2	2560	2560	1280	2560	1280	2560
A/Belgium/G168/2010	2010-02-08	SIAT2	2560	2560	1280	2560	1280	1280
A/Belgium/G170/2010	2010-02-08	SIAT2	1280	1280	1280	2560	640	2560
A/Belgium/G176/2010	2010-02-10	SIAT2	1280	1280	1280	2560	1280	2560
A/Dakar/005/2010	2010-01-17	SIAT3	1280	1280	1280	2560	640	1280
A/Dakar/006/2010	2010-01-17	SIAT3	1280	1280	1280	2560	640	1280
A/Dakar/007/2010	2010-01-17	SIAT3	320	1280	160	640	640	1280
A/Dakar/008/2010	2010-01-17	SIAT3	320	1280	640	1280	640	640
A/Dakar/016/2010	2010-02-09	MDCK2/SIAT1	2560	2560	2560	2560	1280	2560
A/Dakar/018/2010	2010-02-09	MDCK2/SIAT1	2560	2560	2560	2560	1280	2560
A/Dakar/017/2010	2010-02-10	MDCK2/SIAT1	2560	2560	2560	5120	1280	2560
A/Dakar/019/2010	2010-02-16	MDCK2/SIAT1	1280	1280	1280	2560	640	1280
A/Lisboa/1/2010	2010-01-04	MDCK1/SIAT1	1280	1280	1280	2560	640	1280
A/Lisboa/2/2010	2010-01-04	MDCK1/SIAT1	1280	1280	1280	1280	640	1280
A/Lisboa/3/2010	2010-01-18	MDCK1/SIAT1	640	640	1280	1280	640	640
A/Lisboa/4/2010	2010-01-06	MDCK2/SIAT1	1280	1280	1280	1280	640	1280
A/Lisboa/5/2010	2010-01-09	MDCK2/SIAT1	1280	1280	1280	1280	640	1280

Vaccine strain

1. < = 40; 2.Sequences included in HA phylogeny; 3. Additional HA genes sequenced

Table 4. Antigenic analyses of pandemic influenza A(H1N1) viruses

			Haemagglutination inhibition titre[1]					
			Post infection ferret sera					
Viruses	**Collection date**	**Passage History**	**A/Cal 4/09 C4/F14/09**	**A/Cal 7/09 C4/31/09**	**A/Eng 195/09 NIBSC F17/09**	**A/Auck 3/09 C4/17/09**	**A/Bayern 69/09 C4/33/09**	**A/Lviv N6/2009 C4/34/09**
REFERENCE VIRUSES								
A/California/4/2009		C1,E3	1280	1280	1280	2560	1280	2560
A/California/7/2009		E8	640	2560	1280	1280	1280	2560
A/England/195/2009		MDCK5	1280	2560	2560	5120	2560	2560
A/Auckland/3/2009		Ex+3	2560	2560	2560	5120	1280	2560
A/Bayern/69/2009		MDCK4/SIAT2	80	320	80	80	320	320
A/Lviv/N6/2009		MDCK4	320	640	160	160	1280	1280
TEST VIRUSES								
A/Moldova/14/2010	2010-10-01	SIAT3	2560	2560	2560	2560	2560	2560
A/Moldova/8/2010[3]	2010-01-02	SIAT3	2560	2560	2560	2560	1280	2560
A/Moldova/30/2010	2010-01-02	SIAT3	320	640	320	320	640	640
A/Moldova/66/2010	2010-01-03	SIAT3	2560	2560	2560	2560	1280	2560
A/Moldova/63/2010	2010-01-04	SIAT2	2560	5120	2560	2560	2560	2560
A/Moldova/112/2010	2010-01-05	SIAT2	2560	2560	2560	2560	1280	2560
A/Moldova/226/2010	2010-01-11	SIAT2	1280	1280	1280	2560	640	1280
A/Moldova/379/2010	2010-01-20	SIAT3	1280	640	1280	2560	640	1280
A/Moldova/398/2010	2010-01-21	SIAT4	640	640	640	640	640	640
A/Ghana/FS-323/2010	2010-02-26	1st/SIAT1	1280	1280	1280	1280	640	1280
A/Ghana/FS-339/2010	2010-03-01	1st/SIAT1	1280	1280	1280	1280	640	1280
A/Ghana/FS-390/2010	2010-03-09	1st/SIAT1	1280	1280	1280	1280	1280	1280
A/Ghana/FS-411/2010	2010-03-09	1st / SIAT3	640	640	1280	1280	320	640
A/Ghana/FS-1059/2010	2010-03-23	1st/SIAT1	1280	2560	2560	1280	1280	1280
A/Ghana/FS-1307/2010	2010-03-23	1st/SIAT1	1280	1280	1280	1280	640	1280
A/Ghana/FS-1966/2010	2010-04-06	SIAT2	1280	1280	2560	1280	1280	1280
A/Ghana/FS-1974/2010	2010-04-06	SIAT3	160	320	80	160	640	640
A/Ghana/FS-1976/2010[2]	2010-04-06	SIAT3	1280	1280	1280	1280	640	1280
A/Ghana/FS-2135/2010	2010-04-07	SIAT2	1280	1280	1280	2560	640	1280
A/Ukraine/219/2010	2010-01-04	MDCK1/MDCK1	2560	2560	2560	5120	2560	5120
A/Ukraine/131/2010	2010-01-15	p1/MDCK1	2560	2560	2560	5120	1280	2560
A/Ukraine/122/2010	2010-01-15	p1/MDCK1	2560	2560	2560	2560	1280	2560
A/Ukraine/132/2010	2010-01-19	p1/MDCK1	1280	1280	1280	1280	640	2560
A/Ukraine/123/2010	2010-02-14	p1/MDCK1	1280	1280	1280	1280	640	1280
A/Paris/201005/2010	2010-01-04	C1/M1	1280	1280	1280	1280	1280	1280
A/Caen/286/2010	2010-01-05	C2/M1	1280	1280	2560	1280	1280	1280
A/Paris/334/2010	2010-01-18	C1/M1	1280	1280	640	1280	640	1280
A/Basse-Normandie/346/2010	2010-01-18	C1/M1	1280	1280	1280	1280	1280	1280
A/Paris/201031/2010	2010-01-18	C1/M1	2560	1280	2560	2560	1280	2560
A/Centre/356/2010	2010-01-20	C1/M1	2560	2560	2560	2560	1280	2560
A/Paris/370/2010	2010-01-20	C1/M1	1280	1280	2560	1280	1280	1280
A/Lille/533/2010	2010-01-20	C1/M1	2560	2560	2560	2560	1280	2560
A/Paris/432/2010	2010-01-25	C1/M1	2560	1280	2560	2560	1280	2560
A/Paris/425/2010	2010-01-26	C1/M1	2560	1280	2560	2560	1280	1280
A/Nord Pas de Calais/450/2010	2010-01-26	C1/M1	2560	1280	1280	2560	640	1280
A/Picardie/505/2010	2010-01-30	C1/M1	2560	2560	2560	2560	1280	2560
A/Paris/529/2010	2010-02-02	C1/M1	2560	1280	1280	1280	640	1280
A/Lorraine/543/2010	2010-02-02	C1/M1	2560	1280	2560	2560	1280	1280
A/Niger/665/2010	2010-02-05	C2/M1	2560	1280	2560	2560	1280	1280
A/Niger/666/2010	2010-02-05	C1/MDCK2	1280	1280	2560	1280	1280	1280
A/Niger/667/2010	2010-02-05	C1/M1	2560	1280	2560	2560	1280	2560
A/Niger/668/2010	2010-02-05	C1/M1	2560	1280	2560	2560	1280	2560
A/Voronezh/2/2010	2010-01-11	E1/E1	1280	1280	1280	1280	640	640
A/Astrakhan/56/2010	2010-01-11	C2 / MDCK2	320	640	1280	1280	1280	640
A/Smolensk/2/2010	2010-01-25	E2/E1	160	640	160	160	640	1280
A/StPetersburg/202/2010	2010-01-25	E3	1280	1280	2560	1280	1280	2560
A/StPetersburg/123/2010	2010-02-08	E1/E1	1280	1280	1280	2560	640	1280
A/Nizhnii-Novogorod/7/2010	2010-02-15	E3	640	640	1280	1280	640	1280
A/Saratov/6/2010	2010-02-15	E2/E1	1280	1280	1280	1280	640	1280
A/Kursk/1/2010	2010-02-15	E1/E1	1280	1280	1280	1280	1280	1280
A/StPetersburg/4/2010	2010-03-01	E1/E1	640	640	1280	1280	640	1280
A/StPetersburg/11/2010	2010-03-01	E2/E1	640	640	640	1280	640	640
A/StPetersburg/20/2010	2010-03-01	E1/E1	640	640	640	640	320	640
A/StPetersburg/182/2010	2010-03-01	E2/E1	80	320	80	160	640	640
A/Volgda/3/2010	2010-03-15	E2/E1	1280	1280	2560	2560	1280	2560
A/Belgorod/2/2010	2010-03-15	E1/E1	320	640	160	160	1280	1280
A/Madagascar/39/2010	2010-01-04	MDCK2/MDCK1	1280	1280	2560	2560	1280	1280
A/Madagascar/40/2010	2010-01-04	MDCK1/MDCK1	1280	2560	2560	2560	1280	1280
A/Madagascar/95/2010	2010-01-12	MDCK2/MDCK1	320	640	640	1280	320	640
A/Madagascar/294/2010	2010-02-17	MDCK2/MDCK1	1280	1280	1280	2560	1280	1280
A/Madagascar/295/2010	2010-02-17	MDCK2/MDCK1	1280	2560	1280	1280	640	1280
A/Madagascar/290/2010	2010-02-19	MDCK2/MDCK1	1280	1280	1280	2560	640	1280
A/Madagascar/552/2010	2010-02-22	MDCK2/MDCK1	1280	1280	2560	2560	1280	1280
A/Madagascar/557/2010	2010-02-23	MDCK2/MDCK1	1280	1280	2560	2560	1280	1280
A/Madagascar/555/2010	2010-02-24	MDCK2/MDCK1	1280	1280	2560	2560	640	1280
A/Madagascar/4962/2010	2010-07-05	MDCK2	640	1280	1280	1280	640	640
A/St. Petersburg/204/2010	2010-02-08	E2/E3	320	640	160	160	1280	1280
A/NewHampshire/02/2010	2010-02-19	E3/E1	320	640	640	1280	640	640
A/SouthCarolina/02/2010	2010-03-05	E3/E1	1280	1280	2560	2560	1280	2560
A/Texas/03/2010	2010-06-01	MDCK1/M3	2560	2560	2560	5120	1280	2560
A/Puerto Montt/11868/2010	2010-05-14	E3/E3	2560	2560	5120	5120	2560	5120
A/Brisbane/10/2010	2010-04-29	E2/E1	1280	1280	1280	5120	1280	1280
A/Brisbane/12/2010	2010-06-03	E2/E1	1280	5120	1280	2560	1280	2180
A/Christchurch/16/2010	2010-07-12	E2/E1	2560	2560	2560	2560	2560	5120
A/Victoria/502/2010	2010-06-20	E3/E1	1280	2560	1280	2560	1280	2560

Vaccine strain

1. < = 40; 2.Sequences included in HA phylogeny; 3. Additional HA genes sequenced

Table 5. Antigenic analyses of pandemic influenza A(H1N1) viruses

			Haemagglutination inhibition titre[1]					
			Post infection ferret sera					
Viruses	Collection date	Passage History	A/Cal 4/09 C4/F14/09	A/Cal 7/09 F05/10	A/Eng 195/09 NIBSC F17/09	A/Auck 3/09 C4/17/09	A/Bayern 69/09 C4/33/09	A/Lviv N6/2009 C4/34/09
REFERENCE VIRUSES								
A/California/4/2009		C1,E3	1280	1280	1280	1280	640	1280
A/California/7/2009		E8	640	1280	640	1280	640	1280
A/England/195/2009		MDCK5	1280	1280	640	1280	640	1280
A/Auckland/3/2009		Ex+3	1280	1280	1280	1280	640	1280
A/Bayern/69/2009		MDCK4/SIAT2	40	160	80	40	320	160
A/Lviv/N6/2009		MDCK4	160	320	80	80	640	640
TEST VIRUSES								
A/England/8/2010		MDCK2/MDCK1	1280	1280	1280	2560	1280	1280
A/England/9/2010		MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/England/7/2010		MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/England/10/2010		MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/England/6/2010		MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/England/11/2010		MDCK2/MDCK1	1280	1280	2560	2560	1280	1280
A/England/5/2010		MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/England/29/2010		MDCK2/MDCK1	2560	2560	1280	2560	1280	2560
A/England/30/2010		MDCK2/MDCK1	2560	2560	2560	2560	2560	2560
A/England/32/2010		MDCK3/MDCK1	2560	2560	2560	2560	2560	2560
A/England/47/2010		MDCK2/MDCK1	2560	2560	2560	2560	2560	1280
A/England/31/2010	2010-03-17	SIAT1/MDCK1	1280	1280	1280	2560	640	1280
A/Scotland/1/2010	2010-03-17	MDCK1/MDCK1	1280	1280	2560	2560	1280	2560
A/England/50/2010	2010-04-07	SIAT1/MDCK1	2560	1280	2560	2560	1280	1280
A/England/52/2010	2010-04-15	MDCK1/MDCK1	1280	1280	2560	2560	640	1280
A/England/60/2010	2010-05-06	MDCK1/MDCK1	1280	1280	2560	2560	1280	1280
A/England/61/2010	2010-05-21	SIAT1/MDCK1	1280	1280	1280	1280	640	1280
A/Norway/6/2010	2010-01-02	MDCK2/MDCK2	1280	1280	1280	2560	1280	1280
A/Norway83//2010	2010-01-04	MDCK2/MDCK2	1280	1280	2560	2560	640	1280
A/Norway84//2010	2010-01-05	MDCK3/MDCK2	1280	1280	2560	1280	1280	1280
A/Norway194//2010	2010-02-08	MDCK1/MDCK1	640	640	1280	1280	640	1280
A/Norway/238/2010[3]	2010-02-23	MDCK1/MDCK2	640	640	640	1280	320	640
A/Norway247/2010	2010-02-25	MDCK3/MDCK1	1280	1280	1280	2560	1280	2560
A/Norway248/2010	2010-02-25	MDCK3/MDCK1	1280	1280	1280	2560	1280	1280
A/Norway252/2010	2010-02-25	MDCK3/MDCK1	2560	2560	2560	2560	2560	2560
A/Norway308/2010	2010-03-18	MDCK2/MDCK2	2560	5120	5120	5120	1280	2560
A/Norway445/2010[2]	2010-04-22	MDCK3/MDCK1	640	640	640	2560	1280	1280
A/Norway/609/2010	2010-06-12	MDCK1/MDCK2	1280	2560	2560	2560	640	1280
A/Slovakia/1580/2010	2010-03-09	MDCKx/MDCK1	1280	1280	2560	2560	640	1280
A/Slovakia/1581/2010	2010-03-09	MDCKx/MDCK1	1280	2560	2560	2560	640	1280
A/Slovakia/1584/2010	2010-03-09	MDCKx/MDCK1	1280	1280	2560	2560	1280	2560
A/Slovakia/1639/2010	2010-04-06	MDCKx/MDCK1	2560	2560	2560	2560	1280	2560
A/Slovakia/1694/2010	2010-04-06	MDCKx/MDCK1	1280	1280	2560	2560	1280	1280
A/Slovakia/1641/2010	2010-04-06	MDCKx/MDCK1	2560	2560	2560	5120	1280	5120
A/Hong Kong/1884/2010	2010-04-17	MDCK2/MDCK1	1280	1280	1280	1280	640	1280
A/Hong Kong/1881/2010	2010-04-18	MDCK2/MDCK1	1280	1280	1280	1280	640	1280
A/Hong Kong/1885/2010	2010-04-20	MDCK2/MDCK1	640	640	320	640	640	640
A/Hong Kong/1886/2010	2010-04-20	MDCK2/MDCK1	1280	1280	1280	2560	640	1280
A/Hong Kong/2200/2010	2010-07-14	MDCK2/MDCK1	2560	2560	2560	2560	1280	2560
A/Hong Kong2203/2010	2010-07-14	MDCK2/MDCK1	320	640	640	1280	640	640
A/Hong Kong2212/2010	2010-07-16	MDCK2/MDCK1	1280	2560	2560	2560	1280	1280
A/Hong Kong/2213/2010	2010-07-16	MDCK2/MDCK1	1280	2560	1280	1280	640	1280
A/Hong Kong/2200/2010 1B	2010-07-14	E2	2560	2560	2560	2560	1280	2560
A/Hong Kong/2200/2010 2B	2010-07-14	E2	1280	2560	2560	2560	1280	2560
A/Hong Kong/2212/2010 1A	2010-07-16	E2	2560	5120	5120	5120	2560	2560
A/Hong Kong/2212/2010 2A	2010-07-16	E2	2560	2560	5120	2560	2560	2560
A/Stockholm/3/2010	2010-04-29	C0/MDCK1	320	640	640	640	320	640
A/Malmo/2/2010	2010-05-07	C1/MDCK1	1280	1280	1280	2560	640	1280
A/CapeTown/29/2010	2010-06-08	MDCK2/MDCK1	5120	5120	5120	5120	2560	5120
A/Johannesburg/79/2010		MDCK2/MDCK1	640	640	640	1280	640	640
A/Johannesburg/94/2010	2010-07-31	MDCK1/MDCK1	1280	1280	1280	2560	1280	1280
A/Johannesburg/116/2010	2010-08-10	MDCK1/MDCK2	1280	1280	2560	2560	640	1280
A/Johannesburg/117/2010	2010-08-12	MDCK1/MDCK2	1280	1280	1280	1280	640	1280
A/Johannesburg/119/2010	2010-08-19	MDCK1/MDCK2	2560	2560	2560	2560	1280	2560

Vaccine strain

1. < = 40; 2.Sequences included in HA phylogeny; 3. Additional HA genes sequenced

Figure 6. Phylogenetic comparison of influenza A (H1N1pdm) HA1 genes

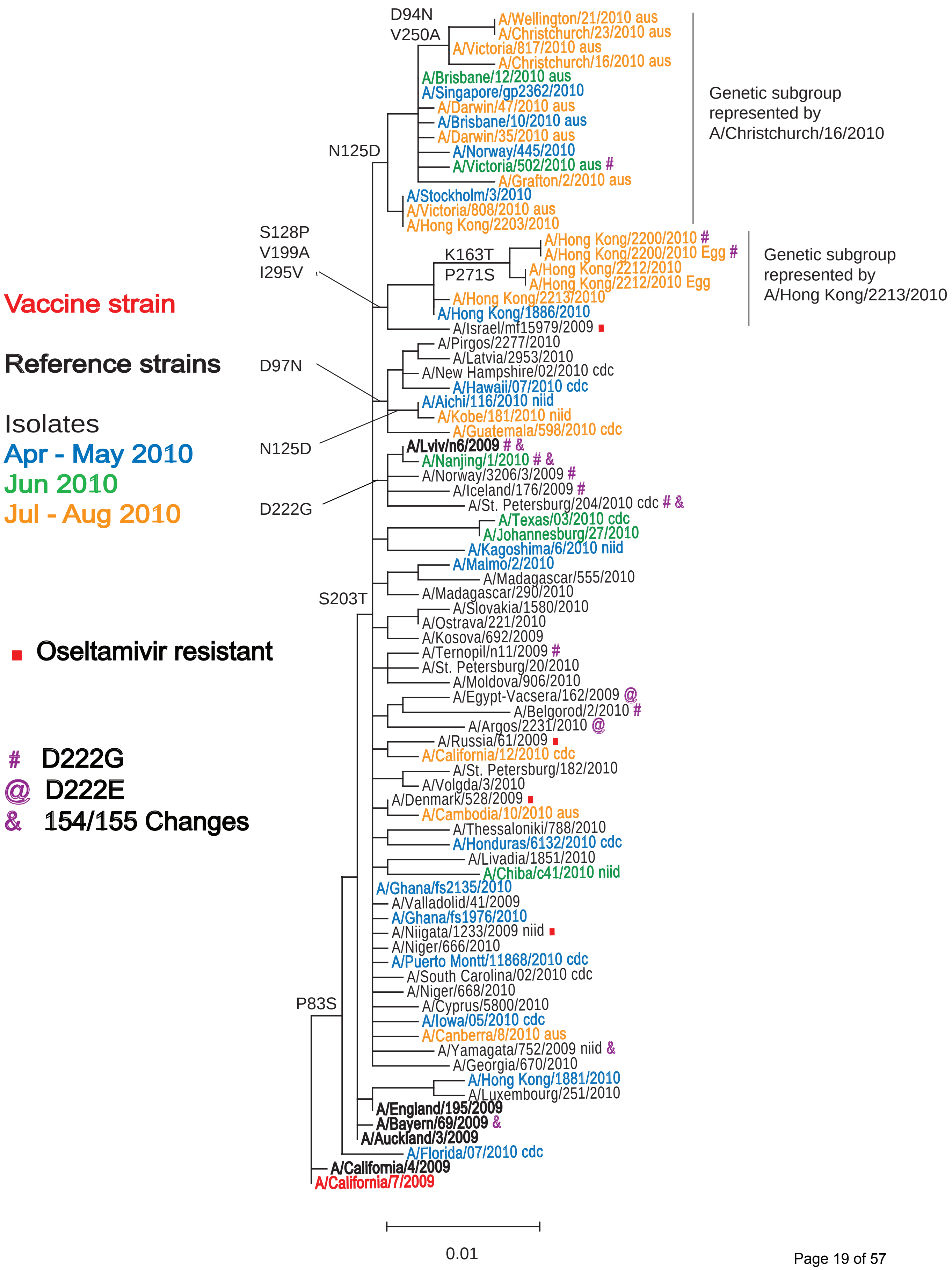


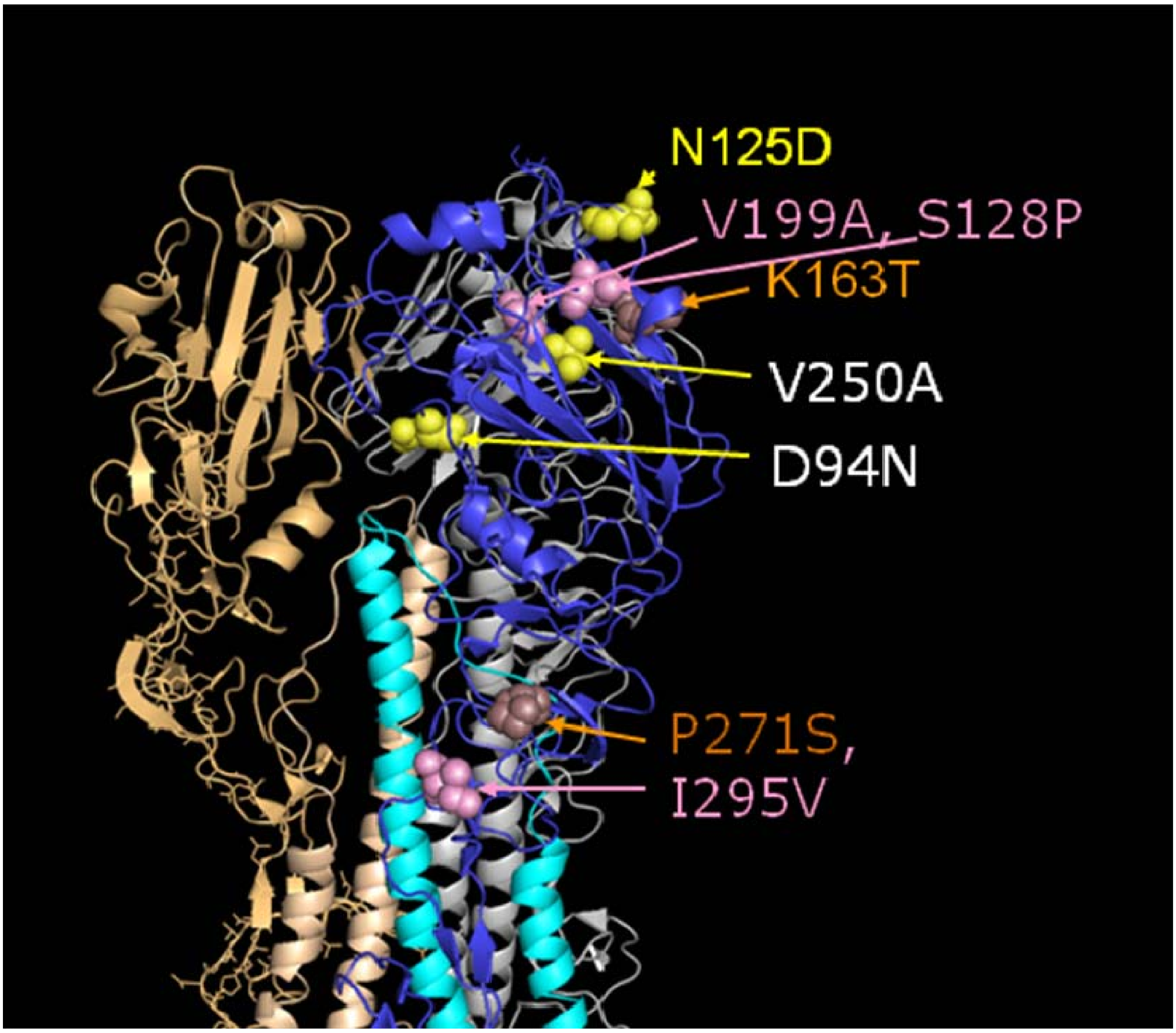
Figure 7. HA1 protein alignment for influenza A (H1N1pdm) lineage viruses

```
                               ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                   5         15         25         35         45         55         65         75         85         95
A/California/7/2009            DTLCIGYHAN NSTDTVDTVL EKNVTVTHSV NLLEDKHNGK LCKLRGVAPL HLGKCNIAGW ILGNPECESL STASSWSYIV ETPSSDNGTC YPGDFIDYEE
A/California/4/2009            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009              .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Bayern/69/2009               .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/England/195/2009             .......... .......... .......... .I........ .......... .......... .......... .......... ..S....... ..........
A/Hong Kong/1881/2010          .......... .......... .......... .I........ .......... .......... .......... .......... ..S....... ..........
A/Yamagata/752/2009            .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Cyprus/5800/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Puerto Montt/11868/2010      .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Niger/666/2010               .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Ghana/fs1976/2010            .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Thessaloniki/788/2010        .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/St. Petersburg/182/2010      .......... .......... .......... .......... .......... .......... .......... .......... ..S...K... ..........
A/Argos/2231/2010              .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Moldova/906/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Slovakia/1580/2010           .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ......N...
A/Madagascar/555/2010          .......... .......... .......... .......... .......... .......... .......... .......... ..S..G.... ..........
A/Johannesburg/27/2010         .......... .......... .......... D......... .......... .......... .......... .......... ..S....... ..........
A/Texas/03/2010                .......... .......... .......... D......... .......... .......... .......... .......... ..S....... ..........
A/St. Petersburg/204/2010      .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Lviv/n6/2009                 .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Kobe/181/2010                .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ......N...
A/Latvia/2953/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ......N...
A/Israel/mf15979/2009          .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Hong Kong/2213/2010          .......... .......... .......... .......... .......... .......... .......... .I........ ..S....... ..........
A/Hong Kong/2200/2010          .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Stockholm/3/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Victoria/502/2010            .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Brisbane/10/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Brisbane/12/2010             .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ..........
A/Christchurch/16/2010         .......... .......... .......... .......... .......... .......... .......... .......... ..S....... ...N......
A/Wellington/21/2010           .......... ....I..... .......... .......... .......... .......... .......... .......... ..S....... ...N......

                               ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                  105        115        125        135        145        155        165        175        185        195
A/California/7/2009            LREQLSSVSS FERFEIFPKT SSWPNHDSNK GVTAACPHAG AKSFYKNLIW LVKKGNSYPK LSKSYINDKG KEVLVLWGIH HPSTSADQQS LYQNADAYVF
A/California/4/2009            .......... .......... .......... .......... .......... .......... .......... .......... .......... ......T...
A/Auckland/3/2009              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Bayern/69/2009               .......... .......... .......... .......... .......... ....E..... .......... .......... .......... ..........
A/England/195/2009             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Yamagata/752/2009            .......... .......... .......... .......... .......... ....E..... .......... R......... .......... ......T...
A/Cyprus/5800/2010             .......... ........X. .......... .......... .......... .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010      .......... .......... ......X... .......... .......... .......... .......... .......... .......... X.........
A/Niger/666/2010               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/fs1976/2010            .......... .......... .......... .......... .......... .......... .I........ .......... .......... ..........
A/Thessaloniki/788/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... I.........
A/St. Petersburg/182/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... I.........
A/Argos/2231/2010              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Moldova/906/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Slovakia/1580/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/555/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/27/2010         .......... .......... .......... .......... .......... .......... .N........ .......... .....T.... ..........
A/Texas/03/2010                .......... .......... .......... .......... .......... .......... .N........ .......... .....T.... ..........
A/St. Petersburg/204/2010      .......... .......... .......... .......... .......... ...E...... .......... .......... .......... ..........
A/Lviv/n6/2009                 .......... .......... .......... .......... .......... ....E..... .......... .......... .......... ..........
A/Kobe/181/2010                .......... .......... ....D..... .......... .......... .......... .......... .......... .......... ..........
A/Latvia/2953/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Israel/mf15979/2009          .......... .....M.... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2213/2010          .......... .......... .......P.. .......... .......... .......... .......... .......... .......... ........A.
A/Hong Kong/2200/2010          .......... .......... .......P.. .......... .......... .......... ..T....... .......... .......... ........A.
A/Stockholm/3/2010             .......... .......... ....D..... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/502/2010            .......... .......... ....D..... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/10/2010             .......... .......... ....D.E... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/12/2010             .......... .......... ....D..... .......... .......... .......... .......... .......... .......... ..........
A/Christchurch/16/2010         .......... .......... ....D..... .......... .......... .........T .......... .......... .......... ..........
A/Wellington/21/2010           .......... .......... ....D..... .......... .......... .......... .......... .......... .......... ..........
```

```
                                ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                    205        215        225        235        245        255        265        275        285        295
A/California/7/2009             VGSSRYSKKF KPEIAIRPKV RDQEGRMNYY WTLVEPGDKI TFEATGNLVV PRYAFAMERN AGSGIIISDT PVHDCNTTCQ TPKGAINTSL PFQNIHPITI
A/California/4/2009             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Bayern/69/2009                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/England/195/2009              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010           .......... .......... .......... .......... .......... .......... .......... .......... .....V.... ..........
A/Yamagata/752/2009             ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Cyprus/5800/2010              ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010       ..T....... .......... ..X....... .......... .......... .......... .......... .......... .......... ..........
A/Niger/666/2010                ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/fs1976/2010             ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Thessaloniki/788/2010         ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/182/2010       ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Argos/2231/2010               ..T....... R......... .E........ .......... .......... .......... .......... .......... .S........ ..........
A/Moldova/906/2010              ..T....... .......... .......... .......... .......... .......... .......... ......A... .......... ..........
A/Slovakia/1580/2010            ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/555/2010           ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/27/2010          ..T....... .......... .......... .......... .......... .......... .......... .I........ .......... ..........
A/Texas/03/2010                 ..T....... .......... .......... .......... .......... ......V... .......... .I........ .......... ..........
A/St. Petersburg/204/2010       ..T....... .......... .G........ .......... .......... .......... .......... .F........ .......... ..........
A/Lviv/n6/2009                  ..T....... .......... .G........ .......... .......... .......... .......... .......... .......... ..........
A/Kobe/181/2010                 ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Latvia/2953/2010              ..T....... ....T..... .......... .......... I......... .......... .......... .......... ..N....... ..........
A/Israel/mf15979/2009           ..T....... .......... .......... .......... .......... .........D .......T.. .......... .......... ..........
A/Hong Kong/2213/2010           ..T....... .......... .......... .......... .......... .......... .......T.. .......... .......... ....V.....
A/Hong Kong/2200/2010           ..T....... .......... .G........ .......... .......... .......... .......... S......... .......... ....V.....
A/Stockholm/3/2010              ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/502/2010             ..T....... .......... .G........ .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/10/2010              ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/12/2010              ..T....... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Christchurch/16/2010          ..T....... .......... .N........ .......... .........A .......... .......... .......... .......... ..........
A/Wellington/21/2010            ..T....... .......... .......... .......... .........A .......... .......... .......... .........H ..........

                                ....|....| ....|....| ....|..
                                    305        315        325
A/California/7/2009             GKCPKYVKST KLRLATGLRN IPSIQSR
A/California/4/2009             .......... .......... .......
A/Auckland/3/2009               .......... .......... V......
A/Bayern/69/2009                .......... .......... V......
A/England/195/2009              .......... .......... V......
A/Hong Kong/1881/2010           .......... .......... V......
A/Yamagata/752/2009             .......... .......... V......
A/Cyprus/5800/2010              .......... .......... V......
A/Puerto Montt/11868/2010       .......... .......... V......
A/Niger/666/2010                .......... .......... V......
A/Ghana/fs1976/2010             .......... .......... V......
A/Thessaloniki/788/2010         .......... .......... .......
A/St. Petersburg/182/2010       .......... .......... .......
A/Argos/2231/2010               ......I... .......... .......
A/Moldova/906/2010              .......... .......... V......
A/Slovakia/1580/2010            .......... .......... .......
A/Madagascar/555/2010           .......... .......... V......
A/Johannesburg/27/2010          .......... .......... V......
A/Texas/03/2010                 .......... .......... V......
A/St. Petersburg/204/2010       .......... .......... .......
A/Lviv/n6/2009                  .......... .......... V......
A/Kobe/181/2010                 .......... .......... V......
A/Latvia/2953/2010              .......... .......... V......
A/Israel/mf15979/2009           .......... .......... V......
A/Hong Kong/2213/2010           .......... .......... V......
A/Hong Kong/2200/2010           .......... .......... V......
A/Stockholm/3/2010              .......... .......... V......
A/Victoria/502/2010             .......... .......... V......
A/Brisbane/10/2010              .......... .......... V......
A/Brisbane/12/2010              .......... .......... V......
A/Christchurch/16/2010          .......... .......... V......
A/Wellington/21/2010            .......... .......... V......
```

Potential N-linked glycosylation sites are highlighted.

Figure 8. H1-HA structure showing amino acid substitutions associated with emerging genetic groups of H1 pandemic virus



Two recently emerged genetic groups (**Figure 6**) each containing a subgroup are shown:

Group	Subgroup
N125D	D94N, V250A
S128P, V199A, I295V	K163T, P271S

Figure 9. Phylogenetic comparison of influenza A (H1N1pdm) NA genes

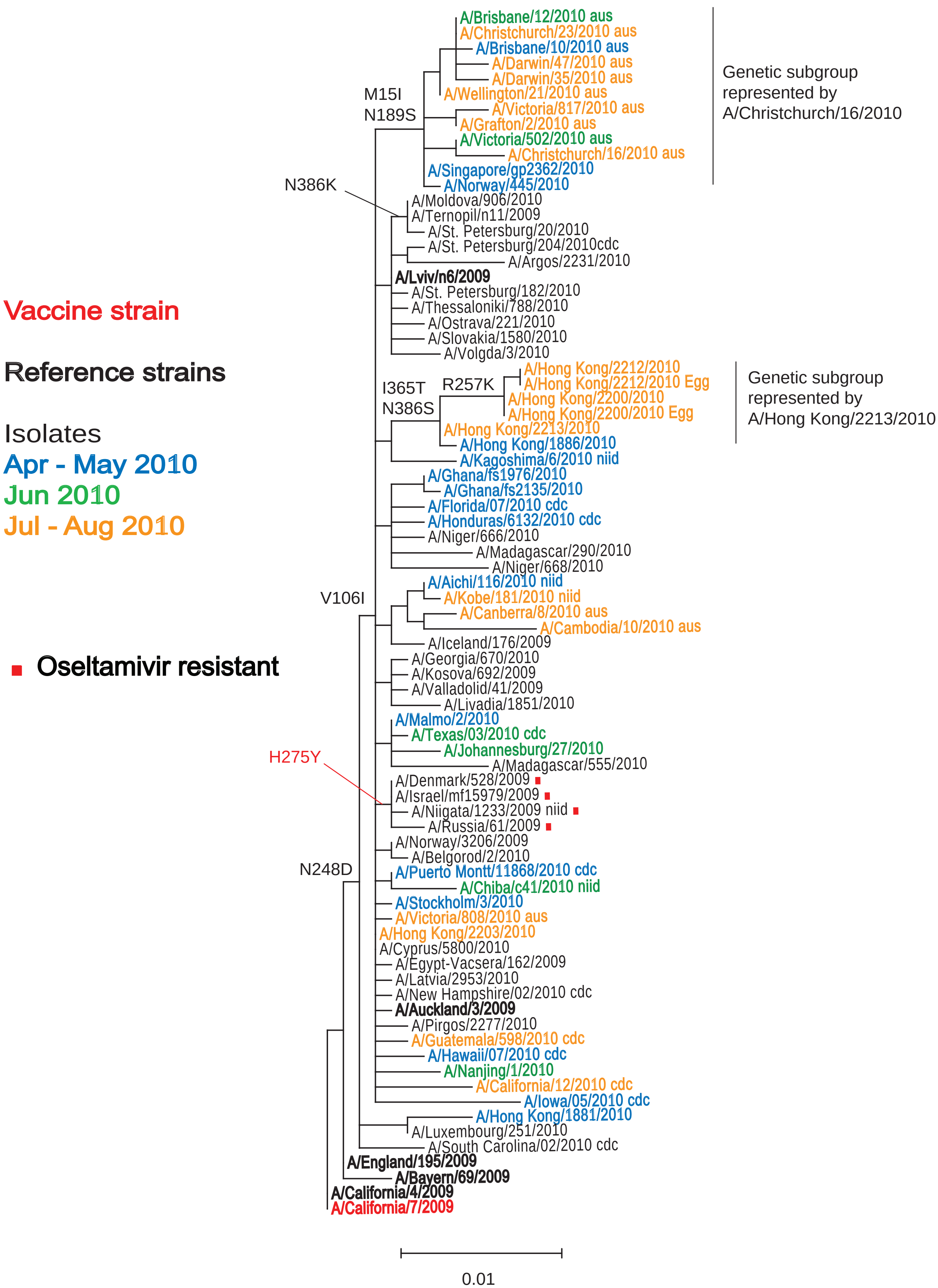


Figure 10. NA protein alignment for influenza A (H1N1pdm) lineage viruses

```
                            ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                5         15         25         35         45         55         65         75         85         95
A/California/7/2009         MNPNQKIITI GSVCMTIGMA NLILQIGNII SIWISHSIQL GNQNQIETCN QSVITYENNT WVNQTYVNIS NTNFAAGQSV VSVKLAGNSS LCPVSGWAIY
A/California/4/2009         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Bayern/69/2009            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/England/195/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010       .......... .......... .......... .......... .......... .......... .......... .......... A......... ..........
A/Cyprus/5800/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Niger/666/2010            .......... .......... .......... .......... .......... .......... .......... ........P. .......... ..........
A/Ghana/fs1976/2010         .......... .......... .......... .......... .......... .......... .......K.. .......... .......... ..........
A/Thessaloniki/788/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/182/2010   .......... .......... .......... .......... .......... ..I....... .......... .......... .......... ..........
A/Argos/2231/2010           .......... .......... .......... .......... .......... .......... .......... ......R... .......... ..........
A/Moldova/906/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Slovakia/1580/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/555/2010       .......... .......... .......... .......... .......... .......... .......... ......E... .......... ..........
A/Johannesburg/27/2010      .......... .......... .......... .......... .......I.. .......... .......... .......... .......... ..........
A/Texas/03/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/204/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Lviv/n6/2009              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Kobe/181/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Latvia/2953/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Israel/mf15979/2009       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2213/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2200/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Stockholm/3/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/502/2010         ...I...... ....I..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/10/2010          .......... ....I..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/12/2010          .......... ....I..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Christchurch/16/2010      ...I...... ....I..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Wellington/21/2010        .......... ....I..... .......... .......... .......... .......... .......... .......... .......... ..........

                            ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              105        115        125        135        145        155        165        175        185        195
A/California/7/2009         SKDNSVRIGS KGDVFVIREP FISCSPLECR TFFLTQGALL NDKHSNGTIK DRSPYRTLMS CPIGEVPSPY NSRFESVAWS ASACHDGINW LTIGISGPDN
A/California/4/2009         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009           .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Bayern/69/2009            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/England/195/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010       .......... .....I.... .......... .......... .......... .......... .......... .......... .......... ..........
A/Cyprus/5800/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010   .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Niger/666/2010            .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/fs1976/2010         .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Thessaloniki/788/2010     .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/182/2010   .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Argos/2231/2010           ....NI.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Moldova/906/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Slovakia/1580/2010        .....I.... .......... .......... .......... .......... .......... .......... .......... .......... .........S
A/Madagascar/555/2010       .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/27/2010      .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Texas/03/2010             .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/204/2010   .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Lviv/n6/2009              .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Kobe/181/2010             .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Latvia/2953/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Israel/mf15979/2009       .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2213/2010       .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2200/2010       .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Stockholm/3/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/502/2010         .....I.... .......... .......... .......... .......... .......... .......... .......... ........S. ..........
A/Brisbane/10/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... ........S. ..........
A/Brisbane/12/2010          .....I.... .......... .......... .......... .......... .......... .......... .......... ........S. ..........
A/Christchurch/16/2010      .....I.... .......... .......... .......... .......... .......... .......... .......... ........S. ..........
A/Wellington/21/2010        .....I.... .......... .......... .......... .......... .......... .......... .......... ........S. ..........

                            ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              205        215        225        235        245        255        265        275        285        295
A/California/7/2009         GAVAVLKYNG IITDTIKSWR NNILRTQESE CACVNGSCFT VMTDGPSNGQ ASYKIFRIEK GKIVKSVEMN APNYHYEECS CYPDSSEITC VCRDNWHGSN
A/California/4/2009         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009           .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Bayern/69/2009            .......... .......... .......... .......... .......... .......... .......... .......... .....G.... ..........
A/England/195/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010       .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Cyprus/5800/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010   .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Niger/666/2010            .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Ghana/fs1976/2010         .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Thessaloniki/788/2010     .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/St. Petersburg/182/2010   .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Argos/2231/2010           .......... .......... .......... .......... .......D.. .......... ..V....... .......... .......... ..........
A/Moldova/906/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Slovakia/1580/2010        .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Madagascar/555/2010       .......... .......... .......... .......... .......D.. ......K... .......... .......... .......... ..........
A/Johannesburg/27/2010      .......... .......... .......... .V........ .......D.. .......... .......... .......... .......... ..........
A/Texas/03/2010             .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/St. Petersburg/204/2010   .......... .......... .......... .......... .......D.. .......... ..V....... .......... .......... ..........
A/Lviv/n6/2009              .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Kobe/181/2010             .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Latvia/2953/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Israel/mf15979/2009       .......... .......... .......... .......... .......D.. .......... .......... ....Y..... .......... ..........
A/Hong Kong/2213/2010       .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Hong Kong/2200/2010       .......... .......... .......... .......... .......D.. ......K... .......... .......... .......... ..........
A/Stockholm/3/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Victoria/502/2010         .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Brisbane/10/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Brisbane/12/2010          .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Christchurch/16/2010      .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
A/Wellington/21/2010        .......... .......... .......... .......... .......D.. .......... .......... .......... .......... ..........
```

```
                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              305        315        325        335        345        355        365        375        385        395
A/California/7/2009       RPWVSFNQNL EYQIGYICSG IFGDNPRPND KTGSCGPVSS NGANGVKGFS FKYGNGVWIG RTKSISSRNG FEMIWDPNGW TGTDNNFSIK QDIVGINEWS
A/California/4/2009       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Auckland/3/2009         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Bayern/69/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/England/195/2009        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1881/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Cyprus/5800/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Puerto Montt/11868/2010 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Niger/666/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/fs1976/2010       .......... .......... .......... .......... .......... .......... .......... .......... .....S.... ..........
A/Thessaloniki/788/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/182/2010 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Argos/2231/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Moldova/906/2010        .......... .......... .......... .......... .......... .......... .......... .......... .....K.... ..........
A/Slovakia/1580/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/555/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/27/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Texas/03/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/St. Petersburg/204/2010 .......... .......... .......... .......... .......... .......... .......... ..I....... .......... ..........
A/Lviv/n6/2009            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Kobe/181/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ...I......
A/Latvia/2953/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Israel/mf15979/2009     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2213/2010     .......... .......... .......... .......... .......... .......... ....T..... .......... .....S.... ..........
A/Hong Kong/2200/2010     .......... .......... .......... .......... .......... .......... ....T..... .......... .....S.... ..........
A/Stockholm/3/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ......K...
A/Victoria/502/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/10/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Brisbane/12/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Christchurch/16/2010    .......... .......... .......... .......... .......... .......... .......... ..I....... .......... ..........
A/Wellington/21/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........

                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              405        415        425        435        445        455        465
A/California/7/2009       GYSGSFVQHP ELTGLDCIRP CFWVELIRGR PKENTIWTSG SSISFCGVNS DTVGWSWPDG AELPFTIDK*
A/California/4/2009       .......... .......... .......... .......... .......... .......... .........*
A/Auckland/3/2009         .......... .......... .......... .......... .......... .......... .....X----
A/Bayern/69/2009          .......... .......... .......... .......... .......... .......... .........*
A/England/195/2009        .......... .......... .......... .......... .......... .......... .......X--
A/Hong Kong/1881/2010     .......... .......... .........Q .......... .......... .......... .........*
A/Cyprus/5800/2010        .......... .......... .......... .......... .......... .......... .........*
A/Puerto Montt/11868/2010 .......... .......... .......... .......... .......... .......... .........*
A/Niger/666/2010          .......... .......... .......... .......... .......... .......... .........*
A/Ghana/fs1976/2010       .......... .......... .........X .......... .......... .......... .........*
A/Thessaloniki/788/2010   .......... .......... .......... .......... .......... .......... .........*
A/St. Petersburg/182/2010 .......... .......... .......... .......... .......... .......... .........*
A/Argos/2231/2010         .......... .......... .......... .......... .......... .......... .........*
A/Moldova/906/2010        .......... .......... .......... .......... .......... .......... .........*
A/Slovakia/1580/2010      .......... .......... .......... .......... .......... .......... ....L....*
A/Madagascar/555/2010     .......... .......... .......... .......... .......... .......... .........*
A/Johannesburg/27/2010    .......... .......... .......... .......... .......... .......... .........*
A/Texas/03/2010           .......... .......... .......... .......... .......... .......... .........*
A/St. Petersburg/204/2010 .......... .......... .......... .......... .......... .......... .........*
A/Lviv/n6/2009            .......... .......... .......... .......... .......... .......... .........*
A/Kobe/181/2010           .......... .......... .......... .......... .......... .......... .....I...*
A/Latvia/2953/2010        .......... .......... .......... .......... .......... .......... .........*
A/Israel/mf15979/2009     .......... .......... .......... .......... .I........ .......... .........*
A/Hong Kong/2213/2010     .......... .......... .......... .......... .......... .......... .........*
A/Hong Kong/2200/2010     .......... .......... .......... .......... .......... .......... .........*
A/Stockholm/3/2010        .......... .......... .......... .......... .......... .......... .........*
A/Victoria/502/2010       .......... .......... .......... .......... .......... .......... .....X----
A/Brisbane/10/2010        .......... ........K. .......... .......... .......... .......... .....X----
A/Brisbane/12/2010        .......... .......... .......... .......... .......... .......... .....X----
A/Christchurch/16/2010    .......... .......... .......... .......... .......... .......... .....X----
A/Wellington/21/2010      .......... .......... .......... .......... .......... .......... .....X----
```

Potential N-linked glycosylation sites are highlighted.

Table 6. Summary of influenza A (H3N2) samples received, collected between January 2010 and present

MONTH Country	H3N2 Number received	 Number propagated	 ≤2 fold*	 4 fold*	 ≥8 fold*
JANUARY					
Germany	1	1	1		
Madagascar	3	0			
FEBRUARY					
Ghana	1	1		1	
Madagascar	4	0			
MARCH					
Ghana	3	3	2		1
Hong Kong	3	3	3		
Madagascar	5	0			
APRIL					
Hong Kong	1	1	1		
Madagascar	2	0			
Netherlands	1	0			
Sweden	2	2	1	1	
MAY					
Madagascar	2	0			
JUNE					
France	1	0			
Hong Kong	1	1			
Madagascar	14	0			
South Africa	1	0			
United Kingdom	1	0			
JULY					
Hong Kong	3	2			
South Africa	6	2**			
Total	**55**	**16**	**8**	**2**	**1**

*** Reduction in HI titre with ferret antiserum raised against A/Perth/16/2009 (egg grown vaccine strain) - results only for those viruses that could be tested in the presence of oseltamivir are presented**
**** failed to haemagglutinate red blood cells and therefore not tested**

Table 7. Antigenic analyses of influenza A H3N2 viruses - Guinea pig RBCs

A) In presence of 20nM Oseltamivir

Viruses	Collection Date	Passage History	A/Wis 67/05 F1/06	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/HK 1985/09 F21/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09	A/HK 34430/09 F4/10	Comments NA	Comments HA
			Haemagglutination inhibition titre[1] Post infection ferret sera								
REFERENCE VIRUSES											
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E7	**1280**	1280	1280	40	<	160	40		
A/Brisbane/10/2007	2007-02-06	E2/E3	2560	**2560**	2560	80	<	160	160		
A/Uruguay/716/2007	2007-06-21	SpfCk1, E3/E3	640	1280	**2560**	<	<	80	40		
A/Hong Kong/1985/2009	2009-04-01	MDCK2/SIAT1	40	80	160	**1280**	640	2560	1280		
A/Perth/16/2009	2009-07-04	E3/E2	<	<	40	640	**640**	640	640		
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	40	640	640	**1280**	1280		
A/Hong Kong/34430/2009	2009-11-22	MDCK2/SIAT2	<	80	160	5120	640	1280	**1280**	D/G mix at 151, binds Ty and GP	
TEST VIRUSES											
A/Hong Kong/1737/2010	2010-03-24	MDCK2/SIAT1	40	80	320	5120	1280	1280	1280	D at 151, GP only	Carries E280A, I230V, D53N
A/Hong Kong/1775/2010	2010-03-28	MDCK2/SIAT1	<	80	160	5120	640	2560	1280	D at 151, binds GP and Ty	Does not carry E280A, I230V, D53N
A/Hong Kong/1837/2010	2010-03-30	MDCK2/SIAT1	40	80	160	5120	640	2560	1280	D/V at 151, binds Ty and GP	Does not carry E280A, I230V, D53N
A/Hong Kong/1888/2010	2010-04-19	MDCK2/SIAT1	160	320	320	5120	1280	2560	2560	D/N at 151 binds Ty and GP	Same HA aa sequence as HK/1737/2010

1. < = <40

Vaccine strain

B) No Oseltamivir

Viruses	Collection Date	Passage History	A/Wis 67/05 F1/06	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/HK 1985/09 F21/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09	A/HK 34430/09 F4/10	Comments NA	Comments HA
			Haemagglutination inhibition titre[1] Post infection ferret sera								
REFERENCE VIRUSES											
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E7	**2560**	1280	2560	<	<	<	<		
A/Brisbane/10/2007	2007-02-06	E2/E3	2560	**2560**	5120	80	80	<	160		
A/Uruguay/716/2007	2007-06-21	SpfCk1, E3/E3	640	1280	**1280**	<	<	<	40		
A/Hong Kong/1985/2009	2009-04-01	MDCK2/SIAT1	<	80	80	**640**	640	160	640		
A/Perth/16/2009	2009-07-04	E3/E2	<	<	40	1280	**640**	160	640		
A/Wisconsin/15/2009	2009-07-06	E2/E3	80	<	<	1280	640	**160**	320		
A/Hong Kong/34430/2009	2009-11-22	MDCK2/SIAT2	40	40	40	160	160	40	**160**	D/G mix at 151, binds Ty and GP	
TEST VIRUSES											
A/Ghana/FS-271/2010	2010-02-22	MDCK1/SIAT1	80	80	80	640	320	160	1280	Binds GP only, D at 151	
A/Ghana/FS-376/2010	2010-03-05	MDCK1/SIAT2	<	40	40	320	160	80	640	Binds GP only, D at 151	
A/Ghana/FS-464/2010	2010-03-15	MDCK1/SIAT1	80	80	160	1280	1280	320	2560	Binds GP only, D at 151	
A/Ghana/FS-467/2010	2010-03-15	MDCK1/SIAT1	80	80	80	1280	640	320	2560	Binds GP only, D at 151	
A/Stockholm/1/2010	2010-04-29	C1/SIAT1	<	160	160	320	320	160	640	Low binding to GP after Os	Same HA aa sequence as HK/1737/2010
A/Stockholm/2/2010	2010-04-30	C1/SIAT1	40	160	80	160	160	80	320	Low binding to GP after Os	Same HA aa sequence as Victoria/208/2009
A/Hong Kong/2097/2010	2010-06-25	MDCK2/SIAT1	<	160	40	160	80	80	320	GP binding sensitive to Os	Same HA aa sequence as HK/1737/2010 also Y94H
A/Hong Kong/2146/2010	2010-07-06	MDCK2/SIAT1	<	160	80	160	160	80	320	GP binding sensitive to Os	Same HA aa sequence as HK/1737/2010 also Y94H

1. < = <40

Vaccine strain

Table 8. Antigenic analyses of influenza A H3N2 viruses - Guinea Pig RBC with 20nM Oseltamivir

Viruses	Collection Date	Passage History	A/Wis 67/05 F1/06	A/Bris 10/07 F29/08	A/Uru 716/07 F26/08	A/HK 1985/09 F21/09	A/Perth 16/09 F25/09	A/Wis 15/09 F24/09	A/HK 34430/09 F4/10	A/Vic 208/09 F7/10	A/Vic 210/09 F11/10	A/Rhode Is 01/10 CDC 77/10
			Haemagglutination inhibition titre[1] — Post infection ferret sera									
REFERENCE VIRUSES												
A/Wisconsin/67/2005	2005-08-31	SpfCk3E3/E8	**2560**	2560	5120	<	<	<	40	80	<	40
A/Brisbane/10/2007	2007-02-06	E2/E3	2560	**5120**	5120	80	40	40	160	160	160	80
A/Uruguay/716/2007	2007-06-21	SpfCK1E3/E3	640	1280	**1280**	<	<	<	40	<	<	40
A/Hong Kong/1985/2009	2009-04-01	MDCK2/SIAT1	<	160	80	**320**	160	80	320	ND	ND	ND
A/Perth/16/2009	2009-07-04	E3/E4	<	<	80	1280	**640**	320	640	640	1280	640
A/Wisconsin/15/2009	2009-07-06	E2/E3	<	<	<	1280	320	**320**	640	<	80	320
A/Hong Kong/34430/2009	2009-11-22	MDCK2/SIAT2	<	80	<	320	80	80	**320**	ND	ND	ND
A/Victoria/208/2009	2009-06-02	E3/E1	80	80	80	640	640	320	1280	**2560**	2560	1280
A/Victoria/210/2009	2009-06-02	E2/E1	80	80	40	1280	640	640	1280	1280	**2560**	640
TEST VIRUSES												
A/Rhode Island/01/2010	2010-01-26	E4/E1	<	<	40	320	320	80	640	80	80	**640**
A/Stockholm/1/2010	2010-04-29	C1/MDCK1	<	<	160	ND	160	320	ND	320	320	ND
A/Stockholm/2/2010	2010-04-30	C1/MDCK1	<	<	80	ND	40	80	ND	80	80	ND
A/Brisbane/11/2010	2010-05-26	E4/E2	<	40	80	320	80	40	320	40	40	ND
A/Alabama/05/2010	2010-07-13	M1/C2/SIAT1	<	<	80	320	320	40	320	40	80	320

1. < = <40; ND = not done

Vaccine strain

Figure 11. Phylogenetic comparison of influenza A (H3N2) HA genes

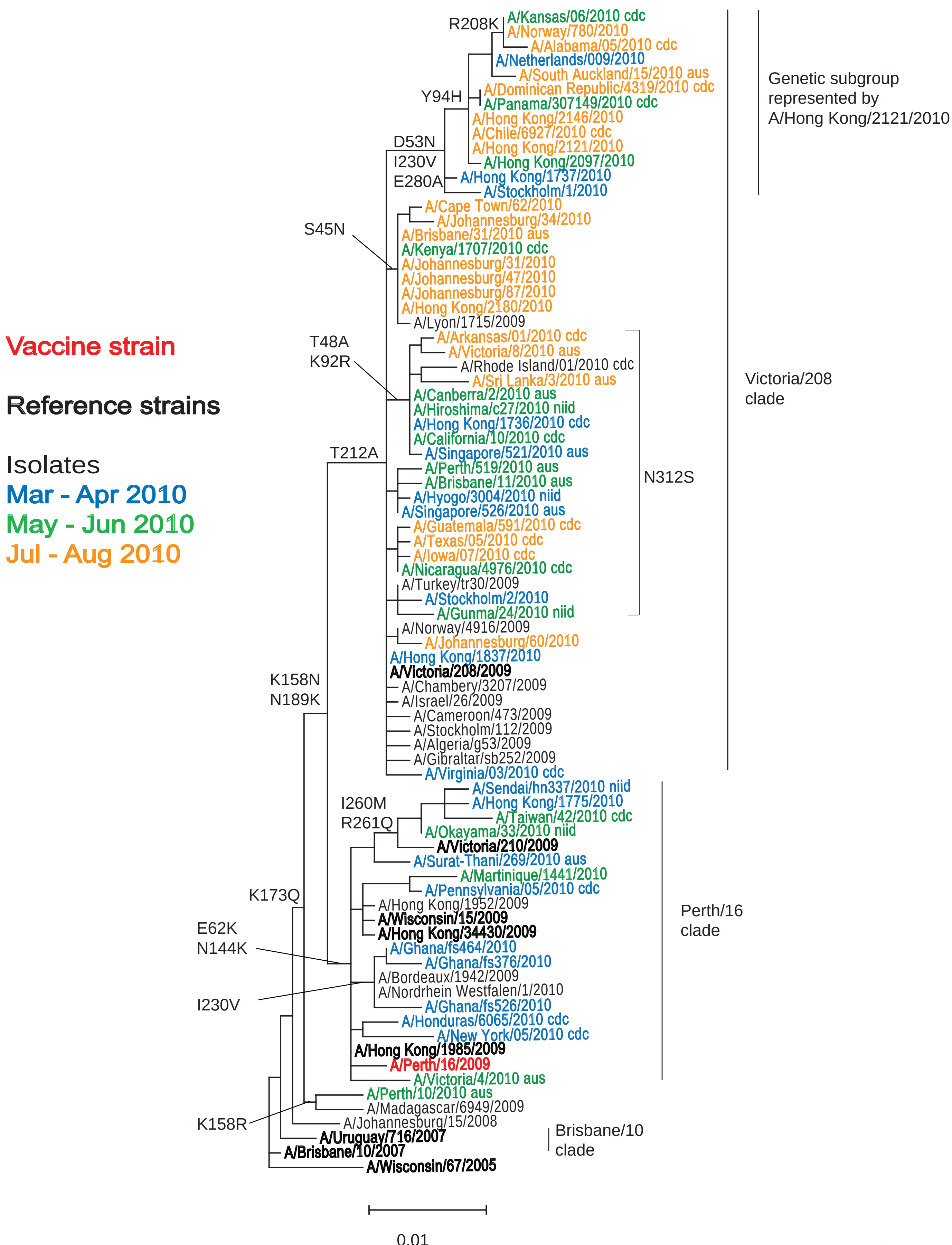


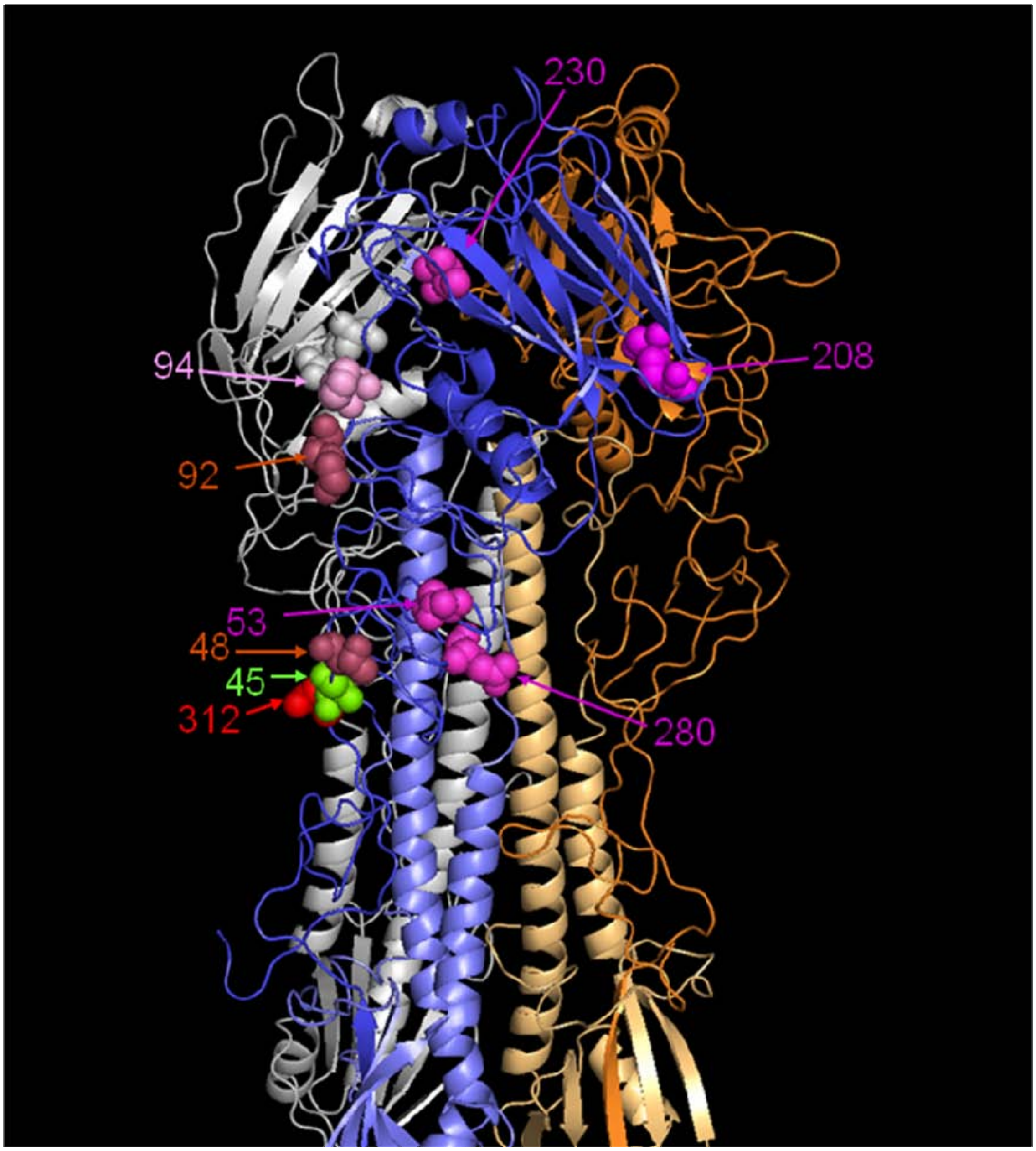
Figure 12. HA1 protein alignment for influenza A (H3N2) viruses

```
                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              5         15         25         35         45         55         65         75         85         95
A/Brisbane/10/2007        QKLPGNDNST ATLCLGHHAV PNGTIVKTIT NDQIEVTNAT ELVQSSSTGE ICDSPHQILD GENCTLIDAL LGDPQCDGFQ NKKWDLFVER SKAYSNCYPY
A/Uruguay/716/2007        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/6949/2009    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/16/2009           .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Hong Kong/1985/2009     .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Ghana/FS526/2010        .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Ghana/FS464/2010        .......... .......... .......... .......... .......... .......... .K........ .......... .......... N.........
A/Hong Kong/34430/2009    .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Wisconsin/15/2009       .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Martinique/1441/2010    .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Victoria/210/2009       .......... .......... .......... .......... ....N..... .......... .K........ .......... .......... ..........
A/Hong Kong/1775/2010     .......... .......... .......... .......... .......... .......... .K........ .......... .......... ..........
A/Gibraltar/SB252/2009    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/208/2009       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/60/2010    .......... .......... .......... .......... ....N..... .......... .......... .......... .......... ..........
A/Stockholm/2/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hyogo/3004/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/519/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hiroshima/C27/2010      .......... .......... .......... .......... .......A.. .......... .......... .......... .......... .R........
A/Rhode Island/01/2010    .......... .......... .......... .......... .......A.. .......... .......... .......... .......... .R........
A/Victoria/8/2010         .......... .......... .......... .......... ....N..A.. .......... .......... .......... .......... .R........
A/Johannesburg/47/2010    .......... .......... .......... .......... ....N..... .......... .......... .......... .......... ..........
A/Johannesburg/34/2010    .......... .......... .......... .......... ....N..... ........I. .......... .......... .......... ..........
A/Stockholm/1/2010        .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ..........
A/Hong Kong/2097/2010     .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ...H......
A/Hong Kong/2146/2010     .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ...H......
A/Panama/307149/2010      .......... .......... .......... .......... .......... ..N....... .......... .......X.. .......... ...H......
A/Netherlands/009/2010    .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ...H......
A/Alabama/05/2010         .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ...H......
A/Norway/780/2010         .......... .......... .......... .......... .......... ..N....... .......... .......... .......... ...H......

                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              105        115        125        135        145        155        165        175        185        195
A/Brisbane/10/2007        DVPDYASLRS LVASSGTLEF NNESFNWTGV TQNGTSSACI RRSNNSFFSR LNWLTHLKFK YPALNVTMPN NEKFDKLYIW GVHHPGTDND QIFLYAQASG
A/Uruguay/716/2007        .......... .......... .......... .......S.. .......... .......... .......... .......... .......... ...P......
A/Madagascar/6949/2009    .......... .......... .......... .......... .......... .......R.. .......... ..Q....... .......... ..........
A/Perth/16/2009           .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ..L.....K. ..........
A/Hong Kong/1985/2009     .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ........K. ..........
A/Ghana/FS526/2010        .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ........K. ..........
A/Ghana/FS464/2010        .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ........K. ..........
A/Hong Kong/34430/2009    .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ........K. ..........
A/Wisconsin/15/2009       .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... ........K. ...P......
A/Martinique/1441/2010    .......... .......... .......... ..D....... .G.K...... .......N.. .......... ..Q....... ........K. ..........
A/Victoria/210/2009       .......... .......... .......... .......... ...K...... .......N.. .......... ..Q....... .....V..K. ..........
A/Hong Kong/1775/2010     .......... .......... .......... .......... ...K...... .......N.. .S........ ..Q....... ........K. ..........
A/Gibraltar/SB252/2009    .......... .......... .......... .......... .G........ .......N.. .......... ..Q....... .I......K. ..........
A/Victoria/208/2009       .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Hong Kong/1837/2010     .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Johannesburg/60/2010    .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Stockholm/2/2010        .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..Y....S..
A/Hyogo/3004/2010         .......... .......... .S........ .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Perth/519/2010          .......... .......... .S........ .......... ...D...... .......N.. .......... ..Q....... ........K. ..........
A/Hiroshima/C27/2010      .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Rhode Island/01/2010    .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ..L....YK. ..........
A/Victoria/8/2010         .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Johannesburg/47/2010    .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Johannesburg/34/2010    .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Stockholm/1/2010        .......... .......... .......... .......... .K........ .......N.. .......... ..Q....... ........K. ..........
A/Hong Kong/2097/2010     .......... .......... .......P.. .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Hong Kong/2146/2010     .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Panama/307149/2010      .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ........A.
A/Netherlands/009/2010    .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Alabama/05/2010         .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
A/Norway/780/2010         .......... .......... .......... .......... .......... .......N.. .......... ..Q....... ........K. ..........
```

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              205        215        225        235        245        255        265        275        285        295
A/Brisbane/10/2007       RITVSTKRSQ QTVIPNIGSR PRVRNIPSRI SIYWTIVKPG DILLINSTGN LIAPRGYFKI RSGKSSIMRS DAPIGKCNSE CITPNGSIPN DKPFQNVNRI
A/Uruguay/716/2007       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/6949/2009   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/16/2009          .......... ...S...... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1985/2009    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS526/2010       .......... .......... .........V .....V.... .......... .......... .......... .......... .......... ..........
A/Ghana/FS464/2010       .......... .......... .........V .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/34430/2009   .......... ..A....... .......... .......... .......... .......... .......... .......... .......... ..........
A/Wisconsin/15/2009      .......... ..A....... .......... .......... .......... .......... .......... .......... .......... ..........
A/Martinique/1441/2010   .......... .AA....... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/210/2009      .......... .......... .X....X... .......... .......... .........M Q......... .......... .......... ..........
A/Hong Kong/1775/2010    .......... .......... .......... .......... .......... .........M Q......... .......... .V........ ..........
A/Gibraltar/SB252/2009   .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/208/2009      .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010    .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/60/2010   .......... .A........ .......... .......... .......... .......... Q......... .......... .......... ..........
A/Stockholm/2/2010       .......... .A........ ....X..... .......... .......... .......... .......... .......... .......... ..........
A/Hyogo/3004/2010        .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/519/2010         .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Hiroshima/C27/2010     .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Rhode Island/01/2010   .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/8/2010        .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/47/2010   .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/34/2010   .......... .A........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Stockholm/1/2010       .......... .A........ L........V .......... .......... .......... .N........ .........A .......... ..........
A/Hong Kong/2097/2010    .......... .A........ .........V .......... .......... .......... .......... .........A .......... ..........
A/Hong Kong/2146/2010    .......... .A........ .........V .......... .......... .......... .......... .........A .......... ..........
A/Panama/307149/2010     .......... .A........ X........V .......... .......... .......... .......... .........A .......... ..........
A/Netherlands/009/2010   .......... .A........ .........V .......... .......... .......... .......... .........A .......... ..........
A/Alabama/05/2010        .......K.. .A...X.... ....D....V .......... .......... .......... .......... .........A .......... ..........
A/Norway/780/2010        .......K.. .A........ .........V .......... .......... .......... .......... .........A .......... ..........

                         ....|....| ....|....| ....|....
                              305        315        325
A/Brisbane/10/2007       TYGACPRYVK QNTLKLATGM RNVPEKQTR
A/Uruguay/716/2007       .......... .......... .........
A/Madagascar/6949/2009   .......... .......... .........
A/Perth/16/2009          .......... .......... .........
A/Hong Kong/1985/2009    .......... .......... .........
A/Ghana/FS526/2010       .......... .......... .........
A/Ghana/FS464/2010       .......... .D........ .........
A/Hong Kong/34430/2009   .......... .......... .........
A/Wisconsin/15/2009      .......... .......... .........
A/Martinique/1441/2010   .......... .......... .........
A/Victoria/210/2009      .......... .......... .........
A/Hong Kong/1775/2010    .......... .......... .........
A/Gibraltar/SB252/2009   .......... .......... .........
A/Victoria/208/2009      .......... .......... .........
A/Hong Kong/1837/2010    .......... .S........ .........
A/Johannesburg/60/2010   .......... .......... .........
A/Stockholm/2/2010       .......... .S........ .........
A/Hyogo/3004/2010        .......... .S........ .........
A/Perth/519/2010         .......... .S........ .........
A/Hiroshima/C27/2010     .......... .S........ .........
A/Rhode Island/01/2010   .......... .S........ .........
A/Victoria/8/2010        .......... .S........ ...Q.....
A/Johannesburg/47/2010   .......... .......... .........
A/Johannesburg/34/2010   .......... .......... .........
A/Stockholm/1/2010       .......... .......... .........
A/Hong Kong/2097/2010    .......... .......... .........
A/Hong Kong/2146/2010    .......... .......... .........
A/Panama/307149/2010     .......... .......... .........
A/Netherlands/009/2010   .......... .......... .........
A/Alabama/05/2010        .......... .......... .........
A/Norway/780/2010        .......... .......... .........
```

Potential N-linked glycosylation sites are highlighted.

Figure 13. H3-HA structure showing amino acid substitutions associated with emerging genetic groups of H3N2 virus



The positions of amino acids defining recently emerged sub-clades (**Figure 11**) within the H3N2 Victoria/208 clade are shown, colour-coded by group. The N312S substitution is common to a number of sub-clades.

Figure 14. Phylogenetic comparison of influenza A (H3N2) NA genes

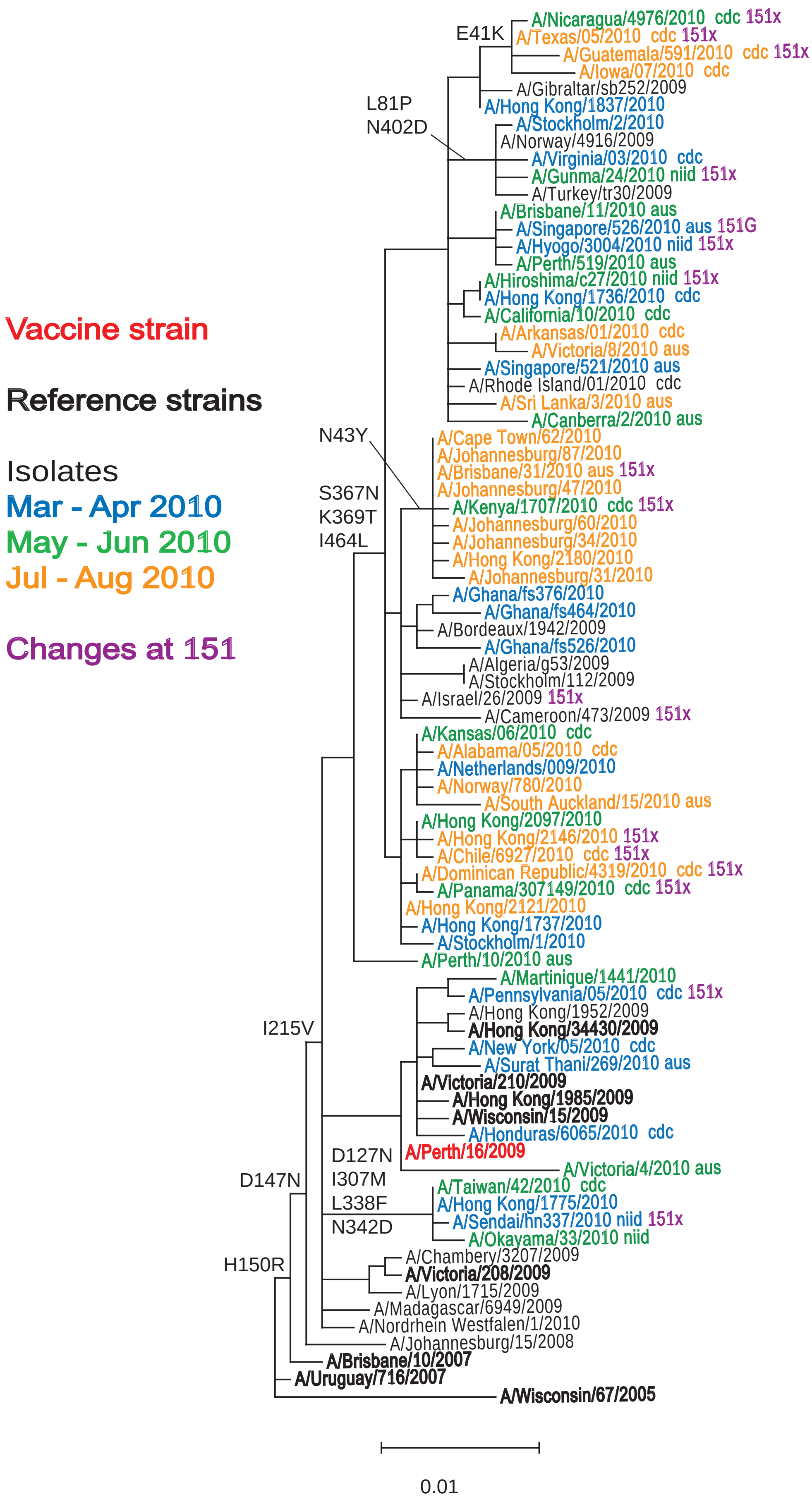


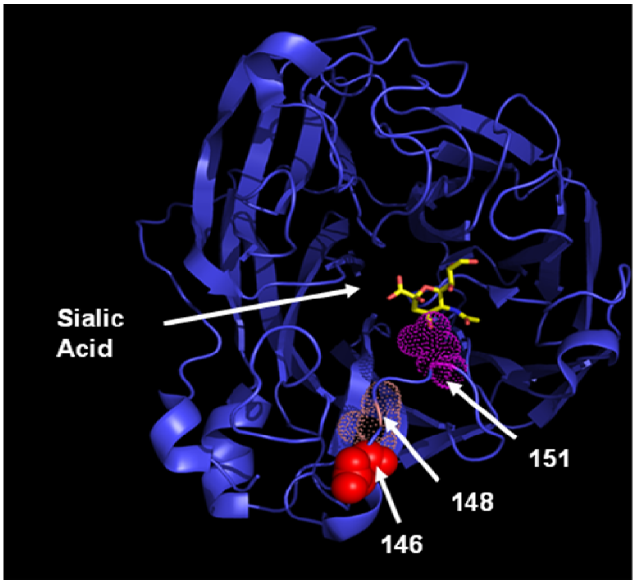
Figure 15. NA protein alignment for influenza A (H3N2) viruses

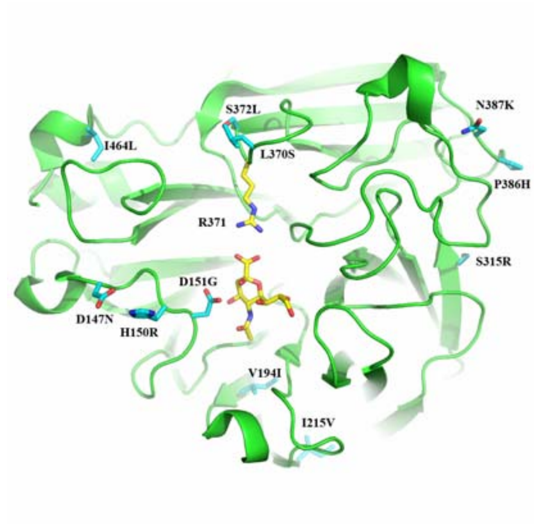
```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                             5         15         25         35         45         55         65         75         85         95
A/Brisbane/10/2007       MNPNQKIITI GSVSLTISTI CFFMQIAILI TTVTLHFKQY EFNSPPNNQV MLCEPTIIER NITEIVYLTN TTIEKEICPK LAEYRNWSKP QCDITGFAPF
A/Uruguay/716/2007       .......... .......... .....T.... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/6949/2009   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/16/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1985/2009    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS526/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS464/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/34430/2009   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Wisconsin/15/2009      .......... .......... .......... .......... .......... .........K .......... .......... .......... ..........
A/Martinique/1441/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/210/2009      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1775/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Gibraltar/SB252/2009   .......... .......... .......... .......... ..T....... .......... .......... .......... .......... H.........
A/Victoria/208/2009      .......... .......... .....T.... .........C .......... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/60/2010   .......... .......... .......... .......... ..Y....... .......... .......... ...R...... .......... ..........
A/Stockholm/2/2010       .......... .......... .......... .......... .......... .......... .......... .......... P......... ..........
A/Hyogo/3004/2010        .......... .......... .......... .......... .......... .......... ........X. .......... .......... ..........
A/Perth/519/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hiroshima/C27/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Rhode Island/01/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/8/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/47/2010   .......... .......... .......... .......... ..Y....... .......... .......... .......... .......... ..........
A/Johannesburg/34/2010   .......... .......... .......... .......... ..Y....... .......... .......... .......... .......... ..........
A/Stockholm/1/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2097/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2146/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Panama/307149/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Netherlands/009/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Alabama/05/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
A/Norway/780/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........

                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                            105        115        125        135        145        155        165        175        185        195
A/Brisbane/10/2007       SKDNSIRLSA GGDIWVTREP YVSCDPDKCY QFALGQGTTL NNVHSNDTVR DRTPYRTLLM NELGVPFHLG TKQVCIAWSS SSCHDGKAWL HVCITGDDKN
A/Uruguay/716/2007       .......... .......... .......... .......... .........H .......... .......... .......... .......... ..........
A/Madagascar/6949/2009   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Perth/16/2009          .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hong Kong/1985/2009    .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Ghana/FS526/2010       .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Ghana/FS464/2010       .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hong Kong/34430/2009   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Wisconsin/15/2009      .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Martinique/1441/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Victoria/210/2009      .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hong Kong/1775/2010    .......... .......... ......N... .......... ......N... .......... .......... .......... .......... ..........
A/Gibraltar/SB252/2009   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Victoria/208/2009      .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010    .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Johannesburg/60/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Stockholm/2/2010       .......... .......... .......... .......... ......NX.. .......... .......... .......... .......... ..........
A/Hyogo/3004/2010        .......... .......... .......... .......... ......N... X......... .......... .......... .......... ..........
A/Perth/519/2010         .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hiroshima/C27/2010     .......... .......... .......... .......... ......N... X......... .......... .......... .......... ..........
A/Rhode Island/01/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Victoria/8/2010        .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Johannesburg/47/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Johannesburg/34/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Stockholm/1/2010       .......... .......... .......... .......... ......NX.. .......... .......... .......... .......... ..........
A/Hong Kong/2097/2010    .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Hong Kong/2146/2010    .......... .......... .......... .......... ......N... X......... .......... .......... .......... ..........
A/Panama/307149/2010     .......... .......... .......... .......... ......N... X......... .......... .......... .......... ..........
A/Netherlands/009/2010   .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Alabama/05/2010        .......... .......... .......... .......... ......N... .......... .......... .......... .......... ..........
A/Norway/780/2010        .......... .......... .......... .......... ......N... .......... .......... .......... T......... ..........

                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                            205        215        225        235        245        255        265        275        285        295
A/Brisbane/10/2007       ATASFIYNGR LVDSIVSWSK EILRTQESEC VCINGTCTVV MTDGSASGKA DTKILFIEEG KIVHTSTLSG SAQHVEECSC YPRYPGVRCV CRDNWKGSNR
A/Uruguay/716/2007       .......... .......... .......... .......... .......... .......... ......I... .......... .......... ..........
A/Madagascar/6949/2009   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/16/2009          .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1985/2009    .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS526/2010       .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS464/2010       .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/34430/2009   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Wisconsin/15/2009      .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Martinique/1441/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/210/2009      .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1775/2010    .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Gibraltar/SB252/2009   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/208/2009      .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010    .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/60/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Stockholm/2/2010       .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hyogo/3004/2010        .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/519/2010         .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hiroshima/C27/2010     .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Rhode Island/01/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Victoria/8/2010        ...L...... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/47/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Johannesburg/34/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Stockholm/1/2010       .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2097/2010    .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/2146/2010    .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Panama/307149/2010     .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Netherlands/009/2010   .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Alabama/05/2010        .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
A/Norway/780/2010        .......... ....V..... .......... .......... .......... .......... .......... .......... .......... ..........
```

```
                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              305        315        325        335        345        355        365        375        385        395
A/Brisbane/10/2007        PIVDINIKDH STVSSYVCSG LVGDTPRKND SSSSSHCLDP NNEEGGHGVK GWAFDDGNDV WMGRTISEKS RLGYETFKVI EGWSNPKSKL QINRQVIVDR
A/Uruguay/716/2007        .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Madagascar/6949/2009    .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Perth/16/2009           .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1985/2009     .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Ghana/FS526/2010        .......... .I........ .......... .......... .......... .......... ......N.T. .......... .........V ..........
A/Ghana/FS464/2010        .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Hong Kong/34430/2009    .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Wisconsin/15/2009       .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Martinique/1441/2010    .......... .I........ .......... ......S... .......... .......... .......... .......... G......... ..........
A/Victoria/210/2009       .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1775/2010     ......M... .I........ .......... ......F... .D........ .......... .......... .......... .......... ..........
A/Gibraltar/SB252/2009    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Victoria/208/2009       .......... .I........ .......... .......... .......... .......... .......... .......... .......... ..........
A/Hong Kong/1837/2010     .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Johannesburg/60/2010    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Stockholm/2/2010        .......... .I........ .......... .......... .....S.... .......... ......N.T. .......... .......... ..........
A/Hyogo/3004/2010         .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Perth/519/2010          .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Hiroshima/C27/2010      .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Rhode Island/01/2010    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Victoria/8/2010         .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Johannesburg/47/2010    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Johannesburg/34/2010    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Stockholm/1/2010        .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Hong Kong/2097/2010     .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Hong Kong/2146/2010     .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Panama/307149/2010      .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Netherlands/009/2010    .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Alabama/05/2010         .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........
A/Norway/780/2010         .......... .I........ .......... .......... .......... .......... ......N.T. .......... .......... ..........

                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                              405        415        425        435        445        455        465
A/Brisbane/10/2007        GNRSGYSGIF SVEGKSCINR CFYVELIRGR KEETEVLWTS NSIVVFCGTS GTYGTGSWPD GADINLMPI*
A/Uruguay/716/2007        .......... .......... .......... .......... .......... .......... .........*
A/Madagascar/6949/2009    .......... .......... .......... ....D..... .......... .......... .........*
A/Perth/16/2009           .......... .......... .......... .......... .......... .......... .........*
A/Hong Kong/1985/2009     .......... .......... .......... .......... .......... .......... .........*
A/Ghana/FS526/2010        .......... .......... .......... .......... .......... .......... ...L.....*
A/Ghana/FS464/2010        .......... .......... .......... .......... .......... .......... ...L.....*
A/Hong Kong/34430/2009    .......... .......... .......... .......... .......... .......... ........V*
A/Wisconsin/15/2009       .......... .......... .......... .......... .......... .......... .........*
A/Martinique/1441/2010    .......... .......... .......... .......... .......... .......... .........*
A/Victoria/210/2009       .......... .......... .......... .......... .......... .......... .........*
A/Hong Kong/1775/2010     .......... .......... .......... .......... .......... .......... .........*
A/Gibraltar/SB252/2009    .......... .......... .......... .......... .......... .......... ...L.....*
A/Victoria/208/2009       .......... .......... .......... .......... .......... .......... .........*
A/Hong Kong/1837/2010     .......... .......... .......... .......... .......... .......... ...L.....*
A/Johannesburg/60/2010    .......... .......... .......... .......... .......... .......... ...L.....*
A/Stockholm/2/2010        .D........ .......... .......... .......... .......... .......... ...L.....*
A/Hyogo/3004/2010         .......... .......... .......... .......... .......... .......... ...L.....*
A/Perth/519/2010          .......... .......... .......... .......... .......... .......... ...L.....*
A/Hiroshima/C27/2010      .......... .......... .......... .......... .......... .......... ...L.....*
A/Rhode Island/01/2010    .......... .......... .......... .......... .......... .......... ...L.....*
A/Victoria/8/2010         .......... .......... .......... .......... .......... .......... ...L.....*
A/Johannesburg/47/2010    .......... .......... .......... .......... .......... .......... ...L.....*
A/Johannesburg/34/2010    .......... .......... .......... .......... .......... .......... ...L.....*
A/Stockholm/1/2010        .......... .......... .......... .......... .......... .......... ...L.....*
A/Hong Kong/2097/2010     .......... .......... .......... .......... .......... .......... ...L.....*
A/Hong Kong/2146/2010     .......... .......... .......... .......... .......... .......... ...L.....*
A/Panama/307149/2010      .......... .......... .......... .......... .......... .......... ...L.....*
A/Netherlands/009/2010    .......... .......... .......... .......... .......... .......... ...L.....*
A/Alabama/05/2010         .......... .......... .......... .......... .......... .......... ...L.....*
A/Norway/780/2010         .......... .......... .......... .......... .......... .......... ...L.....*
```

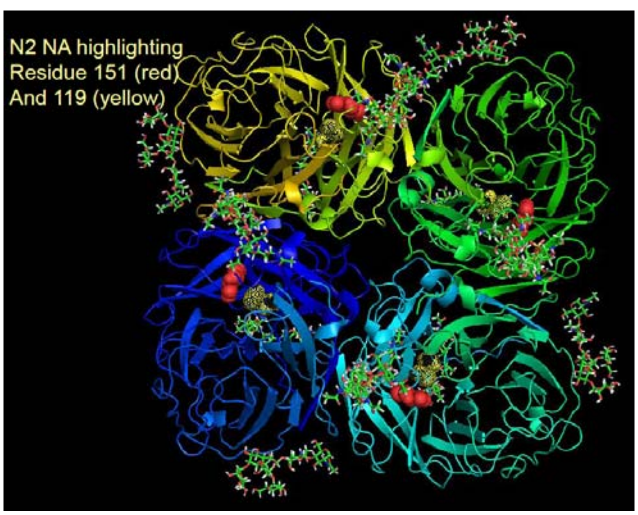
Potential N-linked glycosylation sites are highlighted.

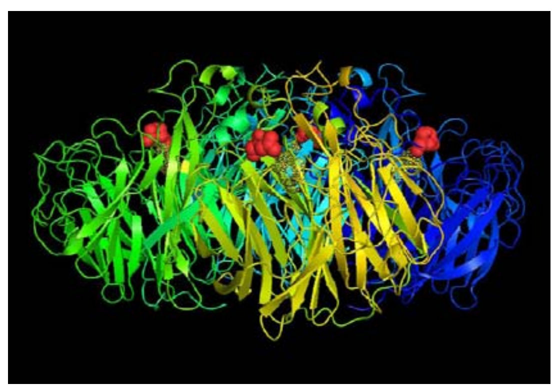
Figure 16. Location of residue 151 in relation to the N2-neuraminidase catalytic site





Sialic acid, in yellow, is located in the X-ray crystal structure of NA of A/England/804/2004 (H3N2) (Collins et al., in preparation). Polymorphisms (amino acid mixtures) have been observed at positions 148 and 151. Substitutions at position 148 can result in loss of an N-linked glycosylation site (N146).





N2 tetramer viewed from above (left panel) showing residue 151 with E119 (E119V confers oseltamivir resistance) shown as a reference residue and the location of N-linked carbohydrates displayed as 'ball and stick' motifs. Residues 119 and 151 only are shown in the side view (right panel).

Table 9. Summary of influenza B viruses (Victoria-lineage) received, collected between January and present

MONTH Country	B Victoria lineage				
	Number received	Number propagated	≤2 fold*	4 fold*	≥8 fold*
JANUARY					
Madagascar	5	1			1
Norway	2	0			
Sweden	1	1			1
United Kingdom	1	0			
FEBRUARY					
Italy	1	1			1
Norway	1	1		1	
Sweden	1	1	1		
Tajikistan	1	0			
MARCH					
France	2	2		1	1
Hong Kong	2	1			1
Italy	8	8		6	2
Madagascar	3	2		2	
Norway	3	0			
Sweden	1	1		1	
Ukraine	12	12			12
United Kingdom	3	3	1	1	1
APRIL					
France	1	1			1
Italy	2	2			2
Madagascar	2	1	1		
Norway	5	2	1	1	
Russia	3	3	2	1	
Sweden	2	1		1	
Ukraine	3	0			
MAY					
France	1	1			1
Norway	5	4		1	3
Russia	3	2	2		
Spain	3	3	1		2
United Kingdom	1	1	1		
JUNE					
France	1	1			1
Norway	1	1			1
South Africa	6	6			6
Spain	3	1			1
JULY					
Hong Kong	2	2		2	
Norway	1	1		1	
South Africa	4	4			4
Spain	1	0			
AUGUST					
South Africa	3	3			3
Total	**100**	**74**	**10**	**19**	**45**

*** Reduction in HI titre with ferret antiserum raised against B/Brisbane/60/2007 (egg grown vaccine strain)**

Table 10. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]							
				Post infection ferret antisera						
			B/Shan[2] 7/97	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/Eng 393/08 F31/08	B/Bris 60/08 F2/09	B/Paris 1762/08 F11/09	B/Mad 7002/09 F2/10	B/HK 514/09 F3/10
REFERENCE VIRUSES										
B/Shandong/7/97		Ex/E1	**2560**	320	160	20	<	<	<	<
B/Malaysia/2506/2004	2004-12-06	E3/E4	1280	**640**	320	40	10	<	<	<
B/Victoria/304/2006	2006-06-06	E2/E4	2560	640	**320**	40	40	<	<	<
B/England/393/2008	2008-08-29	E1/E5	1280	640	160	**320**	160	640	80	40
B/Brisbane/60/2008	2008-08-04	E4/E3	640	160	160	160	**160**	160	80	80
B/Paris/1762/2008	2009-02-09	C2/SIAT2	1280	320	<	<	80	**640**	320	640
B/Madagascar/7002/2009	2009-08-31	MDCK1/SIAT2/MDCK3	1280	40	<	<	80	160	**640**	640
B/Hong Kong/514/2009	2009-10-11	MDCK1/MDCK4	640	40	<	<	80	640	640	**640**
TEST VIRUSES										
B/Madagascar/82/2010[3]	2010-01-12	MDCK1/MDCK1	160	<	<	<	20	40	80	40
B/Norway/198/2010	2010-02-04	MDCK2/MDCK1	640	160	160	80	40	80	160	160
B/Stockholm/2/2010	2010-02-23	C2/MDCK1	640	20	<	<	80	160	320	320
B/Hong Kong/259/2010	2010-03-02	E3/E1	640	80	320	160	640	40	40	40
B/Hong Kong/270/2010	2010-03-03	MDCK2/MDCK1	320	<	<	<	10	20	160	160
B/England/27/2010	2010-03-05	SIAT1/MDCK1	640	20	<	<	80	160	640	320
B/Caen/992/2010	2010-03-09	C1/MDCK1	320	20	<	<	40	160	320	320
B/Odessa/3833/2010	2010-03-12	C2/MDCK1	640	<	<	<	10	40	160	160
B/Madagascar/1178/2010	2010-03-15	MDCK1/MDCK3	640	<	20	<	40	160	320	320
B/Madagascar/1184/2010	2010-03-16	MDCK1/MDCK1	320	10	<	<	40	80	160	160
B/Lorraine/942/2010	2010-03-17	C1/MDCK1	320	20	<	<	20	160	320	320
B/Odessa/3886/2010	2010-03-19	C2/MDCK1	320	<	<	<	10	20	160	160
B/England/56/2010	2010-03-25	SIAT1/MDCK1	640	10	<	<	40	80	320	160
B/England/57/2010	2010-03-25	SIAT1/MDCK1	640	10	<	<	<	80	320	160
B/Stockholm/3/2010	2010-03-29	C0/MDCK1	1280	20	<	<	40	160	160	320
B/Milano/30/2010	2010-03-30	MDCK1/SIAT1	80	<	<	<	<	<	20	40
B/Lorraine/1080/2010	2010-04-01	C1/MDCK1	640	20	<	<	20	160	320	320
B/Milano/34/2010	2010-04-01	MDCK1/SIAT1	160	<	<	<	<	<	20	40
B/Milano/36/2010	2010-04-06	MDCK1/SIAT1	160	<	10	<	10	10	40	80
B/Madagascar/2766/2010	2010-04-07	MDCK1/MDCK1	320	<	20	<	40	80	160	160
B/Ekaterinburg/1/2010	2010-04-09	C2/MDCK1	1280	40	<	<	80	320	640	640
B/Stockholm/5/2010	2010-04-09	C0/MDCK1	640	10	<	<	40	80	80	160
B/Madagascar/2787/2010	2010-04-20	MDCK2/MDCK1	640	<	40	20	80	80	320	320
B/Yamaguchi/1/2010	2010-04-21	MDCK2+2/MDCK1	320	<	<	<	20	80	320	320
B/Norway/459/2010	2010-04-22	MDCK1/MDCK1	1280	20	<	10	80	160	320	320
B/Moscow/1/2010	2010-04-27	C2/MDCK1	640	20	<	<	40	160	160	320
B/Moscow/9/2010	2010-04-27	C2/MDCK2	1280	40	<	<	80	320	160	320
B/England/63/2010	2010-05-02	MDCK1/MDCK1	640	10	<	<	80	<	<	<
B/Pays de Loire/1306/2010	2010-05-17	C1/MDCK1	640	20	<	<	20	160	320	320
B/Moscow/18/2010	2010-05-24	C3/MDCK1	640	40	160	80	80	80	160	320
B/Irkutsk/2/2010	2010-05-24	C3/MDCK2	640	40	160	80	80	80	160	160
B/Rouen/1317/2010	2010-06-04	C1/MDCK1	640	20	<	<	20	160	320	320
B/Hong Kong/1878/2010	2010-07-12	MDCK2/MDCK1	640	20	<	<	40	160	320	320
B/Hong Kong/1889/2010	2010-07-16	MDCK2/MDCK1	640	20	<	<	40	80	320	160

1. < = <10; 2. hyperimmune sheep serum; 3. Sequences included in HA phylogeny

Vaccine strain

Table 11. Antigenic analyses of influenza B viruses (Victoria lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]							
				Post infection ferret antisera						
			B/Shan[2] 7/97	B/Mal 2506/04 F28/05	B/Vic 304/06 F16/06	B/Eng[3] 393/08 F31/08	B/Bris 60/08 F5/09	B/Paris 1762/08 F11/09	B/Mad 7002/09 F2/10	B/HK 514/09 F3/10
REFERENCE VIRUSES										
B/Shandong/7/97		Ex/E1	**2560**	320	320	40	160	<	<	<
B/Malaysia/2506/2004		E3/E4	1280	**640**	160	40	160	<	10	<
B/Victoria/304/2006		E2/E4	640	320	**160**	80	160	<	<	<
B/England/393/2008		E1/E5	160	320	160	**320**	640	80	80	80
B/Brisbane/60/2008		E4/E3	640	640	320	640	**640**	160	80	320
B/Paris/1762/2008		C2/SIAT2	160	80	<	<	40	**160**	160	160
B/Madagascar/7002/2009		MDCK1/SIAT2/MDCK3	160	80	<	<	40	40	**160**	160
B/Hong Kong/514/2009		MDCK1/MDCK4	320	80	<	<	40	80	320	**320**
TEST VIRUSES										
B/Mie/6/2010	2010-03-13	E1+2/E1	320	320	80	<	320	80	20	20
B/Bolivia/104/2010	2010-03-19	X1/E2/E1	320	640	160	80	320	160	160	<
B/Taiwan/11/2010	2010-04-06	E1+2/E1	320	160	40	<	320	80	40	20
B/Norway/481/2010	2010-04-30	MDCK1/MDCK1	320	20	<	40	80	160	320	320
B/Norway/477/2010	2010-05-04	MDCK2/MDCK1	160	20	<	<	40	160	640	320
B/Norway/498/2010	2010-05-06	MDCK1/MDCK1	320	20	<	40	40	160	320	160
B/Norway/495/2010	2010-05-09	MDCK1/MDCK2	160	40	<	40	80	160	320	320
B/Norway/508/2010	2010-05-11	MDCK1/MDCK2	160	20	<	<	40	160	640	160
B/Canarias/RR6769/2010	2010-05-17	MDCK1/MDCK2	320	80	<	<	40	160	160	160
B/Canarias/RR6768/2010	2010-05-17	MDCK1/MDCK3	320	20	<	40	40	160	160	320
B/Canarias/RR6762/2010[4]	2010-05-25	MDCK1/MDCK2	320	80	<	<	160	160	320	640
B/Canarias/RR6757/2010	2010-06-03	MDCK1/MDCK2	320	80	<	<	20	160	320	320
B/Norway/631/2010	2010-06-03	MDCK1/MDCK1	160	20	<	<	40	160	640	160
B/Johannesburg/4/2010	2010-06-07	MDCK1/MDCK2	320	20	<	40	20	160	320	320
B/Johannesburg/14/2010	2010-06-23	MDCK1/MDCK1	320	20	<	<	20	80	80	80
B/Johannesburg/20/2010	2010-06-24	MDCK2/MDCK3	320	20	<	80	20	160	640	320
B/Johannesburg/22/2010	2010-06-28	MDCK2/MDCK1	320	20	<	<	20	160	160	160
B/Johannesburg/28/2010	2010-06-28	MDCK2/MDCK1	640	20	<	<	20	160	160	160
B/Johannesburg/25/2010	2010-06-29	MDCK2/MDCK2	160	20	<	<	40	160	320	320
B/Norway/683/2010	2010-07-05	MDCK1/MDCK1	320	20	<	<	80	160	640	320
B/Johannesburg/93/2010	2010-07-20	MDCK1/MDCK1	640	20	<	<	20	160	160	160
B/CapeTown/95/2010	2010-07-29	MDCK1/MDCK1	640	20	<	<	20	160	160	160
B/Johannesburg/98/2010	2010-07-30	MDCK1/MDCK1	640	20	<	<	20	160	160	160
B/Johannesburg/107/2010	2010-07-31	MDCK1/MDCK1	640	20	<	<	20	80	160	320
B/Johannesburg/101/2010	2010-08-03	MDCK1/MDCK1	640	20	<	<	20	160	160	160
B/Johannesburg/103/2010	2010-08-03	MDCK1/MDCK1	640	20	<	<	20	160	160	160
B/ohannesburg/106/2010	2010-08-03	MDCK1/MDCK1	640	20	<	<	20	80	160	160
B/Panama/307237/2010	2010-08-20	E2/E1	640	320	80	80	320	80	40	<

Vaccine strain

1. < = <10; 2. hyperimmune sheep serum; 3. < = <40; 4. Sequences included in HA phylogeny

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

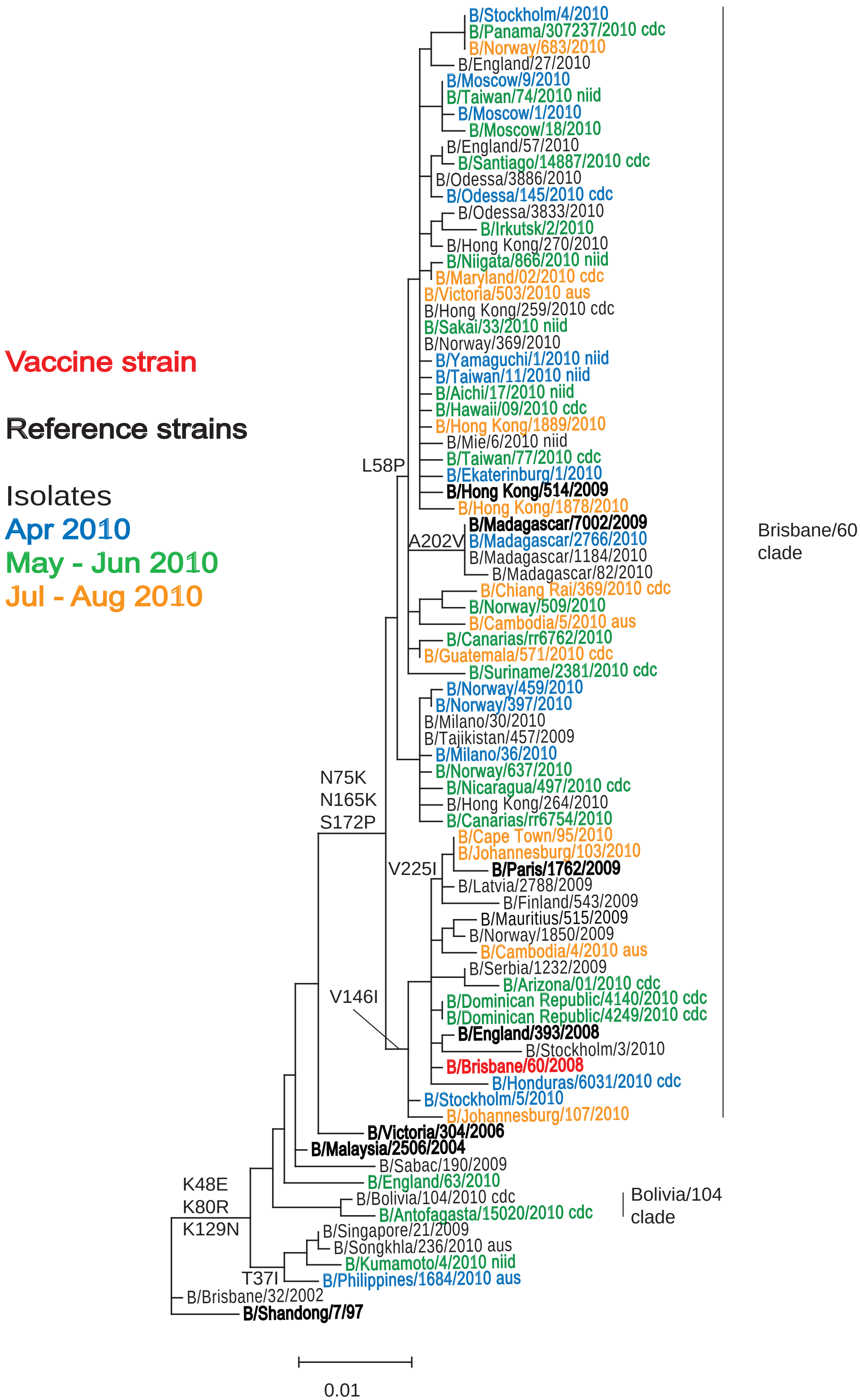


Figure 18. HA1 protein alignment for B-Victoria lineage viruses

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                             5         15         25         35         45         55         65         75         85         95
B/Brisbane/32/2002       DRICTGITSS NSPHVVKTAT QGEVNVTGVI PLTTTPTKSH FANLKGTKTR GKLCPKCLNC TDLDVALGRP KCTGNIPSAK VSILHEVRPV TSGCFPIMHD
B/Kumamoto/4/2010        .......... .......... .......... ......I... .......E.. .......... .......... .........R .......... ..........
B/Singapore/21/2009      .......... .......... .......... ......I... .......E.. .......... .......... .........R .......... ..........
B/Bolivia/104/2010       .......... .......... .......... .......... .......E.. .......... .......... .........I .......... ..........
B/England/63/2010        .......... .......... .......... .......... .......E.. .......... .......... .........R .......... ..........
B/Malaysia/2506/2004     .......... .......... .......... .......... .......E.. .......... .......... .........R .......... ..........
B/Victoria/304/2006      .......... .......... .......... .......... .......E.. .......... .......... .........R .......... ..........
B/Johannesburg/107/2010  .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Brisbane/60/2008       .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/England/393/2008       .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Arizona/01/2010        .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Cambodia/4/2010        .......... .......... .......... .......... .......E.. .....R.... .......... ....K....R .......... ..........
B/Mauritius/515/2009     .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Paris/1762/2009        .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Hong Kong/264/2010     .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Nicaragua/497/2010     .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Milano/36/2010         .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Norway/459/2010        .......... .......... .......... .......... .......E.. .......... .......... ....K....R .......... ..........
B/Canarias/RR6762/2010   .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Cambodia/5/2010        ......V... .......... .......... .......... .......E.. .N.....P.. .......... ....K....R .......... ..........
B/Madagascar/7002/2009   .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Hong Kong/514/2009     .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Mie/6/2010             .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Yamaguchi/1/2010       .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Hong Kong/259/2010     .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Irkutsk/2/2010         .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Odessa/3886/2010       .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/England/57/2010        .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Moscow/18/2010         .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/England/27/2010        .......... .......... .......... ......I... .......E.. .......P.. .......... ....K....R .......... ..........
B/Panama/307237/2010     .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........
B/Stockholm/4/2010       .......... .......... .......... .......... .......E.. .......P.. .......... ....K....R .......... ..........

                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                            105        115        125        135        145        155        165        175        185        195
B/Brisbane/32/2002       RTKIRQLPNL LRGYEHIRLS THNVINAEKA PGGPYKIGTS GSCPNVTNGN GFFATMAWAV PKNDNNKTAT NSLTIEVPYI CTEGEDQITV WGFHSDNEAQ
B/Kumamoto/4/2010        .......... .......... ........N. .......... .......... .......... .......... .......... .......... ........T.
B/Singapore/21/2009      .......... .......... ........N. .......... .......... .......... .......... .......... .......... ........T.
B/Bolivia/104/2010       .......... .......... ........N. .......... .......... .......... ....K.R... .......... .K........ ........T.
B/England/63/2010        .......... .......... ........N. .....R.... .......... .......... .......... .......... .......... ........T.
B/Malaysia/2506/2004     ........K. .......... ........N. .......... .......... .......... .......... .......... .......... ........T.
B/Victoria/304/2006      .......... .......... ........N. .......... .......... .......... .......... .......... .......... ......T.T.
B/Johannesburg/107/2010  .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Brisbane/60/2008       .......... .......... ........N. .......... .....I.... .......... ....K..... .P........ .......... ........T.
B/England/393/2008       .......... .......... ........N. .......... .....I.... .......... ....K..... .P........ .......... ......S.T.
B/Arizona/01/2010        .......... .......... ........N. .......... .....I.... .......... ....K..... .P........ .......... ........T.
B/Cambodia/4/2010        .......... .......... ........N. .......... .....I.... .......... .N..K..... .P........ .......... ........T.
B/Mauritius/515/2009     .......... .......... ........N. .......... .....I.... .......... ....K..... TP........ .......... ........T.
B/Paris/1762/2009        .......... .......... ........N. .......... .....I.... .......... ....K..... .P........ .........I ........T.
B/Hong Kong/264/2010     .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Nicaragua/497/2010     .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........X.
B/Milano/36/2010         .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Norway/459/2010        .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Canarias/RR6762/2010   .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Cambodia/5/2010        .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Madagascar/7002/2009   .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Hong Kong/514/2009     .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Mie/6/2010             .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........X.
B/Yamaguchi/1/2010       .......... .......... ........N. .....R.... .......... .......... ....K..... .P........ .......... ........T.
B/Hong Kong/259/2010     .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........X.
B/Irkutsk/2/2010         .......... .......... ........N. .......... .......... ...E...... ....K..... .P........ .......... ........X.
B/Odessa/3886/2010       .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/England/57/2010        .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Moscow/18/2010         .......... .......... .....I..N. .......... .......... .......... ....K..... .P........ .......... ........X.
B/England/27/2010        .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Panama/307237/2010     .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
B/Stockholm/4/2010       .......... .......... ........N. .......... .......... .......... ....K..... .P........ .......... ........T.
```

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                               205        215        225        235        245        255        265        275        285        295
B/Brisbane/32/2002       MAKLYGDSKP QKFTSSANGV TTHYVSQIGG FPNQTEDGGL PQSGRIVVDY MVQKSGKTGT ITYQRGILLP QKVWCASGRS KVIKGSLPLI GEADCLHEKY
B/Kumamoto/4/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Singapore/21/2009      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bolivia/104/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/63/2010        .V........ .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Malaysia/2506/2004     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Victoria/304/2006      .......... .......... .........D .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/107/2010  .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Brisbane/60/2008       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/393/2008       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Arizona/01/2010        .......... .......... .......V.. .......... .......... .......... .......... .......... .......... ..........
B/Cambodia/4/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mauritius/515/2009     .......... .........A .......... .......... .......... .......... .......... .......... .......... ..........
B/Paris/1762/2009        .......... .......... ....I..... .......... .......... ....P..... .......... .......... .......... ..........
B/Hong Kong/264/2010     .......... .......... .......... .......... .......... .......... .......... .......... .........V ..........
B/Nicaragua/497/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Milano/36/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Norway/459/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Canarias/RR6762/2010   .......... .......... .......... .......... .......... ......R... .......... .......... .......... ..........
B/Cambodia/5/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Madagascar/7002/2009   .V........ .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/514/2009     .......... .......... .......... .......... .......... .......... .......... ........K. .......... ..........
B/Mie/6/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Yamaguchi/1/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/259/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Irkutsk/2/2010         .......... .......... .......... .......... .......... .........I .......... .......... .......... ..........
B/Odessa/3886/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/57/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Moscow/18/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/27/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Panama/307237/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Stockholm/4/2010       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........

                         ....|....| ....|....| ....|....| ....|....| ....|..
                               305        315        325        335        345
B/Brisbane/32/2002       GGLNKSKPYY TGEHAKAIGN CPIWVKTPLK LANGTKYRPP AKLLKER
B/Kumamoto/4/2010        .......... .......... .......... .......... .......
B/Singapore/21/2009      .......... .......... .......... .......... .......
B/Bolivia/104/2010       .......... .......... .......... .......... .......
B/England/63/2010        .......... .......... .......... .......... .......
B/Malaysia/2506/2004     .......... .......... .......... .......... .......
B/Victoria/304/2006      .......... .......... .......... .......... .......
B/Johannesburg/107/2010  .......... .......... .......... .......... .......
B/Brisbane/60/2008       .......... .......... .......... .......... .......
B/England/393/2008       .......... .......... .......... .......... .......
B/Arizona/01/2010        .......... .......... .......... .......... .......
B/Cambodia/4/2010        .......... .......... .......... .......... .......
B/Mauritius/515/2009     .......... .......... .......... .......... .......
B/Paris/1762/2009        .......... .......... .......... .......... .......
B/Hong Kong/264/2010     .......... .......... .......... .......... .......
B/Nicaragua/497/2010     .......... .......... .......... .......... .......
B/Milano/36/2010         .......... .......... .......... .......... .......
B/Norway/459/2010        .......... .......... .......... .......... .......
B/Canarias/RR6762/2010   .......... .......... .......... .......... .......
B/Cambodia/5/2010        .......... .......... .......... .......... .......
B/Madagascar/7002/2009   .......... .......... .......... .......... .......
B/Hong Kong/514/2009     .......... .......... .......... .......... .......
B/Mie/6/2010             .......... .......... ......A... .......... .......
B/Yamaguchi/1/2010       .......... .......... .......... .......... .......
B/Hong Kong/259/2010     .......... .......... .......... .......... .......
B/Irkutsk/2/2010         .......... .......... .......... .......... .......
B/Odessa/3886/2010       .......... .......... .......... .......... .......
B/England/57/2010        .......... .......... .......... .......... .......
B/Moscow/18/2010         .......... .......... .......... .......... .......
B/England/27/2010        .......... .......... .......... .......... .......
B/Panama/307237/2010     .......... .......... .......... .......... .......
B/Stockholm/4/2010       .......... .......... .......... .......... .......
```

Poptential N-linked glycosylation sites are highlighted.

Figure 19. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

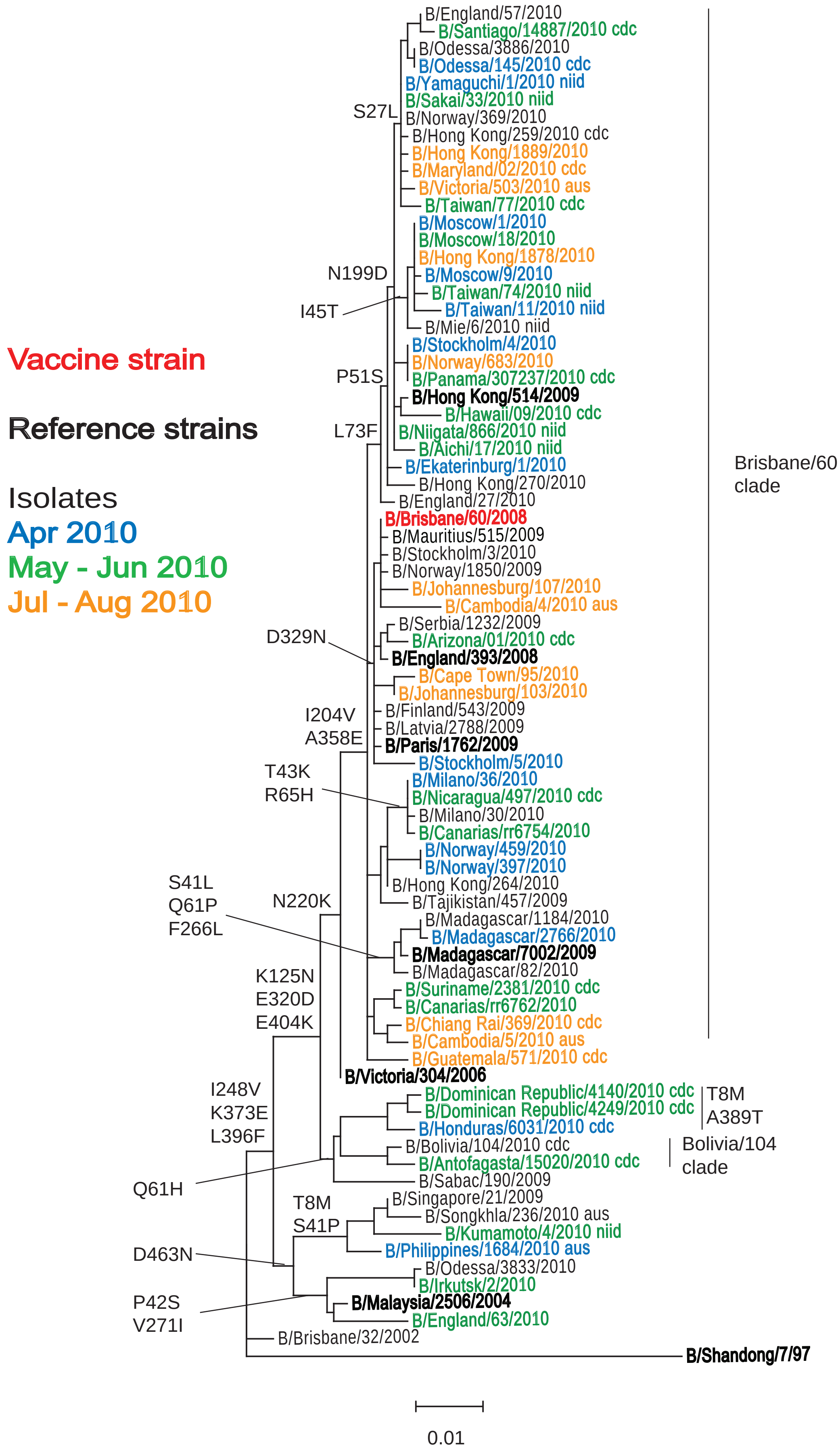


Figure 20. NA protein alignment for B-Victoria lineage viruses

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                             5         15         25         35         45         55         65         75         85         95
B/Brisbane/32/2002       MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SPTEITAPTM PLDCANASNV QAVNRSATKG VTLLLPEPEW TYPRLSCPGS TFQKALLISP
B/Kumamoto/4/2010        .......M.. ....L..... .......... ...L...... P......... .........I .......... .......... .......... ..........
B/Singapore/21/2009      .......M.. ....L..... .......... .......... P......... .......... .......... .......... .......... ..........
B/Bolivia/104/2010       .......... .......... .....V.... ...X...... L......... ....V..... H.....T... .......... .......... ..........
B/England/63/2010        .......... .......... .......... .......... .S........ .......... .......... .......... .......... ..........
B/Malaysia/2506/2004     .......... .......... .......... .......... PS........ .......... .......... .......... .......... ..........
B/Victoria/304/2006      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/107/2010  .......... .......... .......... ....G..... .......... .......... .......... ..F....... .......... ..........
B/Brisbane/60/2008       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/393/2008       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Arizona/01/2010        .......... .......... .......... ....G..... .......... .......... .......... .......... .......... ..........
B/Cambodia/4/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... I.........
B/Mauritius/515/2009     .......... .......... .......... .......... .....I.... .......... .......... .......... .......... ..........
B/Paris/1762/2009        .......... .......... .......... .......... .........T .......... .......... .......... .......... ..........
B/Hong Kong/264/2010     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Nicaragua/497/2010     .......... .......... .......... .......... ..K....... .......... ....H..... .......... .......... ..........
B/Milano/36/2010         .......... .......... .......... .......... ..K....... .......... ....H..... .......... .......... ..........
B/Norway/459/2010        .......... .......... .......... .......... .......... .......... .......... ..F....... .......... ..........
B/Canarias/RR6762/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Cambodia/5/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Madagascar/7002/2009   .......... .......... .......... .......... L......... .......... P......... .......... .......... ..........
B/Hong Kong/514/2009     .......... .......... .......... .......... .......... S......... .......... ..S....... .......... ..........
B/Mie/6/2010             .......... .......... .......... .......... ....T..... S......... .......... ..F....... .......... ..........
B/Yamaguchi/1/2010       .......... .......... ......L... .......... .......... S......... .......... ..F....... .......... ..........
B/Hong Kong/259/2010     .......... .......... ......L... .......... .......... S......... .......... ..F....... .......... ..........
B/Irkutsk/2/2010         .......... .......... .......... .......... .S........ .......... .......... .......... .......... ..........
B/Odessa/3886/2010       .......... .......... ......L... .......... .......... S......... .......... ..F....... .......... ..........
B/England/57/2010        .......... .......... ......L... .......... .......... S......... .......... ..F....... .......... ...R......
B/Moscow/18/2010         .......... .......... .......... .......... ....T..... S......... .......... ..F....... .......... ..........
B/England/27/2010        .......... .......... .......... .......... .......... .......... .......... ..F....... .......... ..........
B/Panama/307237/2010     .......... .......... .......... .......... .......... S......... .......... ..F....... .......... ..........
B/Stockholm/4/2010       .......... .......... .......... .......... .......... S......... .......... ..F....... .......... ..........

                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                            105        115        125        135        145        155        165        175        185        195
B/Brisbane/32/2002       HRFGETKGNS APLIIREPFI ACGPKECKHF ALTHYAAQPG GYYNGTRGDR NKLRHLISVK LGKIPTVENS IFHMAAWSGS ACHDGKEWTY IGVDGPDNNA
B/Kumamoto/4/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Singapore/21/2009      .....A.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bolivia/104/2010       .....IR... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/England/63/2010        .....I.... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Malaysia/2506/2004     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Victoria/304/2006      .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/107/2010  .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Brisbane/60/2008       .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/England/393/2008       .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Arizona/01/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Cambodia/4/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Mauritius/515/2009     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Paris/1762/2009        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/264/2010     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Nicaragua/497/2010     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Milano/36/2010         .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Norway/459/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Canarias/RR6762/2010   .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Cambodia/5/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Madagascar/7002/2009   .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/514/2009     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Mie/6/2010             .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Yamaguchi/1/2010       .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Hong Kong/259/2010     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Irkutsk/2/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Odessa/3886/2010       .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/England/57/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Moscow/18/2010         .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/England/27/2010        .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ..........
B/Panama/307237/2010     .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.
B/Stockholm/4/2010       .......... .......... ....N..... .......... .......... .......... .......... .......... .......... ........D.

                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                            205        215        225        235        245        255        265        275        285        295
B/Brisbane/32/2002       LLKIKYGEAY TDTYHSYANN ILRTQESACN CIGGNCYLMI TDGSASGISE CRFLKIREGR IIKEIFPTGR VKHTEECTCG FASNKTIECA CRDNSYTAKR
B/Kumamoto/4/2010        .........F .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Singapore/21/2009      .......... .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Bolivia/104/2010       .......... .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/England/63/2010        .......... .......... .......... .......... .......V.. .......... .......... T......... .......... ..........
B/Malaysia/2506/2004     .......... .......... .......... .......... .......V.. .......... .......... I......... .......... ..........
B/Victoria/304/2006      .......... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Johannesburg/107/2010  ...V...... .........K .......... .......... .......V.. .......... .......... .......... ....E..... ..........
B/Brisbane/60/2008       ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/England/393/2008       ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Arizona/01/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Cambodia/4/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Mauritius/515/2009     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Paris/1762/2009        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Hong Kong/264/2010     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Nicaragua/497/2010     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Milano/36/2010         ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Norway/459/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Canarias/RR6762/2010   ...V...... .........K .......... .........V .......V.. .......... .......... .......... .......... ..........
B/Cambodia/5/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Madagascar/7002/2009   ...V...... .........K .......... .......... .......V.. .......... .....L.... .......... .......... ..........
B/Hong Kong/514/2009     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Mie/6/2010             ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Yamaguchi/1/2010       ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Hong Kong/259/2010     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Irkutsk/2/2010         .......... .......... .......... .......... .......V.. .......... .......... IQ........ .......... ..........
B/Odessa/3886/2010       ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/England/57/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Moscow/18/2010         ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/England/27/2010        ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Panama/307237/2010     ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Stockholm/4/2010       ...V...... .........K .......... .......... .......V.. .......... .......... .......... .......... ..........
```

```
                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                305        315        325        335        345        355        365        375        385        395
B/Brisbane/32/2002        PFVKLNVETD TAEIRLMCTE TYLDTPRPDD GSITGPCESN GDKGSGGIKG GFVHQRMASK IGRWYSRTMS KTKRMGMGLY VKYDGDPWAD SDALALSGVM
B/Kumamoto/4/2010         .......... .......... .......... .......... .......... .......T.. .......... ..E....... .......... .....F....
B/Singapore/21/2009       .......... .......... .......... .......... .......... .......... .......... ..E....... .......... .....F....
B/Bolivia/104/2010        .......... .........D .......... .......... .N........ .......... .......... ..E....... .......... .....F....
B/England/63/2010         .......... .......... .......... .......... .N........ .......... .......... ..E....... .......... .....F....
B/Malaysia/2506/2004      .......... .......... .......... .......... .......... .......... .......... .......... .......... .....F....
B/Victoria/304/2006       .......... .........D .......... .......... .......... .......... .......... ..E....... .......... .....F....
B/Johannesburg/107/2010   .......... .........D .......... .......... ....G..... .......... .......... ..E....... .......... .....F....
B/Brisbane/60/2008        .......... .........D ........N. .......... .......... .......E.. .......... ..E....... .......... .....F....
B/England/393/2008        .......... .........D ........N. .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Arizona/01/2010         .......... .........D ........N. .......... .......... .......E.. .......... ..E....E.. .......... .....F....
B/Cambodia/4/2010         .......... .........D ........N. .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Mauritius/515/2009      .......... .........D ........N. .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Paris/1762/2009         .......... .........D ........N. .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Hong Kong/264/2010      .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Nicaragua/497/2010      .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Milano/36/2010          .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Norway/459/2010         .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... ....TF....
B/Canarias/RR6762/2010    .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Cambodia/5/2010         ........N. .........D .......... .......... .......... .......E.. .......... ..E....... .......... ....TF....
B/Madagascar/7002/2009    .......... .........D .......... .......... .......... .......E.. .......... ..E....... .........E .....F....
B/Hong Kong/514/2009      .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Mie/6/2010              .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Yamaguchi/1/2010        .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Hong Kong/259/2010      .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Irkutsk/2/2010          .......... .........K .......... .......... .G........ .......... .......... ..E.K..... ...N...... .....F....
B/Odessa/3886/2010        .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/England/57/2010         .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Moscow/18/2010          .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/England/27/2010         .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Panama/307237/2010      .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....
B/Stockholm/4/2010        .......... .........D .......... .......... .......... .......E.. .......... ..E....... .......... .....F....

                          ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|..
                                405        415        425        435        445        455        465
B/Brisbane/32/2002        VSMEEPGWYS FGFEIKDKKC DVPCIGIEMV HDGGKETWHS AATAIYCLMG SGQLLWDTVT GVDMAL*
B/Kumamoto/4/2010         ...K...... .......... .......... .......... .......... .......... ..N...-
B/Singapore/21/2009       ...K...... .......... .......... .......... .......... .......... ..N...*
B/Bolivia/104/2010        ...K...... .......... .......... .......... .......... .......... ......*
B/England/63/2010         .......... .......... .......... .......... .......... .......... ..N...*
B/Malaysia/2506/2004      .......... .......... .......... .......... .......... .......... ..N...*
B/Victoria/304/2006       ...K...... .......... .......... .......... .......... .......... ......*
B/Johannesburg/107/2010   ...K...... .......... .......... .......... .......... .......... ......*
B/Brisbane/60/2008        ...K...... .......... .......... .......... .......... .......... ......*
B/England/393/2008        ...K...... .......... .......... .......... .......... .......... ......*
B/Arizona/01/2010         ...K...... .......... .......... .......... .......... .......... ..N...*
B/Cambodia/4/2010         ...K...... .......... .......... .......... .......... .......... ......*
B/Mauritius/515/2009      ...K...... .......... .......... .......... .......... .......... ......*
B/Paris/1762/2009         ...K...... .......... .......... .......... .......... .......... ......*
B/Hong Kong/264/2010      ...K...... .......... .......... .......... .......... .......... ......*
B/Nicaragua/497/2010      ...K...... .......... .......... .......... .......... .......... ......*
B/Milano/36/2010          ...K...... ...X...... .......... .......... .......... .......... ......*
B/Norway/459/2010         ...K...... .......... .......... .......... .......... .......... ......*
B/Canarias/RR6762/2010    ...K...... .......... .......... .......... .......... .......... ......*
B/Cambodia/5/2010         ...K...... .......... .......... .......... .......... .......... ......*
B/Madagascar/7002/2009    ...K...... .......... .......... .......... .......... .......... ......*
B/Hong Kong/514/2009      ...K...... .......... .......... .......... .......... .......... ......*
B/Mie/6/2010              ...K...... .......... .......... .......... .......... .......... ......-
B/Yamaguchi/1/2010        ...K...... .......... .......... .......... .......... .......... ......-
B/Hong Kong/259/2010      ...K...... .......... .......... .......... .......... .......... ......*
B/Irkutsk/2/2010          I......... .......... .......... .......... .......... .......... ..N.T.*
B/Odessa/3886/2010        ...K...... .......... .......... .......... .......... .......... ......*
B/England/57/2010         ...K...... .......... .......... .......... .......... .......... ......*
B/Moscow/18/2010          ...K...... .......... .......... .......... .......... .......... ......*
B/England/27/2010         ...K...... .......... .......... .......... .......... .......... ......*
B/Panama/307237/2010      ...K...... .....R.... .......... .......... .......... .......... ......*
B/Stockholm/4/2010        ...K...... .....R.... .......... .......... .......... .......... ......X
```

Potential N-linked glycosylation sites are highlighted.

Table 12. Summary of influenza B viruses (Yamagata-lineage) received, collected between January and present

MONTH Country	B Yamagata lineage				
	Number received	Number propagated	≤2 fold*	4 fold*	≥8 fold*
MARCH					
France					
Hong Kong	2	2	2		
Latvia	4	2	2		
APRIL					
Ghana	1	1	1		
Latvia	1	0			
JUNE					
Algeria	1	1	1		
South Africa	1	1	1		
JULY					
Hong Kong	2	2	2		
South Africa	1	1	1		
Total	**13**	**10**	**10**		

* Reduction in HI titre with ferret antiserum raised against B/Florida/4/2006 (egg grown virus)

Table 13. Antigenic analyses of influenza B viruses (Yamagata lineage)

Viruses	Collection date	Passage History	Haemagglutination inhibition titre[1]						
				Post infection ferret sera					
			B/Fl[2] 4/06 Sh 478	B/Eg 144/05 F3/07	B/Fl 4/06 F1/10	B/Bris 3/07 F24/07	B/Eng 145/08 F9/08	B/Vall 18/08 F10/08	B/Ban 3333/07 F21/08
REFERENCE VIRUSES									
B/Egypt/144/2005	2005-05-01	E9	1280	**160**	320	320	40	10	80
B/Florida/4/2006	2006-12-15	E7	**1280**	160	**320**	320	40	10	160
B/Brisbane/3/2007	2007-09-03	E6	1280	160	320	**320**	40	10	160
B/England/145/2008		Ex/E4	160	<	80	<	**80**	<	10
B/Valladolid/18/2008	2008-02-18	MDCK1/MDCK3	1280	160	160	80	160	**640**	160
B/Bangladesh/3333/2007	2007-08-19	E7	1280	80	320	160	40	20	**320**
TEST VIRUSES									
B/Wisconsin/01/2010	2010-02-20	E3/E1	640	40	40	40	40	10	40
B/Hong Kong/345/2010[3]	2010-03-08	MDCK2/MDCK1	1280	160	320	320	320	640	640
B/Hong Kong/398/2010	2010-03-09	MDCK2/MDCK1	1280	160	320	320	320	320	640
B/Latvia/3-1468/2010	2010-03-29	MDCK1/MDCK1	640	80	320	80	160	320	320
B/Latvia/3-1498/2010	2010-03-29	MDCK1/MDCK1	640	80	160	80	160	320	160
B/Ghana/FS-2112/2010	2010-04-07	SIAT1/MDCK1	640	80	160	80	160	320	320
B/Algeria/G486/2010	2010-06-06	SIAT0/MDCK1	640	80	160	80	160	320	160
B/Johannesburg/23/2010	2010-06-28	MDCK2/MDCK1	5120	160	160	320	320	160	320
B/Hong Kong/1832/2010	2010-07-03	MDCK2/MDCK1	2560	160	160	160	160	320	160
B/Johannesburg/40/2010	2010-07-12	MDCK1/MDCK1	5120	160	320	320	320	160	320
B/Hong Kong/1895/2010	2010-07-16	MDCK2/MDCK1	2560	160	160	160	40	320	320

1. < = <10; 2. hyperimmune sheep serum, 3. Sequences included in HA phylogeny

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

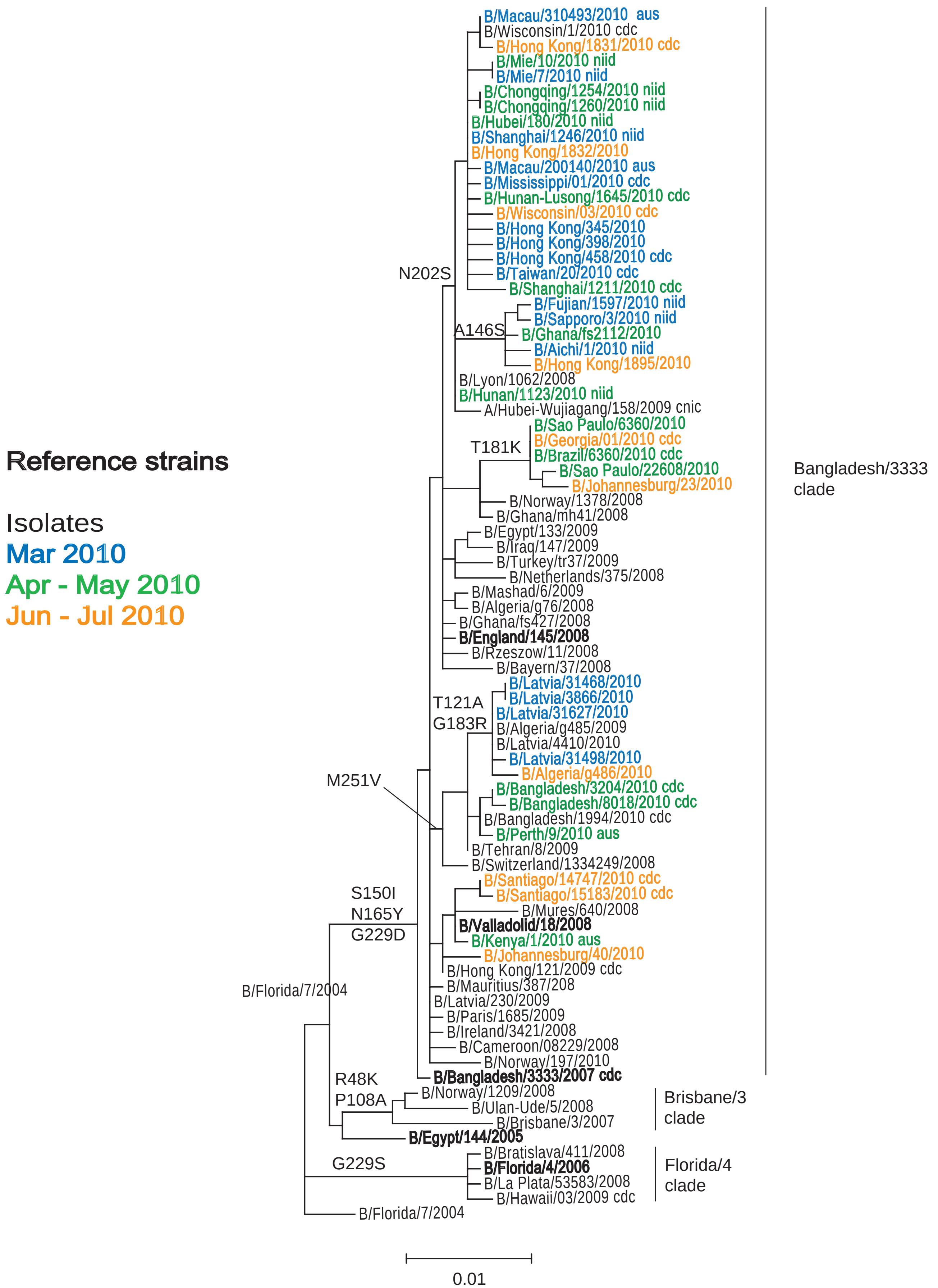


Figure 22. HA1 protein alignment for B-Yamagata lineage viruses

```
                              ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                  5         15         25         35         45         55         65         75         85         95
B/Florida/7/2004              DRICTGITSS NSPHVVKTAT QGEVNVTGVI PLTTTPTKSY FANLKGTRTR GKLCPDCLNC TDLDVALGRP MCVGTTPSAK ASILHEVRPV TSGCFPIMHD
B/Florida/4/2006              .......... .......... .......... .......... .......... .......... .......... .......... .......K.. ..........
B/Egypt/144/2005              .......... .......... .......... .......... .......... .......... .......... .......... ......I... ..........
B/Brisbane/3/2007             .......... .......... .......... .......... .......K.. .......... .......... .......... .......... ..........
B/Ulan-Ude/5/2008             .......... .......... .......... .......... .......K.. .......... .......... .......... .......... ..........
B/Norway/1209/2008            .......... .......... .......... .......... .......K.. .......... .......... .......... .......... ..........
B/Bangladesh/3333/2007        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Valladolid/18/2008          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/40/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Tehran/8/2009               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bangladesh/1994/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Algeria/G486/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31468/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31498/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/145/2008            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mashad/6/2009               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Iraq/147/2009               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Egypt/133/2009              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/MH41/2008             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Turkey/TR37/2009            .......... .......... .......... .......... .......... .......... .......... .......... ......I... ..........
B/Johannesburg/23/2010        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hubei-Wujiagang/158/2009    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1895/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/FS2112/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Sapporo/3/2010              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3981/2010         .......... .......... .......... .......... .......... .......... .......... .......... .........I ..........
B/Hong Kong/3451/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mississippi/01/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Macau/200140/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1832/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mie/10/2010                 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Wisconsin/01/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........

                              ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                 105        115        125        135        145        155        165        175        185        195
B/Florida/7/2004              RTKIRQLPNL LRGYENIRLS TQNVIDAEKA PGGPYRLGTS GSCPNATSKS GFFATMAWAV PKDNNKNATN PLTVEVPYIC TEGEDQITVW GFHSDDKTQM
B/Florida/4/2006              .......... .......... .......... .......... .......... .......... .......... .......... .......... .....N....
B/Egypt/144/2005              .......... .......... .......... .......... .......... .......... .......... .Q........ .......... ..........
B/Brisbane/3/2007             .......A.. .......... .......... .......... .......... .......... .......... .......... .Q........ ..........
B/Ulan-Ude/5/2008             .......A.. .......... .......... .......... .......... .......... .......... .......... .......... .....N....
B/Norway/1209/2008            .......A.. .......... .......... .......... .......... .......... .......... .......... .......... .....N....
B/Bangladesh/3333/2007        .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....X....
B/Valladolid/18/2008          .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Johannesburg/40/2010        .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Tehran/8/2009               .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Bangladesh/1994/2010        .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Algeria/G486/2010           .......... .......... A......... .......... .........I .......... ....Y..... .......... ..R....... .....N....
B/Latvia/31468/2010           .......... .......... A......... .......... .........I .......... ....Y..... .......... ..R....... .....N....
B/Latvia/31498/2010           .......... .......... A......... .......... .........I .......... ....Y..... .......... ..R....... .....N....
B/England/145/2008            .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N.A..
B/Mashad/6/2009               .......... .......... .......... .......... .....T...I .......... ....Y..... .......... .......... .....N....
B/Iraq/147/2009               .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Egypt/133/2009              .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N.I..
B/Ghana/MH41/2008             .......... .......... .......... .......... .........V .......... ....Y..... .......... .......... .....N....
B/Turkey/TR37/2009            .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Johannesburg/23/2010        .......... .......... .......... .......... .........I .......... ....Y..... .......... K......... .....N....
B/Hubei-Wujiagang/158/2009    .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N.N..
B/Hong Kong/1895/2010         .......... .......... .......... .......... .....S...I .......... ....Y..... .......... .......... .....N....
B/Ghana/FS2112/2010           .......... .......... .......... .......... .....S...I .......... ....Y..... .......... .......... .....N....
B/Sapporo/3/2010              .......... .......... .......... .......... .....S...I .......... ....Y..... .......... K......... .....N....
B/Hong Kong/3981/2010         .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Hong Kong/3451/2010         .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Mississippi/01/2010         .......... .......... .......... .......... .........I .......... ....Y..T.. .......... .......... .....N....
B/Macau/200140/2010           .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Hong Kong/1832/2010         .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
B/Mie/10/2010                 .......... .....K.... .......... .......... .........I .......... ....Y..... .......... .......... .....X....
B/Wisconsin/01/2010           .......... .......... .......... .......... .........I .......... ....Y..... .......... .......... .....N....
```

```
                              ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                  205        215        225        235        245        255        265        275        285        295
B/Florida/7/2004              KNLYGDSNPQ KFTSSANGVT THYVSQIGGF PDQTEDGGLP QSGRIVVDYM MQKPGKTGTI VYQRGVLLPQ KVWCASGRSK VIKGSLPLIG EADCLHEKYG
B/Florida/4/2006              .......... .......... ........S. .......... .......... .......... .......... .......... .......... ..........
B/Egypt/144/2005              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Brisbane/3/2007             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ulan-Ude/5/2008             .......... .......... .......... .......... .........I .......... .....I.... .......... .......... ..........
B/Norway/1209/2008            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bangladesh/3333/2007        .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Valladolid/18/2008          .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/40/2010        .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Tehran/8/2009               .......... .......... ........D. .......... .......... V......... .......... .......... .......... ..........
B/Bangladesh/1994/2010        .......... .......... ........D. .......... .......... V......... .......... .......... .......... ..........
B/Algeria/G486/2010           .......... .......... ........D. .......... .......... V.......L. .......... .......... .......... ..........
B/Latvia/31468/2010           .......... .......... ........D. .......... .......... V......... .......... .......... .......... ..........
B/Latvia/31498/2010           .......... .......... ........D. .......... .......... V......... .......... .......... .......... ..........
B/England/145/2008            .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Mashad/6/2009               .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Iraq/147/2009               .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Egypt/133/2009              .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Ghana/MH41/2008             .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Turkey/TR37/2009            .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/23/2010        .......... .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Hubei-Wujiagang/158/2009    .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1895/2010         .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Ghana/FS2112/2010           .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Sapporo/3/2010              .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3981/2010         .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3451/2010         .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Mississippi/01/2010         .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Macau/200140/2010           .S........ .......... ........D. .V........ .......... .......... .......... .......... .......... ..........
B/Hong Kong/1832/2010         .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Mie/10/2010                 .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........
B/Wisconsin/01/2010           .S........ .......... ........D. .......... .......... .......... .......... .......... .......... ..........

                              ....|....| ....|....| ....|....| ....|....| ....|.
                                  305        315        325        335        345
B/Florida/7/2004              GLNKSKPYYT GEHAKAIGNC PIWVKTPLKL ANGTKYRPPA KLLKER
B/Florida/4/2006              .......... .......... .......... .......... ......
B/Egypt/144/2005              .......... .......... .......... .......... ......
B/Brisbane/3/2007             .......... .......... .......... PMT....... ......
B/Ulan-Ude/5/2008             .......... .......... .......... .......... ......
B/Norway/1209/2008            .......... .......... .......... .......... ......
B/Bangladesh/3333/2007        .......... .......... .......... .......... ......
B/Valladolid/18/2008          .......... .......... .......... .......... ......
B/Johannesburg/40/2010        .......... .......... .......... .......... ......
B/Tehran/8/2009               .......... .......... .......... .......... ......
B/Bangladesh/1994/2010        .......... .......... .......... .......... ......
B/Algeria/G486/2010           .......... .......... .......... .......... ......
B/Latvia/31468/2010           .......... .......... .......... .......... ......
B/Latvia/31498/2010           .......... .......... .......... .......... ......
B/England/145/2008            .......... .......... .......... .......... ......
B/Mashad/6/2009               .......... .......... .......... .......... ......
B/Iraq/147/2009               .......... .......... .......... .......... ......
B/Egypt/133/2009              .......... .......... .......... .......... ......
B/Ghana/MH41/2008             .......... .......... .......... .......... ......
B/Turkey/TR37/2009            .......... .......... .......... .......... ......
B/Johannesburg/23/2010        .......... .......... .......... .......... ......
B/Hubei-Wujiagang/158/2009    .......... .......... .......... .......... ......
B/Hong Kong/1895/2010         .......... .......... .......... .......... ......
B/Ghana/FS2112/2010           .......... .......... .......... .......... ......
B/Sapporo/3/2010              .......... .......... .......... .......... ......
B/Hong Kong/3981/2010         .......... .......... .......... .......... ......
B/Hong Kong/3451/2010         .......... .......... .......... .......... ......
B/Mississippi/01/2010         .......... .......... .......... .......... ......
B/Macau/200140/2010           .......... .......... .......... .......... ......
B/Hong Kong/1832/2010         .......... .......... .......... .......... ......
B/Mie/10/2010                 .......... .......... .......... .......... ......
B/Wisconsin/01/2010           .......... .......... .......... .......... ......
```

Figure 23. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

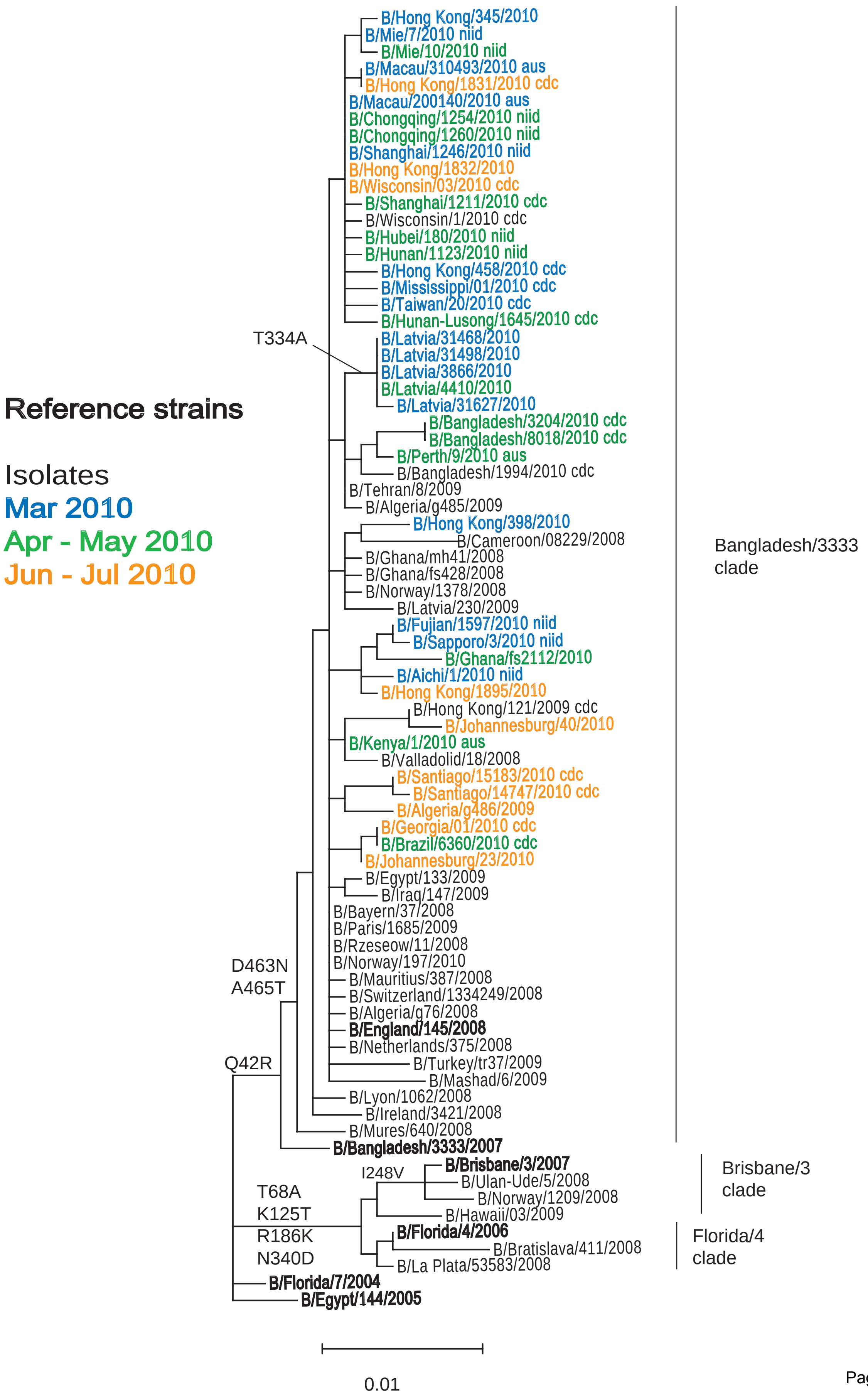


Figure 24. NA protein alignment for B-Yamagata lineage viruses

```
                                ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                    5         15        25        35        45        55        65        75        85        95
B/Florida/7/2004                MLPSTIQTLT LFLTSGGVLL SLYVSASLSY LLYSDILLKF SQTEITAPIM PLDCANASNV QAVNHSATKG VTLLLPEPEW TYPRLSCPGS TFQKALLISP
B/Florida/4/2006                .......... .......... .......... .......... .......... .......... ....R..A.. .......... .......... ..........
B/Egypt/144/2005                .......... .......... .......... ........R. .......... .......... ....R..... .......... .......... ..........
B/Brisbane/3/2007               .......K.. .......... .......... .......... .......... .......... ....R..A.. .......... .......... ..........
B/Ulan-Ude/5/2008               .......... .......... .......... .......... .......... .......... H...R..A.. .......... .......... ..........
B/Norway/1209/2008              .......... .......... .......... .......... .......... .......... ....R..A.E .......... .......... ..........
B/Bangladesh/3333/2007          .......... .......... .......... .......... .R..T..... .......... ....R..... .......... .......... ..........
B/Valladolid/18/2008            .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Johannesburg/40/2010          .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Tehran/8/2009                 .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Bangladesh/1994/2010          .......... .......... .......... .......... .R........ .......... ....R..... ..P....... .......... ..........
B/Algeria/G486/2010             .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Latvia/31468/2010             .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Latvia/31498/2010             .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/England/145/2008              .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Mashad/6/2009                 .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Iraq/147/2009                 .......... .......... .......... .....M.... .R........ .......... ....R..... .......... .......... ..........
B/Egypt/133/2009                .......... .......... .......... .......... .R........ .......... ....R..... ........D. .......... ..........
B/Ghana/MH41/2008               .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Turkey/TR37/2009              .......... .......... .......... .......... .R........ .......... ....R..... ..F....... .......... ..........
B/Johannesburg/23/2010          .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Hong Kong/1895/2010           .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Ghana/FS2112/2010             .......... .......... .......... .......... .R...NT... .......... ....R..... .......... .......... ..........
B/Sapporo/3/2010                .......... .......... .......... .......... .R...N.... .......... ....R..... .......... .......... ..........
B/Hong Kong/3981/2010           .......... .......... .......... .......... .R........ .......... .......... .......... .......... ..........
B/Hong Kong/3451/2010           .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Mississippi/01/2010           .........X .......... .......... .......... .R........ ....V..... ....R..... .......... .......... ..........
B/Macau/200140/2010             .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Hong Kong/1832/2010           .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Mie/10/2010                   .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........
B/Wisconsin/01/2010             .......... .......... .......... .......... .R........ .......... ....R..... .......... .......... ..........

                                ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                    105        115       125       135       145       155       165       175       185       195
B/Florida/7/2004                HRFGETKGNS APLIIREPFI ACGPKECKHF ALTHYAAQPG GYYNGTREDR NKLRHLISVK LGKIPTVENS IFHMAAWSGS ACHDGREWTY IGVDGPDSNA
B/Florida/4/2006                .......... .......... ....T..... .......... .......... .......... .......... .......... .....K.... ..........
B/Egypt/144/2005                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Brisbane/3/2007               .......... .......... ....T..... .......... .......... .......... .......... .......... .....K.... ..........
B/Ulan-Ude/5/2008               .......... .......... ....T..... .......... .......... .......... .......... .......... .....K.... ..........
B/Norway/1209/2008              .......... .......... ....T..... .......... .......... .......... .......... .......... .....K.... ..........
B/Bangladesh/3333/2007          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Valladolid/18/2008            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/40/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Tehran/8/2009                 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bangladesh/1994/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Algeria/G486/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31468/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31498/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/145/2008              .......... .......... .......... .......... .......... .........R .......... .......... .......... ..........
B/Mashad/6/2009                 .......... .......... .......... .......... .......... .........N .......... .......... .......... ..........
B/Iraq/147/2009                 .......... .......... .......... .......... .......... ..M....... .......... .......... .......... ..........
B/Egypt/133/2009                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/MH41/2008               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Turkey/TR37/2009              .......... .........V .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/23/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1895/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/FS2112/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... .......N..
B/Sapporo/3/2010                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3981/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3451/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mississippi/01/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Macau/200140/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1832/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mie/10/2010                   X......... .......... .......... .......... .......... .......... .......... .......... .......... .......G..
B/Wisconsin/01/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........

                                ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                                    205        215       225       235       245       255       265       275       285       295
B/Florida/7/2004                LLKIKYGEAY TDTYHSYAKN ILRTQESACN CIGGDCYLMI TDGPASGISE CRFLKIREGR IIKEIFPTGR VKHTEECTCG FASNKTIECA CRDNSYTAKR
B/Florida/4/2006                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Egypt/144/2005                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Brisbane/3/2007               .......... .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Ulan-Ude/5/2008               .......... .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Norway/1209/2008              ........T. .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Bangladesh/3333/2007          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Valladolid/18/2008            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/40/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Tehran/8/2009                 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bangladesh/1994/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Algeria/G486/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31468/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31498/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/England/145/2008              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mashad/6/2009                 .......... .......... .......... .......... .......V.. .......... .......... .......... .......... ..........
B/Iraq/147/2009                 .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Egypt/133/2009                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/MH41/2008               .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Turkey/TR37/2009              .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/23/2010          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1895/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/FS2112/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Sapporo/3/2010                .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3981/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3451/2010           .......... .......... .......... .......... .......... .......... V......... .......... .......... ..........
B/Mississippi/01/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Macau/200140/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1832/2010           .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mie/10/2010                   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Wisconsin/01/2010             .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
```

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|....|
                               305        315        325        335        345        355        365        375        385        395
B/Florida/7/2004         PFVKLNVETD TAEIRLMCTE TYLDTPRPND GSITGPCESN GDKGSGGIKG GFVHQRMASK IGRWYSRTMS KTKRMGMGLY VKYDGDPWTD SEALALSGVM
B/Florida/4/2006         .......... .......... .......... .........D .......... .......... .......... .......... .......... ..........
B/Egypt/144/2005         .......... .......... .......... .......... .......... .......... .......... ..R....... .......... ..........
B/Brisbane/3/2007        .......... .......... .......... .........D .......... .......... .......... .......... .......... ..........
B/Ulan-Ude/5/2008        .......... .......... .......... .........D .......... .......... .......... .......... .......... ..........
B/Norway/1209/2008       .......... .......... .......... .........D .......... .......... .......... .......... .......... ..........
B/Bangladesh/3333/2007   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Valladolid/18/2008     .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/40/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Tehran/8/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Bangladesh/1994/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Algeria/G486/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Latvia/31468/2010      .......... .......... .......... ...A...... .......... .......... .......... .......... .......... ..........
B/Latvia/31498/2010      .......... .......... .......... ...A...... .......... .......... .......... .......... .......... ..........
B/England/145/2008       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mashad/6/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Iraq/147/2009          .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Egypt/133/2009         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/MH41/2008        .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Turkey/TR37/2009       .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Johannesburg/23/2010   .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1895/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Ghana/FS2112/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Sapporo/3/2010         .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/3981/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... .....F....
B/Hong Kong/3451/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mississippi/01/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Macau/200140/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Hong Kong/1832/2010    .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Mie/10/2010            .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
B/Wisconsin/01/2010      .......... .......... .......... .......... .......... .......... .......... .......... .......... ..........
```

```
                         ....|....| ....|....| ....|....| ....|....| ....|....| ....|....| ....|..
                               405        415        425        435        445        455        465
B/Florida/7/2004         VSMEEPGWYS FGFEIKDKKC DVPCIGIEMV HDGGKTTWHS AATAIYCLMG SGQLLWDTVT GVDMAL*
B/Florida/4/2006         .......... .......... .......... .......... .......... .......... ......*
B/Egypt/144/2005         .......... .......... .......... .......... .......... .......... ......*
B/Brisbane/3/2007        .......... .......... .......... .......... .......... .......... ......*
B/Ulan-Ude/5/2008        .......... .......... .......... .......... .......... .......... ......*
B/Norway/1209/2008       .......... .......... .......... .......... .......... .......... ......*
B/Bangladesh/3333/2007   .......... .......... .......... .......... .......... .......... ......*
B/Valladolid/18/2008     .......... .......... .......... .......... .......... .......... ..N.T.*
B/Johannesburg/40/2010   .......... .......... .......... .......... .......... .......... ..N.T.*
B/Tehran/8/2009          .......... .......... .......... .......... .......... .......... ..N.T.*
B/Bangladesh/1994/2010   .......... .......... .......... .......... .......... .......... ..N.T.*
B/Algeria/G486/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
B/Latvia/31468/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
B/Latvia/31498/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
B/England/145/2008       .......... .......... .......... .......... .......... .......... ..N.T.*
B/Mashad/6/2009          .......... .......... .......... .......... .......... .......... ..N.T.*
B/Iraq/147/2009          .......... .......... .......... .......... .......... .......... ..N.T.*
B/Egypt/133/2009         .......... .......... .......... .......... .......... .......... ..N.T.*
B/Ghana/MH41/2008        .......... .......... .......... .......... .......... .......... ..N.T.*
B/Turkey/TR37/2009       .......... .......... .......... .......... .......... .......... ..N.T.*
B/Johannesburg/23/2010   .......... .......... .......... .......... .......... .......... ..N.T.*
B/Hong Kong/1895/2010    .......... .......... .......... .......... .......... .......... ..N.T.*
B/Ghana/FS2112/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
B/Sapporo/3/2010         .......... .......... .......... .......... .......... .......... ..N.T.*
B/Hong Kong/3981/2010    .......... .......... .......... .......... .......... .......... ..N.T.*
B/Hong Kong/3451/2010    .......... .......... .......... .......... .......... .......... ..N.T.*
B/Mississippi/01/2010    .......... .......... .......... .......... .......... .......... ..N.T.*
B/Macau/200140/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
B/Hong Kong/1832/2010    .......... .......... .......... .......... .......... .......... ..N.T.*
B/Mie/10/2010            .......... .......... .......... .......... .......... .......... ..N.T.*
B/Wisconsin/01/2010      .......... .......... .......... .......... .......... .......... ..N.T.*
```

Potential N-linked glycosylation sites are highlighted.

Figure 25. Influenza Viruses Tested for NI resistance at WHO CC London: Collection Date Time-line, (to 01.08.2010).

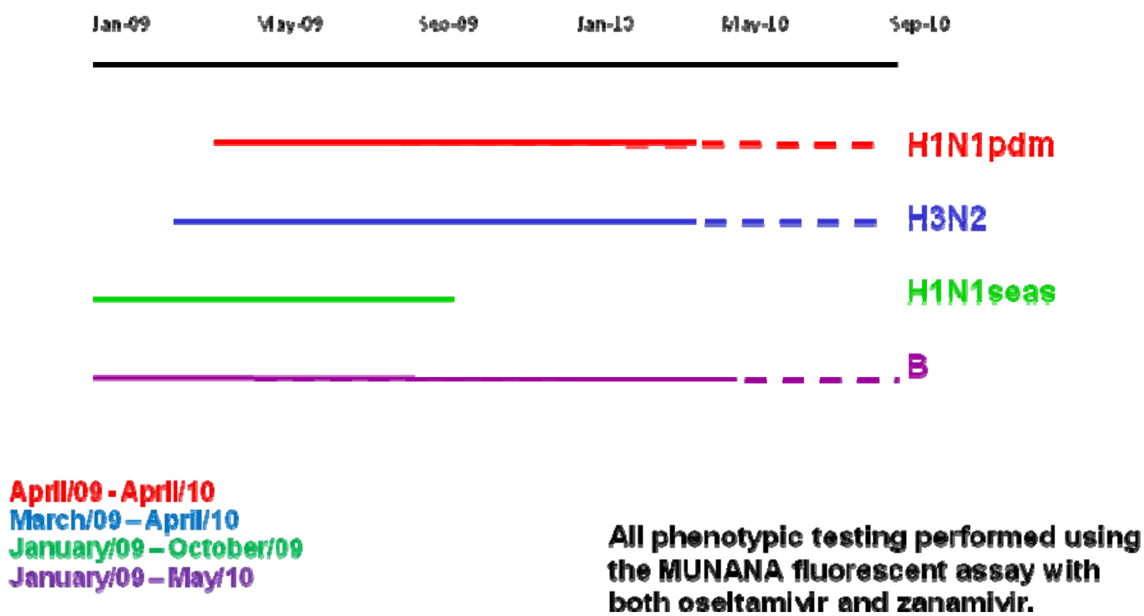


Figure 26. Phenotypic NI susceptibility testing by WHO region.

Region	H1N1 pdm	H3N2	H1N1 seasonal	B	Total
AFRO	146	114	8 (8)	38	306
AMRO		1	2 (2)	1	4
EMRO	80	1		14	95
SEARO	26				26
WPRO	17	21	6 (6)	18	62
5 region subtotal	**269**	**137**	**16 (16)**	**71**	**493**
EURO (East)	338 (2)	10	1 (1)	21	370
EURO (ECDC)	870 (5)	20	5 (5)	8	903
EURO (All)	**1208 (7)**	**30**	**6 (6)**	**29**	**1273**
TOTAL	**1477 (7)**	**167**	**22 (22)**	**100**	**1766**
% oseltamivir resistant	**0.4%**		**100%**		

Numbers in parentheses indicate oseltamivir-resistance (all remain sensitive to zanamivir)

Figure 27. Oseltamivir-resistant H1pdm viruses detected at WHO CC London.

Virus	Collection date	Comment
A/Denmark/528/2009	09/06/2009	First confirmed case – identified by sequence in Denmark, phenotype confirmed at NIMR
A/Israel/MF6290/2009 A/Israel/MF15979/2009	August/2009 November/2009	One case unknown in Israel at time of dispatch
A/Paris/6232/2009 A/Paris/6808/2009	November/2009 November/2009	Sequential samples from same patient, known resistance
A/Belgium/1239/2009	20/11/2009	Unknown in Belgium at time of dispatch
A/Lisboa/171/2009	02/12/2009	Known at time of dispatch, confirmed at NIMR

Figure 28. NA (full-gene) Sequencing at WHO CC London, since January 2009.

NA gene	No sequenced*	Comments
N1pdm**	617	
N1 seasonal	57	This includes samples collected at the end of 2008
N2	213	
Flu B	125	
Total	1012	

* The numbers represent virus isolates and a smaller number of clinical samples. The bulk of the sequences were generated for specimens from laboratories with no/limited sequencing capacity.

** All of these sequences coded for I223 (no substitutions to R or any other amino acids).

Figure 29. H1pdm viruses with reduced susceptibility to Oseltamivir and Zanamivir.

Virus	Collection Date	IC_{50}*		N1 substitutions compared to consensus sequence										
		Ost	Zan	15 M	76 A	77 G	82 S	105 S	199 D	257 R	262 K	263 I	274 Y	396 I
A/Georgia/4032/09	09/12/09	4.85	3.23		T						R			V
A/Livadia/1851/10	26/01/10	6.39	2.96	T			P						H	
A/Argos/2231/10	03/02/10	4.41	1.33			R		N				V		
A/Volgda/3/10	15/03/10	11.80	16.68						G	K				

*** IC_{50} values are normally <2.0 and <1.0 nM for Oseltamivir and Zanamivir respectively.**

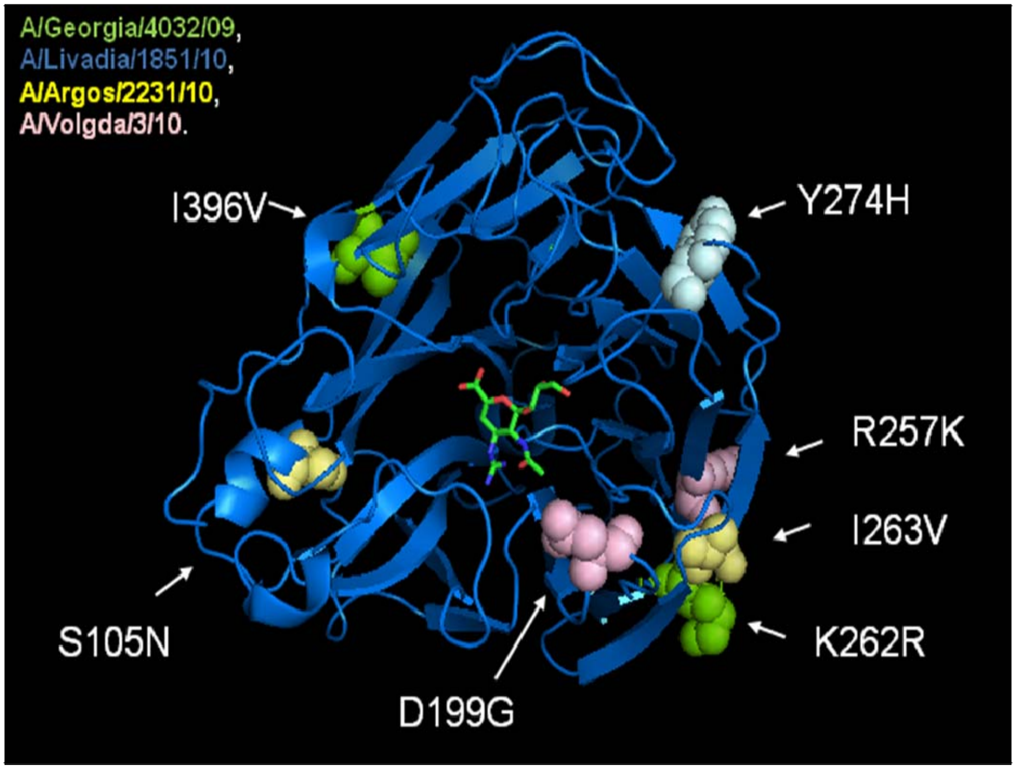
Stoner, TD et al (RG Webster lab): Antiviral Susceptibility of Avian and Swine Influenza of the N1 Neuraminidase Subtype

J. Virol – online ahead of print 21.06.10

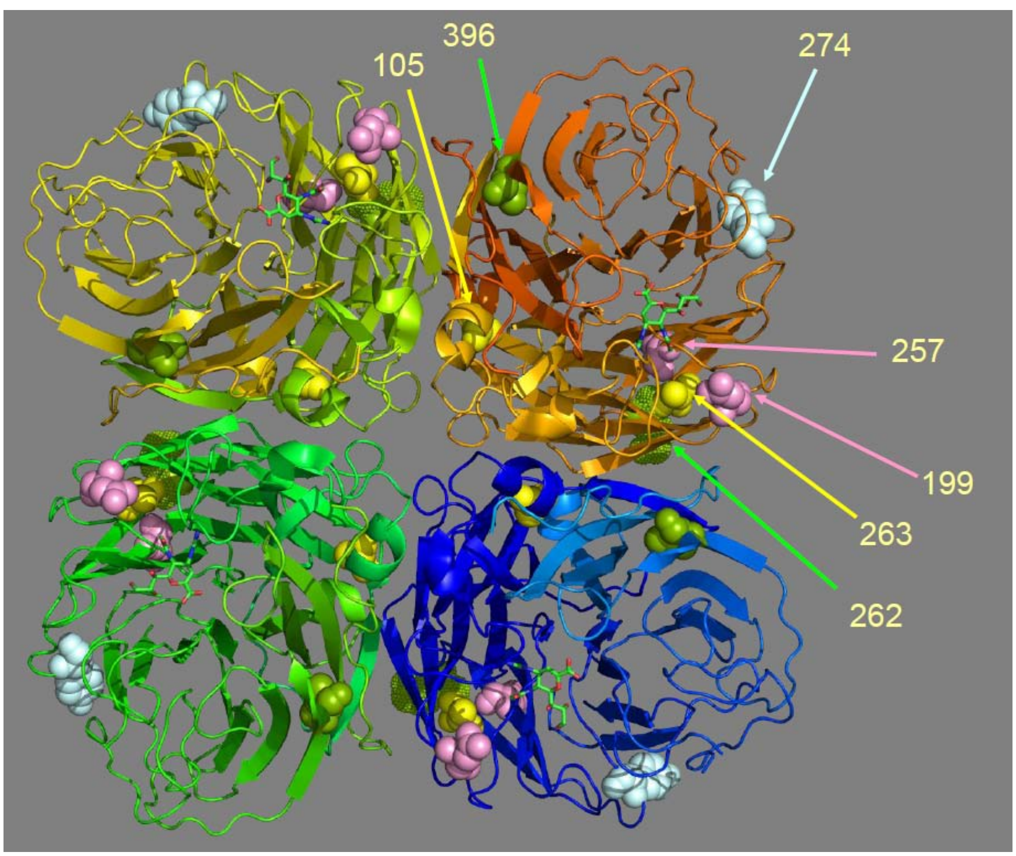
K262R – outlier IC_{50} values – A/Georgia/4032/09 here

G105S – outlier IC_{50} values – A/Argos/2231/10 here (S105N)

Figure 30. Location of NI reduced susceptibility amino acid substitutions on N1 NA.



N1 NA: oseltamivir resistant protein (H275Y) with zanamivir in the active site



Tetrameric view (generated by S. Pennell)